

MMseqs2: The Missing Manual

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Contents

1. MMseqs2 Documentation Overview	5
1.1. What This Manual Assumes	5
1.2. Reading Strategy	5
1.3. Why the Cascade Matters	5
1.4. Source-of-Truth Policy	6
2. Performance Foundations: Algorithmic and Systems Acceleration in MMseqs2	6
2.1. Performance Model	6
2.2. Shared Comparison Backbone	7
2.3. Candidate Generation: K-mer Indexing and Expansion	7
2.4. Diagonal and Ungapped Filters	7
2.5. SIMD Gapped Alignment and Output-Cost Economics	8
2.6. Clustering-Specific Accelerators	8
2.7. Masking and Composition-Bias Tradeoffs	8
2.8. GPU Comparison Backend	8
2.9. Internal Database and Storage Model	9
2.10. Indexing and Data-Access Strategy	9
2.11. Memory Model and Split Tradeoffs	9
2.12. Parallel Execution Model	9
2.13. Practical Tuning Order	10
2.14. Practical Playbooks	10
3. System Map: Cascaded APIs and Module Connections	10
3.1. Layered API Model	10
3.2. Cascade Pattern in Practice	11
3.3. How Submodule Pages Encode This Map	11
3.4. Choosing Your Entry View	11
4. Functional Modules Manual	11
4.1. Functional Groups	12
4.2. Easy Workflows	12
4.3. Search Workflows	19
4.4. Clustering	23
4.5. Prefiltering	30
4.6. Alignment	36
4.7. Profiles	44
4.8. Database Management	53
4.9. Sequence Manipulation	68
4.10. Result Handling	76
4.11. Multi-hit	89
4.12. Taxonomy	94
4.13. Utilities	105
5. Expert Manual	116
5.1. Expert Operating Discipline	116

5.2.	Contract Enforcement in Custom Pipelines	116
5.3.	Reproducibility Controls	116
5.4.	Performance Triage for Advanced Workloads	117
5.5.	MMseqs2 Source Development Guide	117
6.	MMseqs2 Command Reference Index	118
6.1.	Command Group Map	120
6.2.	Command Pages	122
7.	MMseqs2 Dependency Map	344
7.1.	Easy Workflows	345
7.2.	Search Workflows	347
7.3.	Clustering	349
7.4.	Prefiltering	352
7.5.	Alignment	354
7.6.	Profiles	357
7.7.	Database	362
7.8.	Result Handling	370
7.9.	Sequence Manipulation	377
7.10.	Taxonomy	380
7.11.	Multi Hit	385
7.12.	Utilities	387

1. MMseqs2 Documentation Overview

MMseqs2 is designed for a difficult constraint set: very large sequence collections, sensitivity targets close to established alignment tools, and runtime envelopes that still fit production workflows. The project solves this by treating performance as a systems property instead of a single-kernel optimization problem. Workflows, core compute modules, storage format, and split policy are all part of one execution model.

This manual follows that same model. It starts from the big picture, then moves to algorithmic and systems acceleration details, then to functional modules and command-level references. The goal is that a reader can move from a task question (“what should I run?”) to a design question (“why does this run behave this way?”) without switching mental frameworks.

1.1. What This Manual Assumes

The documentation assumes that you will run MMseqs2 as a cascade of stages, not as isolated commands. In practice, prefiltering and indexing choices set the candidate volume and memory behavior, alignment settings control the quality-cost boundary, and result-handling choices decide how downstream interpretation is constrained. If one stage is misconfigured, later stages may still produce outputs, but those outputs may no longer be comparable across runs.

For that reason, this manual repeatedly emphasizes contract stability: database type compatibility, sidecar availability, mode consistency, and reproducible parameter sets. These are not housekeeping details; they are part of algorithmic correctness at scale.

1.2. Reading Strategy

Start with the conceptual chapters before diving into command pages:

1. Read Performance Foundations for algorithmic acceleration, storage/indexing behavior, split policy, and parallel execution tradeoffs.
2. Read System Map to understand the layered API model and command cascade.
3. Use Functional Modules Manual for task-oriented command discovery.
4. Use Command Reference Index and Dependency Map for exact command topology and CLI details.

If you already know your task and only need command selection, you can jump directly to Functional Modules Manual. If you are diagnosing runtime or output drift, read the conceptual chapters first; they explain the cross-stage couplings that usually cause expensive surprises.

1.3. Why the Cascade Matters

A typical MMseqs2 workflow performs database preparation, candidate generation, alignment or rescoring on surviving pairs, and downstream transformation such as clustering, taxonomy assignment, or export. This order is the primary scaling mechanism. Expensive operations are delayed until fast filters shrink the search space.

The practical consequence is that late-stage tuning often has less impact than early-stage tuning. Raising sensitivity or adding richer alignment outputs can be useful, but only after index strategy,

load behavior, and split policy are stable. In large production runs, infrastructure misalignment can dominate wall time long before algorithmic thresholds are the true bottleneck.

1.4. Source-of-Truth Policy

Narrative chapters explain design and usage strategy. Command snapshots in `mmseqs_help_output` define concrete CLI behavior for this repository state. Source code under `MMseqs2/src/` and workflow scripts under `MMseqs2/data/workflow/` remain the final authority when you need implementation-level confirmation.

Use Expert Manual and MMseqs2 Source Development Guide when you need strict reproducibility discipline or source-level debugging and extension guidance.

Note

When narrative guidance and local command snapshots diverge, treat local snapshots and source code as canonical for exact behavior.

2. Performance Foundations: Algorithmic and Systems Acceleration in MMseqs2

MMseqs2 performance is a coupled system. Algorithmic pruning, kernel implementation, storage layout, indexing strategy, split policy, and execution topology all interact. This chapter unifies those layers so tuning decisions are made in the right order and interpreted with the correct assumptions.

2.1. Performance Model

MMseqs2 does not rely on one fast algorithm. It applies a cascade that removes expensive comparisons early, computes detailed alignment only for survivors, and keeps data movement predictable through explicit database contracts.

In practical terms, runtime and output behavior are jointly determined by three levers:

Lever	What It Changes	Typical Mechanisms
Candidate reduction	Number of sequence pairs that reach expensive stages	prefiltering, k-mer matching, diagonal filtering, clustering prefilterers
Per-pair cost	Compute cost of each surviving pair	ungapped rescoring, SIMD alignment modes, backtrace settings
Data-movement cost	Startup and merge overhead around kernels	index reuse, preload mode, split policy, temporary I/O topology

The highest gains usually come from ordering these levers correctly: stabilize contracts and data movement first, then tune candidate volume, then tune expensive alignment settings.

2.2. Shared Comparison Backbone

Search and clustering workflows share a common execution shape. Workflows select paths, pre-filtering reduces candidate pairs, alignment or rescoring computes pair quality, and downstream modules convert those results into task artifacts.

For CPU search paths, the orchestration is explicit in `MMseqs2/data/workflow/blastp.sh`: prefilter then align, with optional iterative merge logic. Clustering orchestration (`MMseqs2/src/workflow/Cluster.cpp` and `MMseqs2/data/workflow/*.sh`) reuses the same primitives and then adds graph construction and cluster assignment.

Because this backbone is shared, upstream prefilter/index decisions usually have larger global impact than late-stage alignment-format tweaks.

2.3. Candidate Generation: K-mer Indexing and Expansion

The prefilter core (`MMseqs2/src/prefiltering/Prefiltering.cpp`, `QueryMatcher.cpp`, `IndexBuilder.cpp`) controls candidate growth through index-driven k-mer retrieval and thresholded similar-k-mer expansion.

For amino-acid searches, similar k-mers are generated from precomputed score matrices (`ExtendedSubstitutionMatrix`, `KmerGenerator`) rather than brute-force enumeration. For nucleotide or exact-match paths, `MMseqs2` can switch to direct exact-k-mer behavior (`takeOnlyBestKmer`) to reduce expansion overhead.

Composition-bias correction in `QueryMatcher::matchQuery` shifts effective k-mer thresholds per position. This couples sensitivity and bias controls: `-s`, `--k-score`, `--alph-size`, `--spaced-kmer-mode`, and bias settings jointly determine index lookup volume and therefore prefilter cost.

When an index DB is reused, metadata compatibility checks (k-mer size, alphabet, spaced-kmer mode, split assumptions) are operationally important. Mismatch is not only a correctness issue; it can change the effective complexity of candidate generation.

2.4. Diagonal and Ungapped Filters

After raw seed hits are collected, `MMseqs2` aggregates by target and diagonal and keeps strongest evidence per target (`QueryMatcher`). This compresses large hit lists into a tractable candidate set.

With diagonal scoring enabled, `UngappedAlignment` computes local ungapped scores on diagonals using SIMD paths (AVX2/SSE fallback in `MMseqs2/src/prefiltering/UngappedAlignment.cpp`). This stage is much cheaper than full Smith-Waterman and often removes weak candidates early enough to dominate total runtime savings.

Performance

If runtime is high, reduce candidate volume and tune ungapped gates before tuning full alignment output richness.

2.5. SIMD Gapped Alignment and Output-Cost Economics

Full alignment (`MMseqs2/src/alignment/Alignment.cpp`) dispatches through `Matcher` to vectorized Striped Smith-Waterman (`StripedSmithWaterman.cpp`) for amino-acid/profile paths and banded nucleotide alignment for nucleotide paths.

The dynamic programming score is only part of cost. Coverage/identity calculations, alternative alignment handling, and backtrace generation can substantially increase per-pair cost and output size. Alignment mode is therefore both a semantic choice and a throughput choice.

Use richer alignment outputs only when downstream interpretation explicitly needs them.

2.6. Clustering-Specific Accelerators

Clustering workflows add graph-level acceleration on top of search-like filtering. Cascaded clustering intentionally starts with cheaper passes and increases sensitivity later (`MMseqs2/src/workflow/Cluster.cpp`). Early passes shrink graph density before expensive stages.

Single-step clustering scripts use redundancy shortcuts such as `clusthash` before full pre-filter+alignment. `Linclust` (`MMseqs2/src/workflow/Linclust.cpp`) applies bounded-k-mer selection plus cheap filters (Hamming and ungapped rescoring) before full local alignment, which is why it scales near-linearly for very large sequence sets.

2.7. Masking and Composition-Bias Tradeoffs

Low-complexity masking and composition-bias correction are both speed and quality controls. Masking reduces repetitive false seed growth during indexing and matching. Composition-bias correction reduces spurious composition-driven matches in prefilter and alignment scoring.

Disabling these controls can improve throughput in some data regimes, but it can also destabilize precision and make run-to-run comparisons less interpretable if mode envelopes are not fixed.

2.8. GPU Comparison Backend

GPU acceleration is implemented in `ungappedprefilter` (`MMseqs2/src/prefiltering/ungappedprefilter.cpp`) with optional persistent server mode in `gpuserver` (`MMseqs2/src/util/gpuserver.cpp`).

In GPU mode, query profiles are prepared on CPU and scanned against GPU-resident target layouts. In server mode, target data remains resident and query exchange uses shared memory, reducing repeated startup overhead for low-latency repeated searches.

This backend targets search workflows; clustering workflows are not GPU-accelerated in the current architecture.

2.9. Internal Database and Storage Model

Most MMseqs2 modules exchange MMseqs2 DB artifacts rather than plain FASTA. Contiguous record files plus explicit index offsets reduce filesystem overhead and enable direct key-based access.

Core file roles are stable: `<db>` stores records, `<db>.index` maps IDs to offsets/lengths, and `<db>.dbtype` encodes contract type. Header and lookup sidecars (`<db>_h`, `<db>.lookup`, taxonomy sidecars) separate sequence-heavy compute paths from annotation-heavy export paths.

This design is why MMseqs2 favors writing new DBs over in-place edits. Offsets are positional contracts; mutating payload layout in place would invalidate index semantics.

Note

Header and sequence separation is intentional. Compute-heavy kernels can avoid header I/O, while export and annotation stages opt in to sidecar reads.

2.10. Indexing and Data-Access Strategy

Precomputed indexes move cost from repeated startup into reusable artifacts. They pay off most when targets are stable across many runs. For short-lived targets, on-the-fly indexing can be cheaper than index management overhead.

Storage topology determines whether index reuse helps or hurts. On slower shared storage, startup can become I/O-bound even if compute kernels are fast. In those environments, local caching, preload controls, and resident-memory behavior may produce larger gains than threshold retuning.

`--db-load-mode` changes startup/throughput tradeoffs and should be selected as part of infrastructure design, not as an afterthought.

2.11. Memory Model and Split Tradeoffs

Prefilter structures are typically the largest memory consumers. Legacy MMseqs2 guidance models major prefilter terms as:

$$M = (7 * N * L + 8 * a^k) \text{ bytes}$$

with N sequence count, L mean length, a alphabet size, and k k-mer length. Operationally, memory grows with database scale plus alphabet/k-mer pointer terms.

When memory is constrained, MMseqs2 splits query or target space. Splitting lowers peak RAM but introduces extra merge and temporary I/O overhead. The penalty is often smaller at high sensitivity (compute-dominated runs) and larger at low sensitivity (I/O-dominated runs).

Set split policy as a resource-envelope decision first. Tune thresholds after split behavior is stable.

2.12. Parallel Execution Model

MMseqs2 parallelism is layered. Intra-node OpenMP paths accelerate kernels inside one process. Inter-node execution (MPI or scheduler partitioning) distributes chunks across machines.

Distributed throughput is constrained by I/O topology as much as CPU count. Target splitting may reduce per-node RAM but can increase merge overhead. Query splitting can simplify scheduling but may duplicate target-side work. Temporary directory placement can become the dominant bottleneck at high split counts.

Warning

Before attributing distributed slowdowns to compute, check shared-disk contention, split count, and merge pressure.

2.13. Practical Tuning Order

A robust tuning order for large runs is:

1. Validate DB contracts and sidecars.
2. Stabilize index and load strategy for your storage topology.
3. Set split-memory and temp-disk policy for resource limits.
4. Align parallel mode with hardware and scheduler realities.
5. Tune sensitivity and filtering thresholds last.

This sequence matches real production bottlenecks: contract and infrastructure instability usually dominates before score-threshold tuning becomes the true limiter.

2.14. Practical Playbooks

For repeated searches against a stable large target, prioritize index reuse and stable preload behavior before changing sensitivity. For memory-bound runs, tune split policy and merge overhead first. For near-identity mapping pipelines, favor ungapped/diagonal-centric paths and avoid expensive output modes that do not change decision quality.

For architecture and layer placement, use System Map; for task-oriented command selection, use Functional Modules Manual; for full topology evidence, use Dependency Map; and for advanced composition and reproducibility discipline, use Expert Manual.

3. System Map: Cascaded APIs and Module Connections

MMseqs2 commands are intentionally layered. Easy workflows and high-level entrypoints decide orchestration, while lower-level modules enforce data contracts and execute the expensive kernels. Understanding this layering is essential for both correctness and performance debugging: the command you invoke is often only the visible front of a deeper module chain.

The generated topology views, Command Reference Index and Dependency Map, provide the complete command-level map. This chapter explains how to interpret that map.

3.1. Layered API Model

At the top, workflow commands prioritize usability and sensible defaults. They hide many internal steps but still inherit all low-level behavior from their called modules. High-level API commands

expose full DB-to-DB task execution and are typically the right interface for reproducible production pipelines.

Mid-level API commands are where core algorithmic work happens: prefiltering, alignment, clustering kernels, and related computational modules. Low-level API commands perform DB creation, conversion, indexing, filtering, and utility operations. These lower layers are frequently used in custom pipelines and debugging workflows where explicit control matters more than convenience.

3.2. Cascade Pattern in Practice

Most workflows follow a common shape: first reduce the candidate space, then score surviving pairs with progressively more expensive methods, then transform outputs into task-specific artifacts. Search and clustering differ in final objectives, but they share this backbone.

That shared backbone creates an important operational rule: changing a high-level command can silently switch lower-level modules and therefore change output semantics, runtime profile, or both. Dependency edges should therefore be read as behavior links, not merely as call graph arrows.

Note

Functional group and API layer are orthogonal. A command can belong to one functional group but live in a different API layer than neighboring commands in that group.

3.3. How Submodule Pages Encode This Map

Each entry in `submodules/*.md` now combines prose and compact metadata. The prose explains what a command does, how it participates in the cascade, and when to use it. The compact metadata table records API layer, category flags, and coupling counts so readers can quickly assess execution context.

To avoid duplication, full upstream/downstream edge lists and script evidence are centralized in Dependency Map.

3.4. Choosing Your Entry View

Start from Functional Modules Manual when your task intent is clear and you need command candidates quickly. Start from Dependency Map when behavior is surprising, performance is unstable, or you are composing a custom low-level pipeline and need exact topology evidence.

Use Performance Foundations with this chapter to connect architecture placement with storage, index, split, and parallel execution consequences.

4. Functional Modules Manual

This section is the task-oriented entry point to MMseqs2 modules. Each functional page groups commands by workflow intent, then links each command into API-layer and dependency context so you can trace from practical usage to implementation topology.

For production-scale work, read Performance Foundations before tuning command-level options. That chapter explains the algorithmic and infrastructure assumptions that dominate runtime and output behavior.

4.1. Functional Groups

Easy Workflows covers shortcut workflows over FASTA/FASTQ inputs for common tasks with minimal setup. Search Workflows covers homology search and mapping entrypoints. Clustering covers cascaded and linear clustering paths plus update operations.

Prefiltering and Alignment cover core compute kernels used inside search and clustering workflows. Profiles covers profile/MSA creation and conversion paths. Database Management covers creation, indexing, splitting, and storage operations required to keep DB contracts valid.

Result Handling, Sequence Manipulation, Taxonomy, Multi-hit, and Utilities cover downstream transforms, specialized analytics, and composition helpers for advanced pipelines. Functional pages answer what to run; generated maps answer how commands connect and where behavior is inherited from upstream orchestration. Use System Map for layered architecture framing, Command Reference Index for per-command CLI pages, and Dependency Map for complete upstream/downstream topology.

4.2. Easy Workflows

High-level shortcuts that operate directly on FASTA/FASTQ inputs and orchestrate MMseqs2 modules with practical defaults.

Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

Performance

In production, tune index/load and split-memory policy before increasing sensitivity. Infrastructure choices usually dominate runtime swings.

4.2.1. easy-cluster

Slower, sensitive clustering.

Workflow-level entrypoint that orchestrates downstream MMseqs2 modules. Design priority is fast onboarding with robust defaults; fine-grained behavior is inherited from downstream workflow modules. Current coupling is 0 upstream caller(s) and 7 downstream call(s).

Aspect	Value
Usage	<code>usage: mmseqs easy-cluster <i:fastaFile1[.gz .bz2]> ...</code>

Aspect	Value
	<i:fastaFileN[.gz .bz2]> <o:clusterPrefix> <tmpDir> [options]
API layer	workflow
Category flags	COMMAND_EASY
Upstream command count	0
Downstream command count	7
Related functional groups	clustering, database, result_handling
References	Full CLI · Dependency entry

4.2.1.1. Key Options

Option	Purpose
- -seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
- -target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
- -k-score	k-mer threshold for generating similar k-mer lists
- -alph-size	Alphabet size (range 2-21)
- -max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
- -split	Split input into N equally distributed chunks. 0: set the best split automatically

4.2.2. easy-linclud

Fast linear time cluster, less sensitive clustering.

Workflow-level entrypoint that orchestrates downstream MMseqs2 modules. Design priority is fast onboarding with robust defaults; fine-grained behavior is inherited from downstream workflow modules. Current coupling is 0 upstream caller(s) and 7 downstream call(s).

Aspect	Value
Usage	usage: mmseqs easy-linclud <i:fastaFile1[.gz .bz2]> ...

Aspect	Value
	<i:fastaFileN[.gz .bz2]> <o:clusterPrefix> <tmpDir> [options]
API layer	workflow
Category flags	COMMAND_EASY
Upstream command count	0
Downstream command count	7
Related functional groups	clustering, database, result_handling
References	Full CLI · Dependency entry

4.2.2.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--alph-size	Alphabet size (range 2-21)
--spaced-kmer-mode	0: use consecutive positions in k-mers; 1: use spaced k-mers
--spaced-kmer-pattern	User-specified spaced k-mer pattern
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probability is above threshold

4.2.3. easy-linsearch

Fast, less sensitive homology search.

Workflow-level entrypoint that orchestrates downstream MMseqs2 modules. Design priority is fast onboarding with robust defaults; fine-grained behavior is inherited from downstream workflow modules. Current coupling is 0 upstream caller(s) and 7 downstream call(s).

Aspect	Value
Usage	usage: mmseqs easy-linsearch <i:queryFastaFile1[.gz .bz2]> ... <i:queryFastaFileN[.gz .bz2]> <i:targetFastaFile[.gz .bz2]> <i:targetDB> <o:alignmentFile> <tmpDir> [options]
API layer	workflow
Category flags	COMMAND_EASY COMMAND_EXPERT
Upstream command count	0
Downstream command count	7
Related functional groups	database, result_handling, search_workflows
References	Full CLI · Dependency entry

4.2.3.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probability is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occur > threshold in a row

4.2.4. easy-proteomecluster

Cluster proteomes and identify reference proteomes.

Workflow-level entrypoint for proteome-level clustering pipelines. This command is visible in MMseqs2/src/MMseqsBase.cpp, but may not be available in every packaged binary build.

Aspect	Value
Usage	usage: mmseqs easy-proteomecluster <inputFileOrDB> <outputPrefixOrFile> <tmpDir> [options] (source-derived synopsis)
API layer	workflow
Category flags	COMMAND_EASY

Run `mmseqs easy-proteomecluster` locally in your target runtime to confirm exact argument support and option availability.

4.2.5. easy-rbh

Find reciprocal best hit.

Workflow-level entrypoint that orchestrates downstream MMseqs2 modules. Design priority is fast onboarding with robust defaults; fine-grained behavior is inherited from downstream workflow modules. Current coupling is 0 upstream caller(s) and 4 downstream call(s).

Aspect	Value
Usage	usage: mmseqs easy-rbh <i:queryFastaFile1[.gz .bz2]> <i:targetFastaFile[.gz .bz2]> <i:targetDB> <o:alignmentFile> <tmpDir> [options]
API layer	workflow
Category flags	COMMAND_EASY
Upstream command count	0
Downstream command count	4
Related functional groups	database, result_handling, search_workflows
References	Full CLI · Dependency entry

4.2.5.1. Key Options

Option	Purpose
<code>--comp-bias-corr</code>	Correct for locally biased amino acid composition (range 0-1)
<code>--comp-bias-corr-scale</code>	Correct for locally biased amino acid composition (range 0-1)

Option	Purpose
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists

4.2.6. easy-search

Sensitive homology search.

Workflow-level entrypoint that orchestrates downstream MMseqs2 modules. Design priority is fast onboarding with robust defaults; fine-grained behavior is inherited from downstream workflow modules. Current coupling is 0 upstream caller(s) and 7 downstream call(s).

Aspect	Value
Usage	usage: mmseqs easy-search <i:queryFastaFile1[.gz .bz2]> ... <i:queryFastaFileN[.gz .bz2]> <i:stdin> <i:targetFastaFile[.gz]> <i:targetDB> <o:alignmentFile> <tmpDir> [options]
API layer	workflow
Category flags	COMMAND_EASY
Upstream command count	0
Downstream command count	7
Related functional groups	database, result_handling, search_workflows
References	Full CLI · Dependency entry

4.2.6.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)

Option	Purpose
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists

4.2.7. easy-taxonomy

Taxonomic classification.

Workflow-level entrypoint that orchestrates downstream MMseqs2 modules. Design priority is fast onboarding with robust defaults; fine-grained behavior is inherited from downstream workflow modules. Current coupling is 0 upstream caller(s) and 11 downstream call(s).

Aspect	Value
Usage	usage: mmseqs easy-taxonomy <i:fastaFile1[.gz .bz2]> ... <i:fastaFileN[.gz .bz2]> <i:targetDB> <o:taxReports> <tmpDir> [options]
API layer	workflow
Category flags	COMMAND_EASY
Upstream command count	0
Downstream command count	11
Related functional groups	database, result_handling, taxonomy, utilities
References	Full CLI · Dependency entry

4.2.7.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists

4.3. Search Workflows

Search and mapping workflows that compose prefiltering, alignment, and result conversion paths under different sensitivity and runtime envelopes.

Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

Performance

In production, tune index/load and split-memory policy before increasing sensitivity. Infrastructure choices usually dominate runtime swings.

4.3.1. **linsearch**

Fast, less sensitive homology search.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is balancing sensitivity, candidate pruning, and alignment cost under explicit memory and split constraints. Current coupling is 2 upstream caller(s) and 9 downstream call(s).

Aspect	Value
Usage	usage: mmseqs linsearch <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]
API layer	high_level_api
Category flags	COMMAND_MAIN COMMAND_EXPERT
Upstream command count	2
Downstream command count	9
Related functional groups	alignment, database, easy_workflows, prefiltering, result_handling, sequence_manipulation, utilities
References	Full CLI · Dependency entry

4.3.1.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probability is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occur > threshold in a row

4.3.2. map

Map nearly identical sequences.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is balancing sensitivity, candidate pruning, and alignment cost under explicit memory and split constraints. Current coupling is 0 upstream caller(s) and 1 downstream call(s).

Aspect	Value
Usage	usage: mmseqs map <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]
API layer	high_level_api
Category flags	COMMAND_MAIN
Upstream command count	0
Downstream command count	1
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.3.2.1. Key Options

Option	Purpose
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists
--alph-size	Alphabet size (range 2-21)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--split	Split input into N equally distributed chunks. 0: set the best split automatically

4.3.3. rbh

Reciprocal best hit search.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is balancing sensitivity, candidate pruning, and alignment cost under explicit memory and split constraints. Current coupling is 1 upstream caller(s) and 6 downstream call(s).

Aspect	Value
Usage	usage: mmseqs rbh <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]
API layer	high_level_api
Category flags	COMMAND_MAIN
Upstream command count	1
Downstream command count	6
Related functional groups	database, easy_workflows, result_handling, utilities
References	Full CLI · Dependency entry

4.3.3.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists

4.3.4. search

Sensitive homology search.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is balancing sensitivity, candidate pruning, and alignment cost under explicit memory and split constraints. Current coupling is 8 upstream caller(s) and 22 downstream call(s).

Aspect	Value
Usage	usage: mmseqs search <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]
API layer	high_level_api
Category flags	COMMAND_MAIN
Upstream command count	8
Downstream command count	22
Related functional groups	alignment, clustering, database, easy_workflows, multi_hit, prefiltering, profiles, result_handling, sequence_manipulation, taxonomy
References	Full CLI · Dependency entry

4.3.4.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists

4.4. Clustering

Modules for cluster construction, incremental updates, and representative selection across cascaded and linear-time strategies.

Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

Performance

In production, tune index/load and split-memory policy before increasing sensitivity. Infrastructure choices usually dominate runtime swings.

4.4.1. clust

Cluster result by Set-Cover/Connected-Component/Greedy-Incremental.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	<code>usage: mmseqs clust <i:sequenceDB> <i:resultDB> <o:clusterDB> [options]</code>
API layer	<code>mid_level_api</code>
Category flags	<code>COMMAND_CLUSTER</code>
Upstream command count	2
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.4.1.1. Key Options

Option	Purpose
<code>--cluster-mode</code>	0: Set-Cover (greedy)
<code>--max-iterations</code>	Maximum depth of breadth first search in connected component clustering
<code>--similarity-type</code>	Type of score used for clustering. 1: alignment score 2: sequence identity
<code>--weights</code>	Weights used for cluster prioritization
<code>--cluster-weight-threshold</code>	Weight threshold used for cluster prioritization
<code>--threads</code>	Number of CPU-cores used (all by default)
<code>--compressed</code>	Write compressed output

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.4.2. cluster

Slower, sensitive clustering.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 2 upstream caller(s) and 18 downstream call(s).

Aspect	Value
Usage	usage: mmseqs cluster <i:sequenceDB> <o:clusterDB> <tmpDir> [options]
API layer	high_level_api
Category flags	COMMAND_MAIN
Upstream command count	2
Downstream command count	18
Related functional groups	alignment, database, easy_workflows, prefiltering, sequence_manipulation, utilities
References	Full CLI · Dependency entry

4.4.2.1. Key Options

Option	Purpose
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists
--alph-size	Alphabet size (range 2-21)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)

Option	Purpose
--split	Split input into N equally distributed chunks. 0: set the best split automatically

4.4.3. clusterupdate

Update previous clustering with new sequences.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 0 upstream caller(s) and 13 downstream call(s).

Aspect	Value
Usage	usage: mmseqs clusterupdate <i:oldSequenceDB> <i:newSequenceDB> <i:oldClustResultDB> <o:newMappedSequenceDB> <o:newClustResultDB> <tmpDir> [options]
API layer	high_level_api
Category flags	COMMAND_MAIN
Upstream command count	0
Downstream command count	13
Related functional groups	database, result_handling, search_workflows, utilities
References	Full CLI · Dependency entry

4.4.3.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)

Option	Purpose
<code>--target-search-mode</code>	target search mode (0: regular k-mer, 1: similar k-mer)
<code>--k-score</code>	k-mer threshold for generating similar k-mer lists

4.4.4. **proteomecluster**

Cluster proteomes and identify reference proteomes.

High-level API command for proteome-level clustering workflows. This command is registered in `MMseqs2/src/MMseqsBase.cpp`, but may be absent from some packaged MMseqs builds.

Aspect	Value
Usage	<code>usage: mmseqs proteomecluster <inputDB> <targetOrResultDB> <outputDB> [tmpDir] [options]</code> (source-derived synopsis)
API layer	<code>high_level_api</code>
Category flags	<code>COMMAND_MAIN</code>

Run `mmseqs proteomecluster` locally in your deployment environment to verify exact usage and supported options.

4.4.5. **clusthash**

Hash-based clustering of equal length sequences.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	<code>usage: mmseqs clusthash <i:sequenceDB> <o:alignmentDB> [options]</code>
API layer	<code>mid_level_api</code>
Category flags	<code>COMMAND_CLUSTER</code>
Upstream command count	1
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.4.5.1. Key Options

Option	Purpose
--alph-size	Alphabet size (range 2-21)
--min-seq-id	List matches above this sequence identity (for clustering) (range 0.0-1.0)
--sub-mat	Substitution matrix file
--max-seq-len	Maximum sequence length
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.4.6. linclust

Fast, less sensitive clustering.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 2 upstream caller(s) and 8 downstream call(s).

Aspect	Value
Usage	usage: mmseqs linclust <i:sequenceDB> <o:clusterDB> <tmpDir> [options]
API layer	high_level_api
Category flags	COMMAND_MAIN
Upstream command count	2
Downstream command count	8
Related functional groups	alignment, database, easy_workflows, prefiltering, utilities
References	Full CLI · Dependency entry

4.4.6.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)

Option	Purpose
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--alph-size	Alphabet size (range 2-21)
--spaced-kmer-mode	0: use consecutive positions in k-mers; 1: use spaced k-mers
--spaced-kmer-pattern	User-specified spaced k-mer pattern
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probability is above threshold

4.4.7. mergeclusters

Merge multiple cascaded clustering steps.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs mergeclusters <i:sequenceDB> <o:clusterDB> <i:clusterDB1> ... <i:clusterDBn> [options]
API layer	mid_level_api
Category flags	COMMAND_CLUSTER
Upstream command count	2
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.4.7.1. Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.4.8. pickconsensusrep

Select new representatives for each cluster based on consensus.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is reducing graph density early, then applying clustering criteria consistently across workflow steps. Current coupling is 0 upstream caller(s) and 7 downstream call(s).

Aspect	Value
Usage	usage: mmseqs pickconsensusrep <inputDB(s)> <outputDB> [options] (source-derived synopsis; run mmseqs pickconsensusrep for exact syntax)
API layer	mid_level_api
Category flags	COMMAND_CLUSTER
Upstream command count	0
Downstream command count	7
Related functional groups	alignment, database, profiles, result_handling, utilities
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.5. Prefiltering

Candidate-generation modules that prune comparison space before expensive alignment kernels.

Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

4.5.1. countkmer

Count k-mers.

Low-level DB or utility command used for composition and contract enforcement. Design priority is minimizing expensive downstream alignments by aggressively pruning unlikely sequence pairs. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs countkmer <DB> [args] [options] (source-derived synopsis; run mmseqs countkmer for exact syntax)
API layer	low_level_api
Category flags	COMMAND_SPECIAL
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.5.2. gappedprefilter

Optimal Smith-Waterman-based prefiltering (slow).

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is minimizing expensive downstream alignments by aggressively pruning unlikely sequence pairs. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs gappedprefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]
API layer	mid_level_api
Category flags	COMMAND_PREFILTER
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.5.2.1. Key Options

Option	Purpose
<code>--comp-bias-corr</code>	Correct for locally biased amino acid composition (range 0-1)
<code>--comp-bias-corr-scale</code>	Correct for locally biased amino acid composition (range 0-1)
<code>--min-ungapped-score</code>	Accept only matches with ungapped alignment score above threshold
<code>--max-seqs</code>	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
<code>--gap-open</code>	Gap open cost
<code>--gap-extend</code>	Gap extension cost
<code>-e</code>	List matches below this E-value (range 0.0-inf)
<code>-c</code>	List matches above this fraction of aligned (covered) residues (see <code>-cov-mode</code>)

4.5.3. kmermatcher

Find bottom-m-hashed k-mer matches within sequence DB.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is minimizing expensive downstream alignments by aggressively pruning unlikely sequence pairs. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs kmermatcher <i:sequenceDB> <o:prefilterDB> [options]
API layer	mid_level_api
Category flags	COMMAND_PREFILTER
Upstream command count	1
Downstream command count	0
Related functional groups	clustering
References	Full CLI · Dependency entry

4.5.3.1. Key Options

Option	Purpose
<code>--alph-size</code>	Alphabet size (range 2-21)

Option	Purpose
--spaced-kmer-mode	0: use consecutive positions in k-mers; 1: use spaced k-mers
--spaced-kmer-pattern	User-specified spaced k-mer pattern
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probablity is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occure > threshold in a rwo
-k	k-mer length (0: automatically set to optimum)

4.5.4. kmersearch

Find bottom-m-hashed k-mer matches between target and query DB.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is minimizing expensive downstream alignments by aggressively pruning unlikely sequence pairs. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs kmersearch <i:queryDB> <i:kmerIndexDB> <o:prefilterDB> [options]
API layer	mid_level_api
Category flags	COMMAND_PREFILTER
Upstream command count	1
Downstream command count	0
Related functional groups	search_workflows
References	Full CLI · Dependency entry

4.5.4.1. Key Options

Option	Purpose
--seed-sub-mat	Substitution matrix file for k-mer generation

Option	Purpose
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probability is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occur > threshold in a row
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
--cov-mode	0: coverage of query and target
-c	List matches above this fraction of aligned (covered) residues (see --cov-mode)

4.5.5. prefilter

Double consecutive diagonal k-mer search.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is minimizing expensive downstream alignments by aggressively pruning unlikely sequence pairs. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs prefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]
API layer	mid_level_api
Category flags	COMMAND_PREFILTER
Upstream command count	3
Downstream command count	0
Related functional groups	clustering, search_workflows, taxonomy
References	Full CLI · Dependency entry

4.5.5.1. Key Options

Option	Purpose
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive

Option	Purpose
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists
--alph-size	Alphabet size (range 2-21)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--split	Split input into N equally distributed chunks. 0: set the best split automatically

4.5.6. ungappedprefilter

Optimal diagonal score search.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is minimizing expensive downstream alignments by aggressively pruning unlikely sequence pairs. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs ungappedprefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]
API layer	mid_level_api
Category flags	COMMAND_PREFILTER
Upstream command count	1
Downstream command count	0
Related functional groups	search_workflows
References	Full CLI · Dependency entry

4.5.6.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)

Option	Purpose
<code>--min-ungapped-score</code>	Accept only matches with ungapped alignment score above threshold
<code>--max-seqs</code>	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
<code>-c</code>	List matches above this fraction of aligned (covered) residues (see <code>-cov-mode</code>)
<code>-e</code>	List matches below this E-value (range 0.0-inf)
<code>--cov-mode</code>	0: coverage of query and target
<code>--taxon-list</code>	Taxonomy ID, possibly multiple values separated by ‘;

4.6. Alignment

Alignment and rescoring modules used after prefiltering to compute pair quality and coordinates.

Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

4.6.1. align

Optimal gapped local alignment.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 5 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs align <i:queryDB> <i:targetDB> <i:resultDB> <o:alignmentDB> [options]
API layer	mid_level_api
Category flags	COMMAND_ALIGNMENT
Upstream command count	5
Downstream command count	0
Related functional groups	clustering, search_workflows
References	Full CLI · Dependency entry

4.6.1.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
-a	Add backtrace string (convert to alignments with mmseqs convertalis module)
--alignment-mode	How to compute the alignment:
--alignment-output-mode	How to compute the alignment:
--wrapped-scoring	Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start
-e	List matches below this E-value (range 0.0-inf)

4.6.2. alignall

Within-result all-vs-all gapped local alignment.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs alignall <i:sequenceDB> <i:resultDB> <o:alignmentDB> [options]
API layer	mid_level_api
Category flags	COMMAND_ALIGNMENT
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.6.2.1. Key Options

Option	Purpose
<code>--comp-bias-corr</code>	Correct for locally biased amino acid composition (range 0-1)
<code>--comp-bias-corr-scale</code>	Correct for locally biased amino acid composition (range 0-1)
<code>--add-self-matches</code>	Artificially add entries of queries with themselves (for clustering)
<code>-a</code>	Add backtrace string (convert to alignments with mmseqs convertalis module)
<code>--alignment-mode</code>	How to compute the alignment:
<code>-e</code>	List matches below this E-value (range 0.0-inf)
<code>--min-seq-id</code>	List matches above this sequence identity (for clustering) (range 0.0-1.0)
<code>--min-aln-len</code>	Minimum alignment length (range 0-INT_MAX)

4.6.3. alignbykmer

Heuristic gapped local k-mer based alignment.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs alignbykmer <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB> [options]
API layer	mid_level_api
Category flags	COMMAND_ALIGNMENT
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.6.3.1. Key Options

Option	Purpose
-k	k-mer length (0: automatically set to optimum)
--spaced-kmer-mode	0: use consecutive positions in k-mers; 1: use spaced k-mers
--spaced-kmer-pattern	User-specified spaced k-mer pattern
--alph-size	Alphabet size (range 2-21)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
-c	List matches above this fraction of aligned (covered) residues (see --cov-mode)
-e	List matches below this E-value (range 0.0-inf)
--cov-mode	0: coverage of query and target

4.6.4. expandaln

Expand an alignment result based on another.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs expandaln <i:queryDB> <i:targetDB> <i:resultDB> <i:resultDB ca3mDB> <o:alignmentDB> [options]
API layer	mid_level_api
Category flags	COMMAND_PROFILE_PROFILE
Upstream command count	1
Downstream command count	0
Related functional groups	search_workflows
References	Full CLI · Dependency entry

4.6.4.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)

Option	Purpose
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--score-bias	Score bias when computing SW alignment (in bits)
-e	List matches below this E-value (range 0.0-inf)
--min-seq-id	List matches above this sequence identity (for clustering) (range 0.0-1.0)
-c	List matches above this fraction of aligned (covered) residues (see -cov-mode)

4.6.5. fwbw

Forward Backward Alignment.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs fwbw <inputDB(s)> <outputDB> [options] (source-derived synopsis; run mmseqs fwbw for exact syntax)
API layer	mid_level_api
Category flags	COMMAND_ALIGNMENT
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.6.6. offsetalignment

Offset alignment by ORF start position.

Low-level DB or utility command used for composition and contract enforcement. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs offsetalignment <i:queryDB> <i:queryOrfDB> <i:targetDB> <i:targetOrfDB> <i:alnDB> <o:alnDB> [options]
API layer	low_level_api
Category flags	COMMAND_RESULT
Upstream command count	3
Downstream command count	0
Related functional groups	clustering, search_workflows
References	Full CLI · Dependency entry

4.6.6.1. Key Options

Option	Purpose
--search-type	Search type 0: auto 1: amino acid, 2: translated, 3: nucleotide, 4: translated nucleotide alignment
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--chain-alignments	Chain overlapping alignments
--merge-query	Combine ORFs/split sequences to a single entry

4.6.7. proteinaln2nucl

Transform protein alignments to nucleotide alignments.

Low-level DB or utility command used for composition and contract enforcement. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs proteinaln2nucl <i:nuclQueryDB> <i:nuclTargetDB> <i:aaQueryDB> <i:aaTargetDB> <i:alnDB> <o:alnDB> [options]
API layer	low_level_api
Category flags	COMMAND_RESULT
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.6.7.1. Key Options

Option	Purpose
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--sub-mat	Substitution matrix file
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.6.8. rescorediagonal

Compute sequence identity for diagonal.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 5 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs rescorediagonal <i:queryDB> <i:targetDB> <i:prefilterDB> <o:resultDB> [options]
API layer	mid_level_api
Category flags	COMMAND_ALIGNMENT
Upstream command count	5
Downstream command count	0

Aspect	Value
Related functional groups	clustering, search_workflows, taxonomy
References	Full CLI · Dependency entry

4.6.8.1. Key Options

Option	Purpose
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--wrapped-scoring	Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start
-e	List matches below this E-value (range 0.0-inf)
-c	List matches above this fraction of aligned (covered) residues (see --cov-mode)
-a	Add backtrace string (convert to alignments with mmseqs convertalis module)
--cov-mode	0: coverage of query and target
--min-seq-id	List matches above this sequence identity (for clustering) (range 0.0-1.0)
--min-aln-len	Minimum alignment length (range 0-INT_MAX)

4.6.9. transitivealign

Transfer alignments via transitivity.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is extracting reliable score/coverage/identity information while controlling DP and backtrace overhead. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs transitivealign <inputDB(s)> <outputDB> [options] (source-derived synopsis; run mmseqs transitivealign for exact syntax)
API layer	mid_level_api
Category flags	COMMAND_ALIGNMENT
Upstream command count	0

Aspect	Value
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.7. Profiles

Profile and MSA modules for profile construction, conversion, and profile-driven search workflows.

Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

4.7.1. convertca3m

Convert a cA3M DB to a result DB.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs convertca3m <inputDB(s)> <outputDB> [options] (source-derived synopsis; run mmseqs convertca3m for exact syntax)
API layer	mid_level_api
Category flags	COMMAND_PROFILE_PROFILE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.7.2. convertmsa

Convert Stockholm/PFAM MSA file to a MSA DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs convertmsa <i:msaFile.sto[.gz]> <o:msaDB> [options]
API layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION
Upstream command count	1
Downstream command count	0
Related functional groups	database
References	Full CLI · Dependency entry

4.7.2.1. Key Options

Option	Purpose
--identifier-field	Field from STOCKHOLM comments for choosing the MSA identifier: 0: ID, 1: AC. If the respective comment does not exist, the name of the first sequence will become the identifier
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.7.3. convertprofiledb

Convert a HH-suite HHM DB to a profile DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs convertprofiledb <DB> [args] [options] (source-derived synopsis; run mmseqs convertprofiledb for exact syn- tax)
API layer	low_level_api

Aspect	Value
Category flags	COMMAND_PROFILE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.7.4. **expand2profile**

Expand an alignment result based on another and create a profile.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs expand2profile <inputDB(s)> <outputDB> [options] (source-derived synopsis; run mmseqs expand2profile for exact syntax)
API layer	mid_level_api
Category flags	COMMAND_PROFILE_PROFILE
Upstream command count	1
Downstream command count	0
Related functional groups	search_workflows
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.7.5. **msa2profile**

Convert a MSA DB to a profile DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs msa2profile <i:msaDB> <o:profileDB> [options]
API layer	low_level_api
Category flags	COMMAND_PROFILE COMMAND_DATABASE_CREATION
Upstream command count	2
Downstream command count	0
Related functional groups	clustering, database
References	Full CLI · Dependency entry

4.7.5.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--match-mode	0: Columns that have a residue in the first sequence are kept, 1: columns that have a residue in --match-ratio of all sequences are kept
--match-ratio	Columns that have a residue in this ratio of all sequences are kept
--pseudo-cnt-mode	use 0: substitution-matrix or 1: context-specific pseudocounts
--pca	Pseudo count admixture strength

4.7.6. msa2result

Convert a MSA DB to a profile DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs msa2result <i:msaDB> <o:seqDB> <o:profileDB> [options]
API layer	low_level_api
Category flags	COMMAND_PROFILE COMMAND_EXPERT
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.7.6.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--match-mode	0: Columns that have a residue in the first sequence are kept, 1: columns that have a residue in --match-ratio of all sequences are kept
--match-ratio	Columns that have a residue in this ratio of all sequences are kept
--pseudo-cnt-mode	use 0: substitution-matrix or 1: context-specific pseudocounts
--pca	Pseudo count admixture strength

4.7.7. pairaln

Pair sequences to match best protein A and B from a species.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs pairaln <DB> [args] [options] (source-derived synopsis; run mmseqs pairaln for exact syntax)
API layer	low_level_api
Category flags	COMMAND_EXPERT
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.7.8. profile2consensus

Extract consensus sequence DB from a profile DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs profile2consensus <DB> [args] [options] (source-derived synopsis; run mmseqs profile2consensus for exact syntax)
API layer	low_level_api
Category flags	COMMAND_PROFILE
Upstream command count	1
Downstream command count	0
Related functional groups	search_workflows
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.7.9. profile2neff

Convert a profile DB to a tab-separated list of Neff scores.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs profile2neff <DB> [args] [options] (source-derived synopsis; run mmseqs profile2neff for exact syntax)
API layer	low_level_api
Category flags	COMMAND_PROFILE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.7.10. profile2pssm

Convert a profile DB to a tab-separated PSSM file.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs profile2pssm <DB> [args] [options] (source-derived synopsis; run mmseqs profile2pssm for exact syntax)
API layer	low_level_api
Category flags	COMMAND_PROFILE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.7.11. profile2repseq

Extract representative sequence DB from a profile DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs profile2repseq <DB> [args] [options] (source-derived synopsis; run mmseqs profile2repseq for exact syntax)
API layer	low_level_api
Category flags	COMMAND_PROFILE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.7.12. result2profile

Compute profile DB from a result DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs result2profile <i:queryDB> <i:targetDB> <i:resultDB> <o:profileDB> [options]
API layer	low_level_api
Category flags	COMMAND_PROFILE
Upstream command count	1
Downstream command count	0
Related functional groups	search_workflows
References	Full CLI · Dependency entry

4.7.12.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
-e	List matches below this E-value (range 0.0-inf)
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--mask-profile	Mask query sequence of profile using tantan [0,1]
--e-profile	Include sequences matches with < E-value thr. into the profile (>=0.0)
--wg	Use global sequence weighting for profile calculation

4.7.13. sequence2profile

Turn sequence into profile by adding context specific pseudo counts.

Low-level DB or utility command used for composition and contract enforcement. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs sequence2profile <DB> [args] [options] (source-derived synopsis; run mmseqs sequence2profile for exact syntax)
API layer	low_level_api
Category flags	COMMAND_PROFILE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.7.14. tsv2exprofiledb

Create an expandable profile db from TSV files.

Mid-level compute module used directly in advanced pipelines and by workflows. Design priority is preserving profile semantics across transforms so search and scoring remain interpretable. Current coupling is 0 upstream caller(s) and 5 downstream call(s).

Aspect	Value
Usage	usage: mmseqs tsv2exprofiledb <i:tsvFilesBase> <o:exprofileDB> [options]
API layer	mid_level_api
Category flags	COMMAND_PROFILE_PROFILE
Upstream command count	0
Downstream command count	5
Related functional groups	database, utilities
References	Full CLI · Dependency entry

4.7.14.1. Key Options

Option	Purpose
--gpu	Use GPU (CUDA) if possible
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.8. Database Management

Database lifecycle modules for creation, indexing, splitting, merging, and contract-preserving transforms.

Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

4.8.1. aliasdb

Create relative symlink of DB to another name in the same folder.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs aliasdb <i:srcDB> <o:dstDB> [options]
API layer	low_level_api
Category flags	COMMAND_STORAGE
Upstream command count	1
Downstream command count	0
Related functional groups	profiles
References	Full CLI · Dependency entry

4.8.1.1. Key Options

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.8.2. concatdbs

Concatenate two DBs, giving new IDs to entries from 2nd DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs concatdbs <i:DB> <i:DB> <o:DB> [options]
API layer	low_level_api
Category flags	COMMAND_SET
Upstream command count	3
Downstream command count	0
Related functional groups	clustering, search_workflows
References	Full CLI · Dependency entry

4.8.2.1. Key Options

Option	Purpose
- -preserve-keys	The keys of the two DB should be distinct, and they will be preserved in the concatenation

Option	Purpose
--take-larger-entry	Only keep the larger entry (dataSize >) in the concatenation, both databases need the same keys in the index
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.8.3. cpdb

Copy a DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs cpdb <i:srcDB> <o:dstDB> [options]
API layer	low_level_api
Category flags	COMMAND_STORAGE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.8.3.1. Key Options

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.8.4. createdb

Convert FASTA/Q file(s) to a sequence DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 8 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs createdb <i:fastaFile1[.gz .bz2]> ... <i:fastaFileN[.gz .bz2]> <i:stdin> <o:sequenceDB> [options]
API layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION
Upstream command count	8
Downstream command count	0
Related functional groups	easy_workflows, multi_hit
References	Full CLI · Dependency entry

4.8.4.1. Key Options

Option	Purpose
--dbtype	Database type 0: auto, 1: amino acid 2: nucleotides
--shuffle	Shuffle input database
--createdb-mode	Createdb mode 0: copy data, 1: soft link data and write new index (works only with single line fasta/q)
--id-offset	Numeric ids in index file are offset by this value
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--write-lookup	write .lookup file containing mapping from internal id, fasta id and file number

4.8.5. createindex

Store precomputed index on disk to reduce search overhead.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 0 upstream caller(s) and 4 downstream call(s).

Aspect	Value
Usage	usage: mmseqs createindex <i:sequenceDB> <tmpDir> [options]
API layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION
Upstream command count	0
Downstream command count	4
Related functional groups	sequence_manipulation
References	Full CLI · Dependency entry

4.8.5.1. Key Options

Option	Purpose
- -seed-sub-mat	Substitution matrix file for k-mer generation
-k	k-mer length (0: automatically set to optimum)
- -alph-size	Alphabet size (range 2-21)
- -comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
- -comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
- -max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
- -mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
- -mask-prob	Mask sequences is probability is above threshold

4.8.6. createlinindex

Create linsearch index.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 2 upstream caller(s) and 4 downstream call(s).

Aspect	Value
Usage	usage: mmseqs createlinindex <i:sequenceDB> <tmpDir> [options]
API layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION COMMAND_EXPERT
Upstream command count	2
Downstream command count	4
Related functional groups	easy_workflows, sequence_manipulation
References	Full CLI · Dependency entry

4.8.6.1. Key Options

Option	Purpose
- -seed-sub-mat	Substitution matrix file for k-mer generation
-k	k-mer length (0: automatically set to optimum)
- -split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
- -alph-size	Alphabet size (range 2-21)
- -mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
- -mask-prob	Mask sequences is probability is above threshold
- -mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
- -mask-n-repeat	Repeat letters that occur > threshold in a row

4.8.7. createsubdb

Create a subset of a DB from list of DB keys.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 5 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs createsubdb <i:subsetFile DB> <i:DB> <o:DB> [options]
API layer	low_level_api
Category flags	COMMAND_SET
Upstream command count	5
Downstream command count	0
Related functional groups	clustering, search_workflows, taxonomy
References	Full CLI · Dependency entry

4.8.7.1. Key Options

Option	Purpose
--subdb-mode	Subdb mode 0: copy data 1: soft link data and write index
--id-mode	Select DB entries based on 0: database keys, 1: FASTA identifiers (.lookup)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.8.8. databases

List and download databases.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 0 upstream caller(s) and 8 downstream call(s).

Aspect	Value
Usage	usage: mmseqs databases <DB> [args] [options] (source-derived synopsis; run mmseqs databases for exact syntax)
API layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION
Upstream command count	0
Downstream command count	8
Related functional groups	profiles, taxonomy, utilities

Aspect	Value
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.8.9. db2tar

Archive contents of a DB to a tar archive.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs db2tar <i:DB> <o:tar[.gz]> [options]
API layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION COMMAND_EXPERT
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.8.9.1. Key Options

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.8.10. lndb

Symlink a DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs lndb <i:srcDB> <o:dstDB> [options]
API layer	low_level_api
Category flags	COMMAND_STORAGE

Aspect	Value
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.8.10.1. Key Options

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.8.11. mergedbs

Merge entries from multiple DBs.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 4 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs mergedbs <i:DB> <o:DB> <i:DB1> ... <i:DBn> [options]
API layer	low_level_api
Category flags	COMMAND_SET
Upstream command count	4
Downstream command count	0
Related functional groups	clustering, search_workflows
References	Full CLI · Dependency entry

4.8.11.1. Key Options

Option	Purpose
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--prefixes	Comma separated list of prefixes for each entry

Option	Purpose
<code>--merge-stop-empty</code>	Don't continue merging entries after an empty entry

4.8.12. mvdb

Move a DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 5 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs mvdb <i:srcDB> <o:dstDB> [options]
API layer	low_level_api
Category flags	COMMAND_STORAGE
Upstream command count	5
Downstream command count	0
Related functional groups	clustering, profiles, search_workflows, taxonomy
References	Full CLI · Dependency entry

4.8.12.1. Key Options

Option	Purpose
<code>-v</code>	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.8.13. renamedbkeys

Create a new DB with original keys renamed.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs renamedbkeys <i:idMapFile stdin> <i:DB> <o:DB> [options]
API layer	low_level_api
Category flags	COMMAND_DB

Aspect	Value
Upstream command count	2
Downstream command count	0
Related functional groups	clustering
References	Full CLI · Dependency entry

4.8.13.1. Key Options

Option	Purpose
--subdb-mode	Subdb mode 0: copy data 1: soft link data and write index
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.8.14. rmdb

Remove a DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 19 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs rmdb <DB> [args] [options] (source-derived synopsis; run mmseqs rmdb for exact syntax)
API layer	low_level_api
Category flags	COMMAND_STORAGE
Upstream command count	19
Downstream command count	0
Related functional groups	clustering, easy_workflows, multi_hit, profiles, search_workflows, taxonomy
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.8.15. splitdb

Split DB into subsets.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs splitdb <i:DB> <o:DB> [options]
API layer	low_level_api
Category flags	COMMAND_SET
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.8.15.1. Key Options

Option	Purpose
--split	Split input into N equally distributed chunks
--split-aa	Try to find the best split boundaries by entry lengths
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.8.16. splitsequence

Split sequences by length.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs splitsequence <i:sequenceDB> <o:sequenceDB> [options]
API layer	low_level_api
Category flags	COMMAND_SEQUENCE
Upstream command count	3

Aspect	Value
Downstream command count	0
Related functional groups	search_workflows
References	Full CLI · Dependency entry

4.8.16.1. Key Options

Option	Purpose
--sequence-overlap	Overlap between sequences
--sequence-split-mode	Sequence split mode 0: copy data, 1: soft link data and write new index,
--headers-split-mode	Header split mode: 0: split position, 1: original header
--max-seq-len	Maximum sequence length
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--create-lookup	Create database lookup file (can be very large)

4.8.17. subtractdbs

Remove all entries from first DB occurring in second DB by key.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs subtractdbs <i:resultDBLeft> <i:resultDBRight> <o:resultDB> [options]
API layer	low_level_api
Category flags	COMMAND_SET
Upstream command count	2
Downstream command count	0
Related functional groups	clustering, search_workflows
References	Full CLI · Dependency entry

4.8.17.1. Key Options

Option	Purpose
-e	List matches below this E-value (range 0.0-inf)
--e-profile	Include sequences matches with < E-value thr. into the profile (>=0.0)
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.8.18. swapdb

Transpose DB with integer values in first column.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 4 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs swapdb <i:resultDB> <o:resultDB> [options]
API layer	low_level_api
Category flags	COMMAND_DB
Upstream command count	4
Downstream command count	0
Related functional groups	clustering, multi_hit, taxonomy
References	Full CLI · Dependency entry

4.8.18.1. Key Options

Option	Purpose
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.8.19. tar2db

Convert content of tar archives to any DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs tar2db <i:tar[.gz]> ... <i:tar[.gz]> <o:resultDB> [options]
API layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION COMMAND_EXPERT
Upstream command count	1
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.8.19.1. Key Options

Option	Purpose
--output-dbtype	Set database type for resulting database: Amino acid sequences 0, Nucl. seq. 1, Profiles 2, Alignment result 5, Clustering result 6, Pre-filtering result 7, Taxonomy result 8, Indexed database 9, cA3M MSAs 10, FASTA or A3M MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional prefiltering result 14, Offsetted headers 15
--tar-include	Include file names based on this regex
--tar-exclude	Exclude file names based on this regex
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.8.20. tsv2db

Convert a TSV file to any DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is keeping MMseqs2 DB contracts valid while avoiding unnecessary I/O and recomputation. Current coupling is 4 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	<code>usage: mmseqs tsv2db <i:tsvFile> <o:resultDB> [options]</code>
API layer	<code>low_level_api</code>
Category flags	<code>COMMAND_DATABASE_CREATION</code> <code> COMMAND_EXPERT</code>
Upstream command count	4
Downstream command count	0
Related functional groups	<code>clustering, multi_hit, profiles</code>
References	Full CLI · Dependency entry

4.8.20.1. Key Options

Option	Purpose
<code>--add-self-matches</code>	Artificially add entries of queries with themselves (for clustering)
<code>--output-dbtype</code>	Set database type for resulting database: Amino acid sequences 0, Nucl. seq. 1, Profiles 2, Alignment result 5, Clustering result 6, Pre-filtering result 7, Taxonomy result 8, Indexed database 9, cA3M MSAs 10, FASTA or A3M MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional prefiltering result 14, Offsetted headers 15
<code>--compressed</code>	Write compressed output
<code>-v</code>	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.9. Sequence Manipulation

Sequence-level transform modules for ORFs, frames, masking, coordinate transforms, and related preprocessing.

Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

Warning

Validate DB-type and sidecar contracts before chaining modules. Most pipeline failures are contract mismatches, not algorithmic defects.

4.9.1. extractalignedregion

Extract aligned sequence region from query.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs extractalignedregion <i:queryDB> <i:targetDB> <i:resultDB> <o:sequenceDB> [options]
API layer	low_level_api
Category flags	COMMAND_SEQUENCE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.9.1.1. Key Options

Option	Purpose
--extract-mode	Extract from 1: Query, 2: Target
--compressed	Write compressed output
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.9.2. extractframes

Extract frames from a nucleotide sequence DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 4 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs extractframes <i:sequenceDB> <o:sequenceDB> [options]
API layer	low_level_api
Category flags	COMMAND_SEQUENCE
Upstream command count	4
Downstream command count	0
Related functional groups	clustering, database, search_workflows
References	Full CLI · Dependency entry

4.9.2.1. Key Options

Option	Purpose
--forward-frames	Comma-separated list of frames on the forward strand to be extracted
--reverse-frames	Comma-separated list of frames on the reverse strand to be extracted
--translation-table	1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3) YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
--translate	Translate ORF to amino acid
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--create-lookup	Create database lookup file (can be very large)

4.9.3. extractorfs

Six-frame extraction of open reading frames.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 6 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	<code>usage: mmseqs extractorfs <i:sequenceDB> <o:sequenceDB> [options]</code>
API layer	<code>low_level_api</code>
Category flags	<code>COMMAND_SEQUENCE</code>
Upstream command count	6
Downstream command count	0
Related functional groups	database, multi_hit, search_workflows, taxonomy
References	Full CLI · Dependency entry

4.9.3.1. Key Options

Option	Purpose
<code>--min-length</code>	Minimum codon number in open reading frames
<code>--max-length</code>	Maximum codon number in open reading frames
<code>--max-gaps</code>	Maximum number of codons with gaps or unknown residues before an open reading frame is rejected
<code>--contig-start-mode</code>	Contig start can be 0: incomplete, 1: complete, 2: both
<code>--contig-end-mode</code>	Contig end can be 0: incomplete, 1: complete, 2: both
<code>--orf-start-mode</code>	Orf fragment can be 0: from start to stop, 1: from any to stop, 2: from last encountered start to stop (no start in the middle)
<code>--forward-frames</code>	Comma-separated list of frames on the forward strand to be extracted
<code>--reverse-frames</code>	Comma-separated list of frames on the reverse strand to be extracted

4.9.4. masksequence

Soft mask sequence DB using tantan.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs masksequence <i:sequenceDB> <o:sequenceDB> [options]
API layer	low_level_api
Category flags	COMMAND_SEQUENCE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.9.4.1. Key Options

Option	Purpose
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probability is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occur > threshold in a row
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.9.5. orftocontig

Write ORF locations in alignment format.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs orftocontig <i:contigsSequenceDB> <i:extractedOrfsHeadersDB> <o:orfsAlignedToContigDB> [options]
API layer	low_level_api
Category flags	COMMAND_SEQUENCE
Upstream command count	1
Downstream command count	0
Related functional groups	multi_hit
References	Full CLI · Dependency entry

4.9.5.1. Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.9.6. recoverlongestorf

Recover longest ORF for taxonomy annotation after elimination.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs recoverlongestorf <i:orfDB> <i:resultDB> <o:tsvFile> [options]
API layer	low_level_api
Category flags	COMMAND_EXPERT
Upstream command count	1
Downstream command count	0
Related functional groups	taxonomy
References	Full CLI · Dependency entry

4.9.6.1. Key Options

Option	Purpose
- -threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.9.7. reverseseq

Reverse (without complement) sequences.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs reverseseq <i:sequenceDB> <o:revSequenceDB> [options]
API layer	low_level_api
Category flags	COMMAND_SEQUENCE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.9.7.1. Key Options

Option	Purpose
- -threads	Number of CPU-cores used (all by default)
- -compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.9.8. translateaa

Translate proteins to lexicographically lowest codons.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs translateaa <i:sequenceDB> <o:sequenceDB> [options]
API layer	low_level_api
Category flags	COMMAND_SEQUENCE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.9.8.1. Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.9.9. translatenucs

Translate nucleotides to proteins.

Low-level DB or utility command used for composition and contract enforcement. Design priority is sequence-space normalization before heavy compute, especially for translated or masked workflows. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs translatenucs <i:sequenceDB> <o:sequenceDB> [options]
API layer	low_level_api
Category flags	COMMAND_SEQUENCE
Upstream command count	1
Downstream command count	0
Related functional groups	multi_hit
References	Full CLI · Dependency entry

4.9.9.1. Key Options

Option	Purpose
--translation-table	1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3) YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
--add-orf-stop	Add stop codon '*' at complete start and end
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)

4.10. Result Handling

Modules that filter, summarize, reshape, and export result databases for downstream analysis.

Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

Warning

Validate DB-type and sidecar contracts before chaining modules. Most pipeline failures are contract mismatches, not algorithmic defects.

4.10.1. convert2fasta

Convert sequence DB to FASTA format.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs convert2fasta <DB> [args] [options] (source-derived synopsis; run mmseqs convert2fasta for exact syntax)
API layer	low_level_api
Category flags	COMMAND_FORMAT_CONVERSION

Aspect	Value
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.10.2. convertalis

Convert alignment DB to BLAST-tab, SAM or custom format.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 4 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs convertalis <i:queryDb> <i:targetDb> <i:alignmentDB> <o:alignmentFile> [options]
API layer	low_level_api
Category flags	COMMAND_FORMAT_CONVERSION
Upstream command count	4
Downstream command count	0
Related functional groups	easy_workflows
References	Full CLI · Dependency entry

4.10.2.1. Key Options

Option	Purpose
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--format-mode	Output format:
--format-output	Choose comma separated list of output columns from: query,tar- get,evalue,gapopen,pident,fident,nident,qs- tart,qend,qlen

Option	Purpose
--translation-table	1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3) YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
--search-type	Search type 0: auto 1: amino acid, 2: translated, 3: nucleotide, 4: translated nucleotide alignment
--sub-mat	Substitution matrix file
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch

4.10.3. createseqfiledb

Create a DB of unaligned FASTA entries.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs createseqfiledb <i:sequenceDB> <i:resultDB> <o:fastaDB> [options]
API layer	low_level_api
Category flags	COMMAND_FORMAT_CONVERSION COMMAND_EXPERT
Upstream command count	2
Downstream command count	0
Related functional groups	easy_workflows
References	Full CLI · Dependency entry

4.10.3.1. Key Options

Option	Purpose
--min-sequences	Minimum number of sequences a cluster may contain
--max-sequences	Maximum number of sequences a cluster may contain

Option	Purpose
--hh-format	Format entries to use with hhsuite (for single-ton clusters)
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.10.4. createtsv

Convert result DB to tab-separated flat file.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs createtsv <i:queryDB> [<i:targetDB>] <i:resultDB> <o:tsvFile> [options]
API layer	low_level_api
Category flags	COMMAND_FORMAT_CONVERSION
Upstream command count	3
Downstream command count	0
Related functional groups	easy_workflows
References	Full CLI · Dependency entry

4.10.4.1. Key Options

Option	Purpose
--first-seq-as-repr	Use the first sequence of the clustering result as representative sequence
--target-column	Select a target column (default 1), 0 if no target id exists
--full-header	Replace DB ID by its corresponding Full Header

Option	Purpose
<code>--idx-seq-src</code>	0: auto, 1: split/translated sequences, 2: input sequences
<code>--threads</code>	Number of CPU-cores used (all by default)
<code>--compressed</code>	Write compressed output
<code>-v</code>	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
<code>--db-output</code>	Return a result DB instead of a text file

4.10.5. **extractdomains**

Extract highest scoring alignment regions for each sequence from BLAST-tab file.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	<code>usage: mmseqs extractdomains <DB> [args] [options] (source-derived synopsis; run mmseqs extractdomains for exact syntax)</code>
API layer	<code>low_level_api</code>
Category flags	<code>COMMAND_SPECIAL</code>
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.10.6. **filterresult**

Pairwise alignment result filter.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs filterresult <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB> [options]
API layer	low_level_api
Category flags	COMMAND_RESULT
Upstream command count	1
Downstream command count	0
Related functional groups	search_workflows
References	Full CLI · Dependency entry

4.10.6.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	keep the query (representative) sequence
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--filter-min-enable	Only filter MSAs with more than N sequences, 0 always filters
--max-seq-id	Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0]
--qid	Reduce diversity of output MSAs using min.seq. identity with query sequences

4.10.7. result2dnamsa

Compute MSA DB with out insertions in the query for DNA sequences.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs result2dnamsa <i:queryDB> <i:targetDB> <i:resultDB> <o:msaDB> [options]
API layer	low_level_api
Category flags	COMMAND_RESULT
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.10.7.1. Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--skip-query	Skip the query sequence

4.10.8. result2flat

Create flat file by adding FASTA headers to DB entries.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs result2flat <i:queryDB> <i:targetDB> <i:resultDB> <o:fastaDB> [options]
API layer	low_level_api
Category flags	COMMAND_FORMAT_CONVERSION COMMAND_EXPERT
Upstream command count	2
Downstream command count	0
Related functional groups	easy_workflows
References	Full CLI · Dependency entry

4.10.8.1. Key Options

Option	Purpose
<code>--use-fasta-header</code>	Use the id parsed from the fasta header as the index key instead of using incrementing numeric identifiers
<code>-v</code>	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.10.9. **result2msa**

Compute MSA DB from a result DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	<code>usage: mmseqs result2msa <i:queryDB> <i:targetDB> <i:resultDB> <o:msaDB> [options]</code>
API layer	<code>low_level_api</code>
Category flags	<code>COMMAND_RESULT</code>
Upstream command count	1
Downstream command count	0
Related functional groups	<code>clustering</code>
References	Full CLI · Dependency entry

4.10.9.1. Key Options

Option	Purpose
<code>--comp-bias-corr</code>	Correct for locally biased amino acid composition (range 0-1)
<code>--comp-bias-corr-scale</code>	Correct for locally biased amino acid composition (range 0-1)
<code>--gap-open</code>	Gap open cost
<code>--gap-extend</code>	Gap extension cost
<code>--filter-msa</code>	Filter msa: 0: do not filter, 1: filter
<code>--filter-min-enable</code>	Only filter MSAs with more than N sequences, 0 always filters

Option	Purpose
<code>--max-seq-id</code>	Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0]
<code>--qid</code>	Reduce diversity of output MSAs using min.seq. identity with query sequences

4.10.10. **result2rbh**

Filter a merged result DB to retain only reciprocal best hits.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs result2rbh <i:resultDB> <o:resultDB> [options]
API layer	low_level_api
Category flags	COMMAND_RESULT
Upstream command count	1
Downstream command count	0
Related functional groups	search_workflows
References	Full CLI · Dependency entry

4.10.10.1. Key Options

Option	Purpose
<code>--threads</code>	Number of CPU-cores used (all by default)
<code>--compressed</code>	Write compressed output
<code>-v</code>	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.10.11. **result2repseq**

Get representative sequences from result DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs result2repseq <i:sequenceDB> <i:resultDB> <o:sequenceDb> [options]
API layer	low_level_api
Category flags	COMMAND_RESULT
Upstream command count	3
Downstream command count	0
Related functional groups	clustering, easy_workflows
References	Full CLI · Dependency entry

4.10.11.1. Key Options

Option	Purpose
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.10.12. result2stats

Compute statistics for each entry in a DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs result2stats <i:queryDB> <i:targetDB> <i:resultDB> <o:statsDB> [options]
API layer	low_level_api
Category flags	COMMAND_RESULT
Upstream command count	2
Downstream command count	0
Related functional groups	multi_hit, search_workflows

Aspect	Value
References	Full CLI · Dependency entry

4.10.12.1. Key Options

Option	Purpose
--stat	One of: linecount, mean, min, max, doolittle, charges, seqlen, firstline
--tsv	Return output in TSV format
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.10.13. sortresult

Sort a result DB in the same order as the prefilter or align module.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs sortresult <i:resultbDB> <o:resultDB> [options]
API layer	low_level_api
Category flags	COMMAND_RESULT
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.10.13.1. Key Options

Option	Purpose
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.10.14. summarizealis

Summarize alignment result to one row (uniq. cov., cov., avg. seq. id.).

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs summarizealis <i:alignmentDB> <o:summerizedDB> [options]
API layer	low_level_api
Category flags	COMMAND_RESULT
Upstream command count	1
Downstream command count	0
Related functional groups	easy_workflows
References	Full CLI · Dependency entry

4.10.14.1. Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.10.15. summarizeheaders

Summarize FASTA headers of result DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs summarizeheaders <DB> [args] [options] (source-derived synopsis; run mmseqs summarizeheaders for exact syn- tax)
API layer	low_level_api
Category flags	COMMAND_SPECIAL
Upstream command count	0

Aspect	Value
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.10.16. summarizeresult

Extract annotations from alignment DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs summarizeresult <i:alignmentDB> <o:alignmentDB> [options]
API layer	low_level_api
Category flags	COMMAND_RESULT
Upstream command count	2
Downstream command count	0
Related functional groups	easy_workflows
References	Full CLI · Dependency entry

4.10.16.1. Key Options

Option	Purpose
-a	Add backtrace string (convert to alignments with mmseqs convertalis module)
-c	List matches above this fraction of aligned (covered) residues (see -cov-mode)
--overlap	Maximum overlap of covered regions
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.10.17. swapresults

Transpose prefilter/alignment DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is transforming outputs without silently changing scoring semantics inherited from upstream modules. Current coupling is 4 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs swapresults <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB> [options]
API layer	low_level_api
Category flags	COMMAND_RESULT
Upstream command count	4
Downstream command count	0
Related functional groups	easy_workflows, search_workflows
References	Full CLI · Dependency entry

4.10.17.1. Key Options

Option	Purpose
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
-e	List matches below this E-value (range 0.0-inf)
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--sub-mat	Substitution matrix file
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch

4.11. Multi-hit

Set-based modules that aggregate sequence-level evidence into per-set statistics and outputs.

Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

4.11.1. **besthitperset**

For each set of sequences compute the best element and update p-value.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is robust set-level aggregation over sequence-level evidence with transparent statistical behavior. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	<code>usage:</code> <code>mmseqs besthitperset <i:targetSetDB></code> <code><i:resultDB> <o:resultDB> [options]</code>
API layer	<code>high_level_api</code>
Category flags	<code>COMMAND_MULTIHIT</code>
Upstream command count	1
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.11.1.1. Key Options

Option	Purpose
<code>--simple-best-hit</code>	Update the p-value by a single best hit, or by best and second best hits
<code>--threads</code>	Number of CPU-cores used (all by default)
<code>--compressed</code>	Write compressed output
<code>-v</code>	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.11.2. **combinepvalperset**

For each set compute the combined p-value.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is robust set-level aggregation over sequence-level evidence with transparent statistical behavior. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	<code>usage:</code> <code>mmseqs combinepvalperset <i:querySetDB></code> <code><i:targetSetDB> <i:resultDB> <o:pvalDB></code> <code>[options]</code>

Aspect	Value
API layer	high_level_api
Category flags	COMMAND_MULTIHIT
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.11.2.1. Key Options

Option	Purpose
--alpha	Set alpha for combining p-values during aggregation
--aggregation-mode	Combined P-values computed from 0: multi-hit, 1: minimum of all P-values, 2: product-of-P-values, 3: truncated product
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.11.3. mergeresultsbyset

Merge results from multiple ORFs back to their respective contig.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is robust set-level aggregation over sequence-level evidence with transparent statistical behavior. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs mergeresultsbyset <i:setDB> <i:DB> <o:DB> [options]
API layer	high_level_api
Category flags	COMMAND_MULTIHIT
Upstream command count	2
Downstream command count	0
Related functional groups	taxonomy

Aspect	Value
References	Full CLI · Dependency entry

4.11.3.1. Key Options

Option	Purpose
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.11.4. multihitdb

Create sequence DB for multi hit searches.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is robust set-level aggregation over sequence-level evidence with transparent statistical behavior. Current coupling is 0 upstream caller(s) and 8 downstream call(s).

Aspect	Value
Usage	usage: mmseqs multihitdb <i:fastaFile1[.gz bz2]> ... <i:fastaFileN[.gz bz2]> <o:setDB> <tmpDir> [options]
API layer	high_level_api
Category flags	COMMAND_MULTIHIT
Upstream command count	0
Downstream command count	8
Related functional groups	database, result_handling, sequence_manipulation, utilities
References	Full CLI · Dependency entry

4.11.4.1. Key Options

Option	Purpose
--dbtype	Database type 0: auto, 1: amino acid 2: nucleotides
--shuffle	Shuffle input database

Option	Purpose
--createdb-mode	Createdb mode 0: copy data, 1: soft link data and write new index (works only with single line fasta/q)
--id-offset	Numeric ids in index file are offset by this value
--min-length	Minimum codon number in open reading frames
--max-length	Maximum codon number in open reading frames
--max-gaps	Maximum number of codons with gaps or unknown residues before an open reading frame is rejected
--contig-start-mode	Contig start can be 0: incomplete, 1: complete, 2: both

4.11.5. multihitsearch

Search with a grouped set of sequences against another grouped set.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is robust set-level aggregation over sequence-level evidence with transparent statistical behavior. Current coupling is 0 upstream caller(s) and 4 downstream call(s).

Aspect	Value
Usage	usage: mmseqs multihitsearch <i:querySetDB> <i:targetSetDB> <o:resultDB> <tmpDir> [options]
API layer	high_level_api
Category flags	COMMAND_MULTIHIT
Upstream command count	0
Downstream command count	4
Related functional groups	database, search_workflows
References	Full CLI · Dependency entry

4.11.5.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists

4.12. Taxonomy

Modules for taxonomy database preparation, assignment, filtering, and report generation.

Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

Warning

Validate DB-type and sidecar contracts before chaining modules. Most pipeline failures are contract mismatches, not algorithmic defects.

4.12.1. addtaxonomy

Add taxonomic labels to result DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs addtaxonomy <DB> [args] [options] (source-derived synopsis; run mmseqs addtaxonomy for exact syntax)
API layer	low_level_api
Category flags	COMMAND_TAXONOMY COMMAND_EXPERT
Upstream command count	1
Downstream command count	0
Related functional groups	easy_workflows
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.12.2. **aggregatetax**

Aggregate multiple taxon labels to a single label.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs aggregatetax <DB> [args] [options] (source-derived synopsis; run mmseqs aggregatetax for exact syntax)
API layer	low_level_api
Category flags	COMMAND_TAXONOMY
Upstream command count	1
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.12.3. **aggregatetaxweights**

Aggregate multiple taxon labels to a single label.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs aggregatetaxweights <DB> [args] [options] (source-derived synopsis; run mmseqs aggregatetaxweights for exact syntax)
API layer	low_level_api
Category flags	COMMAND_TAXONOMY
Upstream command count	1
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.12.4. createbintaxmapping

Create binary taxonomy mapping from tabular taxonomy mapping.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs createbintaxmapping <DB> [args] [options] (source-derived synopsis; run mmseqs createbintaxmapping for exact syntax)
API layer	low_level_api
Category flags	COMMAND_TAXONOMY COMMAND_EXPERT
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.12.5. createbintaxonomy

Create binary taxonomy from NCBI input.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs createbintaxonomy <DB> [args] [options] (source-derived synopsis; run mmseqs createbintaxonomy for exact syntax)
API layer	low_level_api
Category flags	COMMAND_TAXONOMY COMMAND_EXPERT
Upstream command count	1
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.12.6. createdmptaxonomy

Create dmp files from binary taxonomy.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs createdmptaxonomy <DB> [args] [options] (source-derived synopsis; run mmseqs createdmptaxonomy for exact syntax)
API layer	low_level_api
Category flags	COMMAND_TAXONOMY COMMAND_EXPERT
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.12.7. createtaxdb

Add taxonomic labels to sequence DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 1 downstream call(s).

Aspect	Value
Usage	usage: mmseqs createtaxdb <i:sequenceDB> <tmpDir> [options]
API layer	low_level_api
Category flags	COMMAND_TAXONOMY
Upstream command count	1
Downstream command count	1
Related functional groups	database
References	Full CLI · Dependency entry

4.12.7.1. Key Options

Option	Purpose
--ncbi-tax-dump	tax dump directory. The tax dump can be downloaded here “ftp://ftp.ncbi.nlm.nih.gov/pub/taxonomy/taxdump.tar.gz”
--tax-mapping-file	File to map sequence identifier to taxonomical identifier
--tax-mapping-mode	Map taxonomy based on sequence database 0: .lookup file 1: .source file
--tax-db-mode	Create taxonomy database as: 0: .dmp flat files (human readable) 1: binary dump (faster readin)
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.12.8. filtertaxdb

Filter taxonomy result database.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs filtertaxdb <i:targetDB> <i:taxDB> <o:taxDB> [options]
API layer	low_level_api
Category flags	COMMAND_TAXONOMY
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.12.8.1. Key Options

Option	Purpose
--taxon-list	Taxonomy ID, possibly multiple values separated by ‘,’
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.12.9. filtertaxseqdb

Filter taxonomy sequence database.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs filtertaxseqdb <i:taxSeqDB> <o:taxSeqDB> [options]
API layer	low_level_api
Category flags	COMMAND_TAXONOMY
Upstream command count	0

Aspect	Value
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.12.9.1. Key Options

Option	Purpose
--taxon-list	Taxonomy ID, possibly multiple values separated by ‘,’
--subdb-mode	Subdb mode 0: copy data 1: soft link data and write index
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.12.10. lca

Compute the lowest common ancestor.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 2 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs lca <i:targetDB> <i:resultDB> <o:taxaDB> [options]
API layer	low_level_api
Category flags	COMMAND_TAXONOMY
Upstream command count	2
Downstream command count	0
Related functional groups	easy_workflows
References	Full CLI · Dependency entry

4.12.10.1. Key Options

Option	Purpose
--lca-ranks	Add column with specified ranks (',' separated)
--blacklist	Comma separated list of ignored taxa in LCA computation
--tax-lineage	0: don't show, 1: add all lineage names, 2: add all lineage taxids
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.12.11. lcaalign

Efficient gapped alignment for lca computation.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs lcaalign <i:queryDB> <i:targetDB> <i:resultDB> <o:alignmentDB> [options]
API layer	low_level_api
Category flags	COMMAND_TAXONOMY
Upstream command count	1
Downstream command count	0
Related functional groups	search_workflows
References	Full CLI · Dependency entry

4.12.11.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)

Option	Purpose
<code>--add-self-matches</code>	Artificially add entries of queries with themselves (for clustering)
<code>-a</code>	Add backtrace string (convert to alignments with mmseqs convertalis module)
<code>--alignment-mode</code>	How to compute the alignment:
<code>--alignment-output-mode</code>	How to compute the alignment:
<code>--wrapped-scoring</code>	Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start
<code>-e</code>	List matches below this E-value (range 0.0-inf)

4.12.12. **majoritylca**

Compute the lowest common ancestor using majority voting.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs majoritylca <DB> [args] [options] (source-derived synopsis; run mmseqs majoritylca for exact syntax)
API layer	low_level_api
Category flags	COMMAND_TAXONOMY COMMAND_EXPERT
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.12.13. **nrtotaxmapping**

Create taxonomy mapping for NR database.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs nrtotaxmapping <DB> [args] [options] (source-derived synopsis; run mmseqs nrtotaxmapping for exact syntax)
API layer	low_level_api
Category flags	COMMAND_SPECIAL
Upstream command count	1
Downstream command count	0
Related functional groups	database
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.12.14. taxonomy

Taxonomic classification.

High-level API command for end-to-end DB workflows with explicit controls. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 2 upstream caller(s) and 15 downstream call(s).

Aspect	Value
Usage	usage: mmseqs taxonomy <i:queryDB> <i:targetDB> <o:taxaDB> <tmpDir> [options]
API layer	high_level_api
Category flags	COMMAND_MAIN
Upstream command count	2
Downstream command count	15
Related functional groups	alignment, database, easy_workflows, multi_hit, prefiltering, search_workflows, sequence_manipulation, utilities
References	Full CLI · Dependency entry

4.12.14.1. Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)

Option	Purpose
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists

4.12.15. taxonomyreport

Create a taxonomy report in Kraken or Krona format.

Low-level DB or utility command used for composition and contract enforcement. Design priority is consistent taxonomy mapping and aggregation semantics across assignment and reporting modules. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs taxonomyreport <i:seqTaxDB> <i:taxResultDB/resultDB/sequenceDB> <o:taxonomyReport> [options]
API layer	low_level_api
Category flags	COMMAND_TAXONOMY COMMAND_FORMAT_CONVERSION
Upstream command count	1
Downstream command count	0
Related functional groups	easy_workflows
References	Full CLI · Dependency entry

4.12.15.1. Key Options

Option	Purpose
--report-mode	Taxonomy report mode 0: Kraken 1: Krona

Option	Purpose
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.13. Utilities

General-purpose helpers and special-purpose composition commands used in advanced pipelines.

Note

This page is task-oriented. Detailed call topology is centralized in the Dependency Map to avoid repeating large edge lists.

4.13.1. apply

Execute given program on each DB entry.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs apply <i:DB> <o:DB> -- program [args...] [options]
API layer	low_level_api
Category flags	COMMAND_DB
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.13.1.1. Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.13.2. compress

Compress DB entries.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs compress <i:DB> <o:DB> [options]
API layer	low_level_api
Category flags	COMMAND_STORAGE
Upstream command count	1
Downstream command count	0
Related functional groups	profiles
References	Full CLI · Dependency entry

4.13.2.1. Key Options

Option	Purpose
- -threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.13.3. convertkb

Convert UniProtKB data to a DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs convertkb <DB> [args] [options] (source-derived synopsis; run mmseqs convertkb for exact syntax)
API layer	low_level_api
Category flags	COMMAND_SPECIAL
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.13.4. convertblastdb

Convert BLAST database to a sequence DB.

Low-level conversion utility registered in MMseqs2/src/MMseqsBase.cpp. In some packaged builds this command is not exposed in the public CLI, so treat availability as build-dependent.

Aspect	Value
Usage	usage: mmseqs convertblastdb <DB> [args] [options] (source-derived synopsis)
API layer	low_level_api
Category flags	COMMAND_EXPERT

Run `mmseqs convertblastdb` locally on your target installation to confirm whether this command is compiled and enabled.

4.13.5. decompress

Decompress DB entries.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs decompress <i:DB> <o:DB> [options]
API layer	low_level_api
Category flags	COMMAND_STORAGE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.13.5.1. Key Options

Option	Purpose
- -threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.13.6. diffseqdbs

Compute diff of two sequence DBs.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 1 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs diffseqdbs <DB> [args] [options] (source-derived synopsis; run mmseqs diffseqdbs for exact syntax)
API layer	low_level_api
Category flags	COMMAND_SPECIAL
Upstream command count	1
Downstream command count	0
Related functional groups	clustering
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.13.7. filterdb

DB filtering by given conditions.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 8 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs filterdb <i:resultDB> <o:resultDB> [options]
API layer	low_level_api
Category flags	COMMAND_DB
Upstream command count	8
Downstream command count	0
Related functional groups	clustering, easy_workflows, multi_hit, search_workflows, taxonomy
References	Full CLI · Dependency entry

4.13.7.1. Key Options

Option	Purpose
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--filter-expression	Specify a mathematical expression to filter lines
--filter-column	column
--column-to-take	column to take in join mode. If -1, the whole line is taken
--filter-regex	Regex to select column (example float: [0-9]*([0-9]+)? int:[1-9]{1}[0-9])
--positive-filter	Used in conjunction with -filter-file. If true, out = in filter ; if false, out = in - filter
--filter-file	Specify a file that contains the filtering elements
--beats-first	Filter by comparing each entry to the first entry

4.13.8. gff2db

Extract regions from a sequence database based on a GFF3 file.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs gff2db <DB> [args] [options] (source-derived synopsis; run mmseqs gff2db for exact syntax)
API layer	low_level_api
Category flags	COMMAND_SPECIAL
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.13.9. gpuserver

Start a GPU server.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs gpuserver <i:DB> [options]
API layer	low_level_api
Category flags	COMMAND_STORAGE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.13.9.1. Key Options

Option	Purpose
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--prefilter-mode	prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped
--gpu	Use GPU (CUDA) if possible
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch

4.13.10. maskbygff

Mask out sequence regions in a sequence DB by features selected from a GFF3 file.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs maskbygff <DB> [args] [options] (source-derived synopsis; run mmseqs maskbygff for exact syntax)

Aspect	Value
API layer	low_level_api
Category flags	COMMAND_SPECIAL
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.13.11. prefixid

For each entry in a DB prepend the entry key to the entry itself.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 3 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs prefixid <i:DB> <o:DB> [options]
API layer	low_level_api
Category flags	COMMAND_DB
Upstream command count	3
Downstream command count	0
Related functional groups	clustering, database
References	Full CLI · Dependency entry

4.13.11.1. Key Options

Option	Purpose
--prefix	Use this prefix for all entries
--mapping-file	Specify a file that translates the keys of a DB to new keys, TSV format
--tsv	Return output in TSV format
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.13.12. **setextendeddbtype**

Write an extended DB.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs setextendeddbtype <i:DB> [options]
API layer	low_level_api
Category flags	COMMAND_DB
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.13.12.1. Key Options

Option	Purpose
--extended-dbtype	Set extended dbtype 1: compressed, 2: need src, 4: context pseudoe cnts
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.13.13. **suffixid**

For each entry in a DB append the entry key to the entry itself.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs suffixid <i:resultDB> <o:resultDB> [options]

Aspect	Value
API layer	low_level_api
Category flags	COMMAND_DB
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.13.13.1. Key Options

Option	Purpose
--prefix	Use this prefix for all entries
--mapping-file	Specify a file that translates the keys of a DB to new keys, TSV format
--tsv	Return output in TSV format
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.13.14. summarizetabs

Extract annotations from HHblits BLAST-tab-formatted results.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs summarizetabs <DB> [args] [options] (source-derived synopsis; run mmseqs summarizetabs for exact syntax)
API layer	low_level_api
Category flags	COMMAND_SPECIAL
Upstream command count	0
Downstream command count	0

Aspect	Value
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

No local option snapshot was parsed for this command. Use the Full CLI reference page for details.

4.13.15. touchdb

Preload DB into memory (page cache).

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs touchdb <i:DB> [options]
API layer	low_level_api
Category flags	COMMAND_STORAGE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.13.15.1. Key Options

Option	Purpose
- -threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.13.16. unpackdb

Unpack a DB into separate files.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs unpackdb <i:DB> <o:outDir> [options]

Aspect	Value
API layer	low_level_api
Category flags	COMMAND_STORAGE
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.13.16.1. Key Options

Option	Purpose
--unpack-name-mode	Name unpacked files by 0: DB key, 1: accession (through .lookup)
--unpack-suffix	File suffix for unpacked files.
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

4.13.17. view

Print DB entries given in `-id-list` to stdout.

Low-level DB or utility command used for composition and contract enforcement. Design priority is composability and operational control for custom pipelines and debugging workflows. Current coupling is 0 upstream caller(s) and 0 downstream call(s).

Aspect	Value
Usage	usage: mmseqs view <i:DB> [options]
API layer	low_level_api
Category flags	COMMAND_DB
Upstream command count	0
Downstream command count	0
Related functional groups	No direct cross-group coupling detected in the current dependency map.
References	Full CLI · Dependency entry

4.13.17.1. Key Options

Option	Purpose
--id-list	Entries to be printed separated by ‘;
--id-mode	Select DB entries based on 0: database keys, 1: FASTA identifiers (.lookup)
--idx-entry-type	0: sequence, 1: src sequence, 2: header, 3: src header
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

5. Expert Manual

This chapter covers advanced MMseqs2 operation for users who build custom pipelines, run large production workloads, or debug behavior across layered workflows. It assumes familiarity with System Map and Performance Foundations.

5.1. Expert Operating Discipline

At expert scale, the common failure mode is not command syntax. It is uncontrolled assumptions. Pipelines drift when DB contracts change silently, when modes are mixed across experiments, or when topology changes are not tracked.

A disciplined workflow starts by fixing contracts and mode envelopes before threshold tuning. DB type compatibility, sidecar completeness, and mode stability should be treated as required preconditions for meaningful performance or quality comparison.

5.2. Contract Enforcement in Custom Pipelines

Every module boundary should be treated as an explicit contract surface. Structural contracts define whether downstream modules can parse and trust the input DB type and index structures. Metadata contracts define whether required sidecars (headers, lookup maps, taxonomy files, mapping tables) are present. Semantic contracts define whether output fields still mean the same thing after mode and filter choices.

A frequent anti-pattern is adjusting scoring thresholds while one of these contracts is already broken. That often yields plausible-looking output with unstable interpretation.

Warning

When a downstream result looks wrong, verify structural, metadata, and semantic contracts first. Threshold tuning should be the last step.

5.3. Reproducibility Controls

Reproducibility requires pinning an execution envelope, not only command names. You should lock alignment and rescoring modes, filter gates, profile-construction controls, taxonomy extrac-

tion/aggregation policy, and runtime envelope settings such as split mode, load mode, and parallel topology.

In practice, most hard-to-explain result deltas come from one unpinned control in that envelope. Record these controls with every benchmark or production run so regression triage is possible.

5.4. Performance Triage for Advanced Workloads

When large jobs become slow or unstable, triage in infrastructure-first order: startup/index behavior, split and merge overhead, distributed I/O topology, then mode/filter output inflation. This ordering matches how MMseqs2 workloads typically fail at scale.

Performance

In production, index/load/split decisions usually produce larger runtime swings than late-stage threshold changes.

5.5. MMseqs2 Source Development Guide

This section is for MMseqs2 source development. The authoritative behavior definition lives in MMseqs2/ source files; documentation markdown is a derived layer for users.

5.5.1. Source-First Trace Strategy

When you change a command, start from command registration in MMseqs2/src/MMseqsBase.cpp and confirm declarations in MMseqs2/src/CommandDeclarations.h. This tells you visibility, category flags, and baseline intent.

When you change workflow behavior, inspect MMseqs2/src/workflow/*.cpp and the corresponding script glue in MMseqs2/data/workflow/*.sh. Most user-visible behavior differences come from this orchestration boundary rather than isolated kernel changes.

When you change algorithmic behavior, move to the corresponding kernel directories: prefiltering, alignment, clustering, lincust, taxonomy, and multihit. Then verify where those kernels are invoked from workflow entrypoints so you can reason about downstream impact.

5.5.2. Source Tree Responsibilities

MMseqs2/src/workflow/ is the orchestration layer for end-to-end workflows. MMseqs2/src/prefiltering/, MMseqs2/src/alignment/, MMseqs2/src/clustering/, and MMseqs2/src/lincust/ host core compute modules. MMseqs2/src/taxonomy/ and MMseqs2/src/multihit/ cover specialized assignment and set-aggregation logic.

MMseqs2/src/util/ and MMseqs2/src/commons/ provide DB transforms, exports, shared parameter logic, and low-level I/O abstractions. If behavior shifts unexpectedly across many commands, inspect commons first; it is often the shared root cause.

5.5.3. Practical Debug Workflow

Use a repeatable debug loop:

1. Locate command registration and category metadata.
2. Trace workflow composition and parameter propagation.
3. Inspect the target kernel implementation.
4. Validate DB type and sidecar boundary assumptions.
5. Re-check output semantics in result/export transforms.

Representative commands for this loop:

- `rg -n '<command>' MMseqs2/src/MMseqsBase.cpp`
- `rg -n 'createParameterString\(\par\.' MMseqs2/src/workflow/*.cpp`
- `rg -n '<symbol>' MMseqs2/src/{prefiltering,alignment,clustering,linclust,taxonomy,multihit}/*.cpp`
- `rg -n 'DBReader|DBWriter|DBTYPE' MMseqs2/src/{workflow,util,commons}/*.{h,cpp}`
- `rg -n 'MMSEQS|RUNNER|\$\{.*_PAR\}' MMseqs2/data/workflow/*.sh`

After source validation, regenerate documentation artifacts so command pages and module pages remain synchronized with implementation.

Use Performance Foundations for storage/index/split/parallel mechanics, System Map and Dependency Map for architecture and topology debugging, Functional Modules Manual for task-first command selection, and this section (MMseqs2 Source Development Guide) for source-first tracing paths.

6. MMseqs2 Command Reference Index

This index is generated from command metadata, dependency topology, and local CLI help snapshots where available. Use it to move quickly from command name to functional placement and detailed page content.

Metric	Value
Total visible commands	128
Commands with help snapshots	97
Commands with source-derived fallback pages	31

Note

Some commands currently use source-derived fallback text because local help snapshots were not present in `mmseqs_help_output`.

Command	Snapshot status
<code>addtaxonomy</code>	source-derived fallback
<code>aggregatetax</code>	source-derived fallback
<code>aggregatetaxweights</code>	source-derived fallback

Command	Snapshot status
convert2fasta	source-derived fallback
convertca3m	source-derived fallback
convertkb	source-derived fallback
convertprofiledb	source-derived fallback
countkmer	source-derived fallback
createbintaxmapping	source-derived fallback
createbintaxonomy	source-derived fallback
createdmptaxonomy	source-derived fallback
databases	source-derived fallback
diffseqdbs	source-derived fallback
expand2profile	source-derived fallback
extractdomains	source-derived fallback
fwbw	source-derived fallback
gff2db	source-derived fallback
majoritylca	source-derived fallback
maskbygff	source-derived fallback
nrtotaxmapping	source-derived fallback
pairaln	source-derived fallback
pickconsensusrep	source-derived fallback
profile2consensus	source-derived fallback
profile2neff	source-derived fallback
profile2pssm	source-derived fallback
profile2repseq	source-derived fallback
rmdb	source-derived fallback
sequence2profile	source-derived fallback
summarizeheaders	source-derived fallback
summarizetabs	source-derived fallback
transitivealign	source-derived fallback

Primary topology view: Dependency map.

6.1. Command Group Map

Group	Command count	Commands
easy_workflows	6	easy-cluster, easy-linclud, easy-linsearch, easy-rbh, easy-search, easy-taxonomy
search_workflows	4	linsearch, map, rbh, search
clustering	7	clust, cluster, clusterupdate, clusthash, linclud, mergeclusters, pickconsensusrep
prefiltering	6	countkmer, gappedprefilter, kmermatcher, kmersearch, prefilter, ungappedprefilter
alignment	9	align, alignall, alignbykmer, expandaln, fwbw, offsetalignment, proteinaln2nucl, rescorediagonal, transitivealign
profiles	14	convertca3m, convertmsa, convertprofiledb, expand2profile, msa2profile, msa2result, pairaln, profile2consensus, profile2neff, profile2pssm, profile2repseq, result2profile, sequence2profile, tsv2expfiledb
database	20	aliasdb, concatdbs, cpdb, createdb, createindex, createlinindex, createsubdb, databases, db2tar, lndb, mergedbs,

Group	Command count	Commands
		mvdb, renamedbkeys, rmdb, splitdb, splitsequence, subtractdb, swapdb, tar2db, tsv2db
result_handling	17	convert2fasta, convertalis, createseqfiledb, createtsv, extractdomains, filterresult, result2dnamsa, result2flat, result2msa, result2rbh, result2repseq, result2stats, sortresult, summarizealis, summarizeheaders, summarizeresult, swapresults
sequence_manipulation	9	extractalignedregion, extractframes, extractorfs, masksequence, orftocontig, recoverlongestorf, reverseseq, translateaa, translatenucs
taxonomy	15	addtaxonomy, aggregatetax, aggregatetaxweights, createbintaxmapping, createbintaxonomy, createdmptaxonomy, createtaxdb, filtertaxdb, filtertaxseqdb, lca, lcaalign, majoritylca, nrtotaxmapping, taxonomy, taxonomyreport
multi_hit	5	besthitperset, combinepvalperset, mergeresultsbyset, multihitdb, multihitsearch

Group	Command count	Commands
utilities	16	apply, compress, convertkb, decompress, diffseqdbs, filterdb, gff2db, gpuserver, maskbygff, prefixid, setextendeddbtype, suffixid, summarizetabs, touchdb, unpackdb, view

6.2. Command Pages

Full command pages follow below. Each page keeps local usage/options snapshots plus dependency links.

6.2.1. addtaxonomy

Add taxonomic labels to result DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

usage: mmseqs addtaxonomy <DB> [args] [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs addtaxonomy locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.2. **aggregatetax**

Aggregate multiple taxon labels to a single label.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

```
usage: mmseqs aggregatetax <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs aggregatetax` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.3. **aggregatetaxweights**

Aggregate multiple taxon labels to a single label.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

```
usage: mmseqs aggregatetaxweights <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs aggregatetaxweights` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.4. `aliasdb`

Create relative symlink of DB to another name in the same folder.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

usage: `mmseqs aliasdb <i:srcDB> <o:dstDB> [options]`

Key Options

Option	Purpose
<code>-v</code>	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: `mmseqs aliasdb <i:srcDB> <o:dstDB> [options]`

By Milot Mirdita <milot@mirdita.de>

options: common:

`-v INT` Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

6.2.5. alignall

Within-result all-vs-all gapped local alignment.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

Usage

usage: mmseqs alignall <i:sequenceDB> <i:resultDB> <o:alignmentDB> [options]

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
-a	Add backtrace string (convert to alignments with mmseqs convertalis module)
--alignment-mode	How to compute the alignment:
-e	List matches below this E-value (range 0.0-inf)
--min-seq-id	List matches above this sequence identity (for clustering) (range 0.0-1.0)
--min-aln-len	Minimum alignment length (range 0-INT_MAX)
--seq-id-mode	0: alignment length 1: shorter, 2: longer sequence
-c	List matches above this fraction of aligned (covered) residues (see --cov-mode)
--cov-mode	0: coverage of query and target

Option	Purpose
--score-bias	Score bias when computing SW alignment (in bits)

Full CLI Help Snapshot

```
usage: mmseqs alignall <i:sequenceDB> <i:resultDB> <o:alignmentDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
(range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with themselves
(for clustering) [0]
align:
  -a BOOL                      Add backtrace string (convert to alignments with
mmseqs convertalis module) [0]
  --alignment-mode INT          How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id [0]
  -e DOUBLE                    List matches below this E-value (range 0.0-inf)
[1.000E-03]
  --min-seq-id FLOAT           List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
  --min-aln-len INT            Minimum alignment length (range 0-INT_MAX) [0]
  --seq-id-mode INT            0: alignment length 1: shorter, 2: longer sequence
[0]
  -c FLOAT                    List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.000]
  --cov-mode INT               0: coverage of query and target
                                1: coverage of target
                                2: coverage of query
                                3: target seq. length has to be at least x% of
query length
                                4: query seq. length has to be at least x% of
target length
                                5: short seq. needs to be at least x% of the other
seq. length [0]
  --score-bias FLOAT           Score bias when computing SW alignment (in bits)
[0.000]
  --gap-open TWIN              Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN            Gap extension cost [aa:1,nucl:2]
  --zdrop INT                  Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
profile:
```

```

--pca                      Pseudo count admixture strength []
--pcb                      Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
common:
--sub-mat TWIN             Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--max-seq-len INT          Maximum sequence length [65535]
--db-load-mode INT         Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--threads INT              Number of CPU-cores used (all by default) [10]
--compressed INT           Write compressed output [0]
-v INT                     Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.6. alignbykmer

Heuristic gapped local k-mer based alignment.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

Usage

```
usage: mmseqs alignbykmer <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB>
[options]
```

Key Options

Option	Purpose
-k	k-mer length (0: automatically set to optimum)
--spaced-kmer-mode	0: use consecutive positions in k-mers; 1: use spaced k-mers
--spaced-kmer-pattern	User-specified spaced k-mer pattern
--alph-size	Alphabet size (range 2-21)

Option	Purpose
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
-c	List matches above this fraction of aligned (covered) residues (see --cov-mode)
-e	List matches below this E-value (range 0.0-inf)
--cov-mode	0: coverage of query and target
--min-seq-id	List matches above this sequence identity (for clustering) (range 0.0-1.0)
--min-aln-len	Minimum alignment length (range 0-INT_MAX)
--gap-open	Gap open cost
--gap-extend	Gap extension cost

Full CLI Help Snapshot

Rescore diagonals.

usage: mmseqs alignbykmer <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB>
[options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

-k INT k-mer length (0: automatically set to optimum) [0]

--spaced-kmer-mode INT 0: use consecutive positions in k-mers; 1: use

spaced k-mers [1]

--spaced-kmer-pattern STR User-specified spaced k-mer pattern []

--alph-size TWIN Alphabet size (range 2-21) [aa:21,nuc:5]

--add-self-matches BOOL Artificially add entries of queries with themselves
(for clustering) [0]

align:

-c FLOAT List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.000]

-e DOUBLE List matches below this E-value (range 0.0-inf)
[1.000E-03]

--cov-mode INT 0: coverage of query and target
1: coverage of target
2: coverage of query
3: target seq. length has to be at least x% of query

length 4: query seq. length has to be at least x% of target

length 5: short seq. needs to be at least x% of the other

seq. length [0]

--min-seq-id FLOAT List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]


```

--min-aln-len INT           Minimum alignment length (range 0-INT_MAX) [0]
--gap-open TWIN             Gap open cost [aa:11,nucl:5]
--gap-extend TWIN          Gap extension cost [aa:1,nucl:2]
common:
--sub-mat TWIN              Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--threads INT               Number of CPU-cores used (all by default) [10]
--compressed INT            Write compressed output [0]
-v INT                      Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
3: +info [3]
expert:
--filter-hits BOOL          Filter hits by seq.id. and coverage [0]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

```

6.2.7. align

Optimal gapped local alignment.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 5 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

Usage

usage: mmseqs align <i:queryDB> <i:targetDB> <i:resultDB> <o:alignmentDB> [options]

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
-a	Add backtrace string (convert to alignments with mmseqs convertalis module)

Option	Purpose
--alignment-mode	How to compute the alignment:
--alignment-output-mode	How to compute the alignment:
--wrapped-scoring	Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start
-e	List matches below this E-value (range 0.0-inf)
--min-seq-id	List matches above this sequence identity (for clustering) (range 0.0-1.0)
--min-aln-len	Minimum alignment length (range 0-INT_MAX)
--seq-id-mode	0: alignment length 1: shorter, 2: longer sequence
--alt-ali	Show up to this many alternative alignments

Full CLI Help Snapshot

```
usage: mmseqs align <i:queryDB> <i:targetDB> <i:resultDB> <o:alignmentDB>
[options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr> & Maria Hauser
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
(range 0-1) [1]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid composition
(range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
align:
  -a BOOL                      Add backtrace string (convert to alignments with
mmseqs convertalis module) [0]
  --alignment-mode INT          How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id [0]
  --alignment-output-mode INT   How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment
                                5: score only (output) cluster format [0]
  --wrapped-scoring BOOL        Double the (nucleotide) query sequence during the
scoring process to allow wrapped diagonal scoring around end and start [0]
```

-e DOUBLE List matches below this E-value (range 0.0-inf)
 [1.000E-03]
 --min-seq-id FLOAT List matches above this sequence identity (for
 clustering) (range 0.0-1.0) [0.000]
 --min-aln-len INT Minimum alignment length (range 0-INT_MAX) [0]
 --seq-id-mode INT 0: alignment length 1: shorter, 2: longer sequence
 [0]
 --alt-ali INT Show up to this many alternative alignments [0]
 -c FLOAT List matches above this fraction of aligned
 (covered) residues (see --cov-mode) [0.000]
 --cov-mode INT 0: coverage of query and target
 1: coverage of target
 2: coverage of query
 3: target seq. length has to be at least x% of
 query length
 4: query seq. length has to be at least x% of
 target length
 5: short seq. needs to be at least x% of the other
 seq. length [0]
 --max-rejected INT Maximum rejected alignments before alignment
 calculation for a query is stopped [2147483647]
 --max-accept INT Maximum accepted alignments before alignment
 calculation for a query is stopped [2147483647]
 --score-bias FLOAT Score bias when computing SW alignment (in bits)
 [0.000]
 --realign BOOL Compute more conservative, shorter alignments
 (scores and E-values not changed) [0]
 --realign-score-bias FLOAT Additional bias when computing realignment
 [-0.200]
 --realign-max-seqs INT Maximum number of results to return in realignment
 [2147483647]
 --corr-score-weight FLOAT Weight of backtrace correlation score that is
 added to the alignment score [0.000]
 --gap-open TWIN Gap open cost [aa:11,nucl:5]
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]
 --zdrop INT Maximal allowed difference between score values
 before alignment is truncated (nucleotide alignment only) [40]
 profile:
 --pca Pseudo count admixture strength []
 --pcb Pseudo counts: Neff at half of maximum admixture
 (range 0.0-inf) []
 common:
 --sub-mat TWIN Substitution matrix file
 [aa:blosom62.out,nucl:nucleotide.out]
 --max-seq-len INT Maximum sequence length [65535]
 --db-load-mode INT Database preload mode 0: auto, 1: fread, 2: mmap,
 3: mmap+touch [0]
 --threads INT Number of CPU-cores used (all by default) [10]
 --compressed INT Write compressed output [0]

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.8. apply

Execute given program on each DB entry.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

usage: mmseqs apply <i:DB> <o:DB> -- program [args...] [options]

Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: mmseqs apply <i:DB> <o:DB> -- program [args...] [options]

By Milot Mirdita <milot@mirdita.de>

options: common:

--threads INT Number of CPU-cores used (all by default) [10]
--compressed INT Write compressed output [0]
-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

examples:

Gather all sequences from a cluster DB
mmseqs createseqfiledb sequenceDB clusterDB unalignedDB --min-sequences 2
Build MSAs with Clustal-Omega
mmseqs apply unalignedDB msadb -- clustalo -i - -o stdout --threads=1

```
# Count lines in each DB entry inefficiently (result2stats is way faster)
mmseqs apply DB wcDB -- awk '{ counter++; } END { print counter; }'
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.9. besthitperset

For each set of sequences compute the best element and update p-value.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family aggregates sequence-level hits into set-level statistics and decisions. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: multi_hit.

Usage

```
usage: mmseqs besthitperset <i:targetSetDB> <i:resultDB> <o:resultDB> [options]
```

Key Options

Option	Purpose
--simple-best-hit	Update the p-value by a single best hit, or by best and second best hits
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs besthitperset <i:targetSetDB> <i:resultDB> <o:resultDB> [options]
```

By Ruoshi Zhang, Clovis Norroy & Milot Mirdita <milot@mirdita.de>

options: misc:

```
--simple-best-hit BOOL Update the p-value by a single best hit, or by best and
second best hits [1]
```

common:

```
--threads INT Number of CPU-cores used (all by default) [10]
```

```
--compressed INT Write compressed output [0]
```

```
-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
```

```
+info [3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.10. cluster

Slower, sensitive clustering.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 2 upstream caller(s) and 18 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

Usage

usage: mmseqs cluster <i:sequenceDB> <o:clusterDB> <tmpDir> [options]

Key Options

Option	Purpose
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists
--alph-size	Alphabet size (range 2-21)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--split	Split input into N equally distributed chunks. 0: set the best split automatically
--split-mode	0: split target db; 1: split query db; 2: auto, depending on main memory

Option	Purpose
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)

Full CLI Help Snapshot

```
usage: mmseqs cluster <i:sequenceDB> <o:clusterDB> <tmpDir> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr> & Lars von den Driesch
options: prefilter:
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                    Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [4.000]
  -k INT                      k-mer length (0: automatically set to optimum)
[0]
  --target-search-mode INT     target search mode (0: regular k-mer, 1:
similar k-mer) [0]
  --k-score TWIN              k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
  --alph-size TWIN            Alphabet size (range 2-21) [aa:21,nucl:5]
  --max-seqs INT              Maximum results per query sequence allowed to
pass the prefilter (affects sensitivity) [20]
  --split INT                 Split input into N equally distributed chunks.
0: set the best split automatically [0]
  --split-mode INT            0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
  --split-memory-limit BYTE    Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
  --comp-bias-corr INT         Correct for locally biased amino acid
composition (range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid
composition (range 0-1) [1.000]
  --diag-score BOOL           Use ungapped diagonal scoring during prefilter
[1]
  --exact-kmer-matching INT    Extract only exact k-mers for matching (range
0-1) [0]
  --mask INT                  Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [1]
  --mask-prob FLOAT           Mask sequences is probablity is above threshold
[0.900]
  --mask-lower-case INT        Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
  --mask-n-repeat INT          Repeat letters that occure > threshold in a two
```

```

[0]
--min-ungapped-score INT          Accept only matches with ungapped alignment
score above threshold [15]
--add-self-matches BOOL          Artificially add entries of queries with
themselves (for clustering) [0]
--spaced-kmer-mode INT            0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
--spaced-kmer-pattern STR         User-specified spaced k-mer pattern []
--local-tmp STR                   Path where some of the temporary files will be
created []
align:
-c FLOAT                          List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.800]
--cov-mode INT                    0: coverage of query and target
                                1: coverage of target
                                2: coverage of query
                                3: target seq. length has to be at least x% of
query length
                                4: query seq. length has to be at least x% of
target length
                                5: short seq. needs to be at least x% of the
other seq. length [0]
-a BOOL                           Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]
--alignment-mode INT              How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment [3]
--alignment-output-mode INT       How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment
                                5: score only (output) cluster format [0]
--wrapped-scoring BOOL            Double the (nucleotide) query sequence during
the scoring process to allow wrapped diagonal scoring around end and start [0]
-e DOUBLE                         List matches below this E-value (range 0.0-inf)
[1.000E-03]
--min-seq-id FLOAT                List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
--min-aln-len INT                 Minimum alignment length (range 0-INT_MAX) [0]
--seq-id-mode INT                 0: alignment length 1: shorter, 2: longer
sequence [0]
--alt-ali INT                     Show up to this many alternative alignments [0]
--max-rejected INT                Maximum rejected alignments before alignment
calculation for a query is stopped [2147483647]

```



```

--max-accept INT           Maximum accepted alignments before alignment
calculation for a query is stopped [2147483647]
--score-bias FLOAT         Score bias when computing SW alignment (in
bits) [0.000]
--realign BOOL             Compute more conservative, shorter alignments
(scores and E-values not changed) [0]
--realign-score-bias FLOAT Additional bias when computing realignment
[-0.200]
--realign-max-seqs INT     Maximum number of results to return in
realignment [2147483647]
--corr-score-weight FLOAT  Weight of backtrace correlation score that is
added to the alignment score [0.000]
--gap-open TWIN            Gap open cost [aa:11,nucl:5]
--gap-extend TWIN          Gap extension cost [aa:1,nucl:2]
--zdrop INT                Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
clust:
--cluster-mode INT         0: Set-Cover (greedy)
                           1: Connected component (BLASTclust)
                           2,3: Greedy clustering by sequence length
(CDHIT) [0]
--max-iterations INT       Maximum depth of breadth first search in
connected component clustering [1000]
--similarity-type INT       Type of score used for clustering. 1: alignment
score 2: sequence identity [2]
--single-step-clustering BOOL Switch from cascaded to simple clustering
workflow [0]
--cluster-steps INT         Cascaded clustering steps from 1 to -s [3]
--cluster-reassign BOOL     Cascaded clustering can cluster sequence that
do not fulfill the clustering criteria.
                           Cluster reassignment corrects these errors [0]
kmermatcher:
--weights STR              Weights used for cluster prioritization []
--cluster-weight-threshold FLOAT Weight threshold used for cluster prioritization
[0.900]
--kmer-per-seq INT         k-mers per sequence [21]
--kmer-per-seq-scale TWIN   Scale k-mer per sequence based on sequence
length as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
--adjust-kmer-len BOOL     Adjust k-mer length based on specificity (only
for nucleotides) [0]
--hash-shift INT           Shift k-mer hash initialization [67]
--include-only-extendable BOOL Include only extendable [0]
--ignore-multi-kmer BOOL   Skip k-mers occurring multiple times (>=2) [0]
profile:
--pca                      Pseudo count admixture strength []
--pcb                      Pseudo counts: Neff at half of maximum
admixture (range 0.0-inf) []
misc:
--taxon-list STR           Taxonomy ID, possibly multiple values separated

```

```

by ',' []
--rescore-mode INT                                Rescore diagonals with:
                                                    0: Hamming distance
                                                    1: local alignment (score only)
                                                    2: local alignment
                                                    3: global alignment
                                                    4: longest alignment fulfilling window quality

criterion [0]
common:
--sub-mat TWIN                                    Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--max-seq-len INT                                Maximum sequence length [65535]
--db-load-mode INT                                Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
--threads INT                                    Number of CPU-cores used (all by default) [10]
--compressed INT                                Write compressed output [0]
-v INT                                            Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--remove-tmp-files BOOL                          Delete temporary files [0]
--force-reuse BOOL                                Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
--mpi-runner STR                                Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
expert:
--filter-hits BOOL                                Filter hits by seq.id. and coverage [0]
--sort-results INT                                Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]

```

examples:

```

# Cascaded clustering of FASTA file
mmseqs cluster sequenceDB clusterDB tmp

```

```

#                               --cov-mode
# Sequence                     0    1    2
# Q: MAVGTACRPA 60% IGN 60%
# T: -AVGTAC-- 60% 100% IGN
# Cutoff -c 0.7  -    +    -
#           -c 0.6  +    +    +

```

```

# Cascaded clustering with reassignment
# - Corrects criteria-violations of cascaded merging
# - Produces more clusters and is a bit slower
mmseqs cluster sequenceDB clusterDB tmp --cluster-reassign

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)
- Hauser M, Steinegger M, Soding J: MMseqs software suite for fast and deep

clustering and searching of large protein sequence sets. *Bioinformatics*, 32(9), 1323-1330 (2016)

- Steinegger M, Soding J: Clustering huge protein sequence sets in linear time. *Nature Communications*, 9(1), 2542 (2018)

6.2.11. clusterupdate

Update previous clustering with new sequences.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 0 upstream caller(s) and 13 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

Usage

```
usage:      mmseqs      clusterupdate      <i:oldSequenceDB>      <i:newSequenceDB>
<i:oldClustResultDB>      <o:newMappedSequenceDB>      <o:newClustResultDB>      <tmpDir>
[options]
```

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists

Option	Purpose
--alph-size	Alphabet size (range 2-21)
--split	Split input into N equally distributed chunks. 0: set the best split automatically
--split-mode	0: split target db; 1: split query db; 2: auto, depending on main memory
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory

Full CLI Help Snapshot

```
usage: mmseqs clusterupdate <i:oldSequenceDB> <i:newSequenceDB>
<i:oldClustResultDB> <o:newMappedSequenceDB> <o:newClustResultDB> <tmpDir>
[options]
  By Clovis Galiez & Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid
composition (range 0-1) [1]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid
composition (range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [4.000]
  -k INT                       k-mer length (0: automatically set to optimum)
[0]
  --target-search-mode INT      target search mode (0: regular k-mer, 1:
similar k-mer) [0]
  --k-score TWIN               k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
  --alph-size TWIN             Alphabet size (range 2-21) [aa:21,nucl:5]
  --split INT                  Split input into N equally distributed chunks.
0: set the best split automatically [0]
  --split-mode INT             0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
  --split-memory-limit BYTE     Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
  --diag-score BOOL            Use ungapped diagonal scoring during prefilter
[1]
  --exact-kmer-matching INT     Extract only exact k-mers for matching (range
0-1) [0]
  --mask INT                   Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [1]
  --mask-prob FLOAT            Mask sequences is probablity is above threshold
```

[0.900]

--mask-lower-case INT Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region [0]

--mask-n-repeat INT Repeat letters that occur > threshold in a row [0]

--min-ungapped-score INT Accept only matches with ungapped alignment score above threshold [15]

--spaced-kmer-mode INT 0: use consecutive positions in k-mers; 1: use spaced k-mers [1]

--spaced-kmer-pattern STR User-specified spaced k-mer pattern []

--local-tmp STR Path where some of the temporary files will be created []

--disk-space-limit BYTE Set max disk space to use for reverse profile searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the temp folder [0]

align:

-a BOOL Add backtrace string (convert to alignments with mmseqs convertalis module) [0]

--alignment-mode INT How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment [3]

--alignment-output-mode INT How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment
5: score only (output) cluster format [0]

--wrapped-scoring BOOL Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start [0]

-e DOUBLE List matches below this E-value (range 0.0-inf) [1.000E-03]

--min-seq-id FLOAT List matches above this sequence identity (for clustering) (range 0.0-1.0) [0.000]

--min-aln-len INT Minimum alignment length (range 0-INT_MAX) [0]

--seq-id-mode INT 0: alignment length 1: shorter, 2: longer sequence [0]

--alt-ali INT Show up to this many alternative alignments [0]

-c FLOAT List matches above this fraction of aligned (covered) residues (see --cov-mode) [0.000]

--cov-mode INT 0: coverage of query and target
1: coverage of target
2: coverage of query
3: target seq. length has to be at least x% of query length
4: query seq. length has to be at least x% of

target length 5: short seq. needs to be at least x% of the other seq. length [0]

--max-rejected INT Maximum rejected alignments before alignment calculation for a query is stopped [2147483647]

--max-accept INT Maximum accepted alignments before alignment calculation for a query is stopped [2147483647]

--score-bias FLOAT Score bias when computing SW alignment (in bits) [0.000]

--realign BOOL Compute more conservative, shorter alignments (scores and E-values not changed) [0]

--realign-score-bias FLOAT Additional bias when computing realignment [-0.200]

--realign-max-seqs INT Maximum number of results to return in realignment [2147483647]

--corr-score-weight FLOAT Weight of backtrace correlation score that is added to the alignment score [0.000]

--gap-open TWIN Gap open cost [aa:11,nucl:5]

--gap-extend TWIN Gap extension cost [aa:1,nucl:2]

--zdrop INT Maximal allowed difference between score values before alignment is truncated (nucleotide alignment only) [40]

--exhaustive-search-filter INT Filter result during search: 0: do not filter, 1: filter [0]

clust:

--cluster-mode INT 0: Set-Cover (greedy)
1: Connected component (BLASTclust)
2,3: Greedy clustering by sequence length

(CDHIT) [0]

--max-iterations INT Maximum depth of breadth first search in connected component clustering [1000]

--similarity-type INT Type of score used for clustering. 1: alignment score 2: sequence identity [2]

--single-step-clustering BOOL Switch from cascaded to simple clustering workflow [0]

--cluster-steps INT Cascaded clustering steps from 1 to -s [3]

--cluster-reassign BOOL Cascaded clustering can cluster sequence that do not fulfill the clustering criteria.
Cluster reassignment corrects these errors [0]

kmermatcher:

--weights STR Weights used for cluster prioritization []

--cluster-weight-threshold FLOAT Weight threshold used for cluster prioritization [0.900]

--kmer-per-seq INT k-mers per sequence [21]

--kmer-per-seq-scale TWIN Scale k-mer per sequence based on sequence length as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]

--adjust-kmer-len BOOL Adjust k-mer length based on specificity (only for nucleotides) [0]

--hash-shift INT Shift k-mer hash initialization [67]

--include-only-extendable BOOL Include only extendable [0]

```

--ignore-multi-kmer BOOL      Skip k-mers occurring multiple times (>=2) [0]
profile:
--pca                          Pseudo count admixture strength []
--pcb                          Pseudo counts: Neff at half of maximum
admixture (range 0.0-inf) []
--mask-profile INT             Mask query sequence of profile using tantan
[0,1] [1]
--e-profile DOUBLE             Include sequences matches with < E-value thr.
into the profile (>=0.0) [1.000E-03]
--wg BOOL                      Use global sequence weighting for profile
calculation [0]
--filter-msa INT               Filter msa: 0: do not filter, 1: filter [1]
--filter-min-enable INT        Only filter MSAs with more than N sequences, 0
always filters [0]
--max-seq-id FLOAT             Reduce redundancy of output MSA using max.
pairwise sequence identity [0.0,1.0] [0.900]
--qid STR                      Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
                                Alternatively, can be a list of multiple
thresholds:
                                E.g.: 0.15,0.30,0.50 to defines filter buckets
of [0.15-0.30] and [0.30-0.50] [0.0]
--qsc FLOAT                   Reduce diversity of output MSAs using min.
score per aligned residue with query sequences [-50.0,100.0] [-20.000]
--cov FLOAT                   Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
--diff INT                    Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
--pseudo-cnt-mode INT          use 0: substitution-matrix or 1: context-
specific pseudocounts [0]
--profile-output-mode INT       Profile output mode: 0: binary log-odds 1:
human-readable frequencies [0]
--num-iterations INT           Number of iterative profile search iterations
[1]
--exhaustive-search BOOL        For bigger profile DB, run iteratively the
search by greedily swapping the search results [0]
--lca-search BOOL              Efficient search for LCA candidates [0]
misc:
--taxon-list STR               Taxonomy ID, possibly multiple values separated
by ',' []
--prefilter-mode INT            prefilter mode: 0: kmer/ungapped 1: ungapped,
2: nofilter, 3: ungapped&gapped [0]
--rescore-mode INT              Rescore diagonals with:
                                0: Hamming distance
                                1: local alignment (score only)
                                2: local alignment
                                3: global alignment
                                4: longest alignment fulfilling window quality
criterion [0]

```

```

--allow-deletion BOOL          Allow deletions in a MSA [0]
--min-length INT               Minimum codon number in open reading frames
[30]
--max-length INT               Maximum codon number in open reading frames
[32734]
--max-gaps INT                 Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT        Contig start can be 0: incomplete, 1: complete,
2: both [2]
--contig-end-mode INT          Contig end can be 0: incomplete, 1: complete,
2: both [2]
--orf-start-mode INT           Orf fragment can be 0: from start to stop, 1:
from any to stop, 2: from last encountered start to stop (no start in the middle)
[1]
--forward-frames STR           Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR           Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT         1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16)
CHLOROPHYCEAN_MITOCHONDRIAL, 21) TREMATODE_MITOCHONDRIAL, 22)
SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31)
BLASTOCRITHIDIA [1]
--translate INT                Translate ORF to amino acid [0]
--use-all-table-starts BOOL    Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT                Numeric ids in index file are offset by this
value [0]
--sequence-overlap INT          Overlap between sequences [0]
--sequence-split-mode INT       Sequence split mode 0: copy data, 1: soft link
data and write new index, [1]
--headers-split-mode INT        Header split mode: 0: split position, 1:
original header [0]
--search-type INT               Search type 0: auto 1: amino acid, 2:
translated, 3: nucleotide, 4: translated nucleotide alignment [0]
--start-sens FLOAT              Start sensitivity [4.000]
--sens-steps INT                Number of search steps performed from --start-
sens to -s [1]
--translation-mode INT          Translation AA seq from nucleotide by 0: ORFs,
1: full reading frames [0]
--use-seq-id BOOL               Sequence ID (Uniprot, GenBank, ...) is used for

```



```

identifying matches between the old and the new DB [0]
--recover-deleted BOOL          Find and recover deleted sequences during
updating of clustering [0]
common:
--sub-mat TWIN                  Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--max-seq-len INT               Maximum sequence length [65535]
--db-load-mode INT              Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
--threads INT                   Number of CPU-cores used (all by default) [10]
--compressed INT                Write compressed output [0]
-v INT                          Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--mpi-runner STR                Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
--force-reuse BOOL              Reuse tmp file in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL         Delete temporary files [0]
expert:
--filter-hits BOOL              Filter hits by seq.id. and coverage [0]
--sort-results INT              Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--create-lookup INT             Create database lookup file (can be very large)
[0]
--chain-alignments INT          Chain overlapping alignments [0]
--merge-query INT               Combine ORFs/split sequences to a single entry
[1]
--strand INT                    Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]

examples:
# Update clustering workflow
# Perform initial clustering of 5000 sequences
mmseqs createdb <(head -n 10000 examples/DB.fasta) sequenceDB
mmseqs cluster sequenceDB clusterDB tmp

# Use-case 1: Update by only adding sequences
mmseqs createdb examples/QUERY.fasta addedSequenceDB
mmseqs concatdbs sequenceDB addedSequenceDB allSequenceDB
mmseqs concatdbs sequenceDB_h addedSequenceDB_h allSequenceDB_h
mmseqs clusterupdate sequenceDB allSequenceDB clusterDB newSequenceDB
newClusterDB tmp

# Use-case 2: Update clustering with deletions)
# Create a FASTA file missing 500 of the original sequences and 2500 new ones
mmseqs createdb <(tail -n +1001 examples/DB.fasta | head -n 15000)
updateSequenceDB
mmseqs clusterupdate sequenceDB updateSequenceDB clusterDB newSequenceDB
newClusterDB tmp

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Hauser M, Steinegger M, Soding J: MMseqs software suite for fast and deep clustering and searching of large protein sequence sets. *Bioinformatics*, 32(9), 1323-1330 (2016)

6.2.12. clusthash

Hash-based clustering of equal length sequences.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

Usage

usage: mmseqs clusthash <i:sequenceDB> <o:alignmentDB> [options]

Key Options

Option	Purpose
--alph-size	Alphabet size (range 2-21)
--min-seq-id	List matches above this sequence identity (for clustering) (range 0.0-1.0)
--sub-mat	Substitution matrix file
--max-seq-len	Maximum sequence length
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs clusthash <i:sequenceDB> <o:alignmentDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --alph-size TWIN      Alphabet size (range 2-21) [aa:3,nucl:5]
align:
  --min-seq-id FLOAT    List matches above this sequence identity (for clustering)
                        (range 0.0-1.0) [0.990]
common:
  --sub-mat TWIN        Substitution matrix file
                        [aa:blosum62.out,nucl:nucleotide.out]
  --max-seq-len INT      Maximum sequence length [65535]
  --db-load-mode INT     Database preload mode 0: auto, 1: fread, 2: mmap, 3:
                        mmap+touch [0]
  --threads INT          Number of CPU-cores used (all by default) [10]
  --compressed INT       Write compressed output [0]
  -v INT                 Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
                        +info [3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
    for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
    (2017)
```

6.2.13. clust

Cluster result by Set-Cover/Connected-Component/Greedy-Incremental.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

Usage

```
usage: mmseqs clust <i:sequenceDB> <i:resultDB> <o:clusterDB> [options]
```

Key Options

Option	Purpose
--cluster-mode	0: Set-Cover (greedy)
--max-iterations	Maximum depth of breadth first search in connected component clustering

Option	Purpose
--similarity-type	Type of score used for clustering. 1: alignment score 2: sequence identity
--weights	Weights used for cluster prioritization
--cluster-weight-threshold	Weight threshold used for cluster prioritization
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs clust <i:sequenceDB> <i:resultDB> <o:clusterDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr> & Lars von den Driesch &
  Maria Hauser
options: clust:
  --cluster-mode INT          0: Set-Cover (greedy)
                              1: Connected component (BLASTclust)
                              2,3: Greedy clustering by sequence length

(CDHIT) [0]
  --max-iterations INT       Maximum depth of breadth first search in
connected component clustering [1000]
  --similarity-type INT       Type of score used for clustering. 1: alignment
score 2: sequence identity [2]
kmermatcher:
  --weights STR              Weights used for cluster prioritization []
  --cluster-weight-threshold FLOAT Weight threshold used for cluster prioritization
[0.900]
common:
  --threads INT              Number of CPU-cores used (all by default) [10]
  --compressed INT           Write compressed output [0]
  -v INT                     Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
  - Hauser M, Steinegger M, Soding J: MMseqs software suite for fast and deep
clustering and searching of large protein sequence sets. Bioinformatics, 32(9),
1323-1330 (2016)
```

6.2.14. combinepvalperset

For each set compute the combined p-value.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family aggregates sequence-level hits into set-level statistics and decisions. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: multi_hit.

Usage

```
usage: mmseqs combinepvalperset <i:querySetDB> <i:targetSetDB> <i:resultDB>
<o:pvalDB> [options]
```

Key Options

Option	Purpose
--alpha	Set alpha for combining p-values during aggregation
--aggregation-mode	Combined P-values computed from 0: multi-hit, 1: minimum of all P-values, 2: product-of-P-values, 3: truncated product
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs combinepvalperset <i:querySetDB> <i:targetSetDB> <i:resultDB>
<o:pvalDB> [options]
  By Ruoshi Zhang, Clovis Norroy & Milot Mirdita <milot@mirdita.de>
options: misc:
  --alpha FLOAT          Set alpha for combining p-values during aggregation
  [1.000]
  --aggregation-mode INT  Combined P-values computed from 0: multi-hit, 1:
  minimum of all P-values, 2: product-of-P-values, 3: truncated product [0]
common:
  --threads INT          Number of CPU-cores used (all by default) [10]
  --compressed INT       Write compressed output [0]
  -v INT                 Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
  +info [3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
```

for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.15. compress

Compress DB entries.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

usage: mmseqs compress <i:DB> <o:DB> [options]

Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: mmseqs compress <i:DB> <o:DB> [options]

By Milot Mirdita <milot@mirdita.de>

options: common:

--threads INT Number of CPU-cores used (all by default) [10]
-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.16. concatdbs

Concatenate two DBs, giving new IDs to entries from 2nd DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

```
usage: mmseqs concatdbs <i:DB> <i:DB> <o:DB> [options]
```

Key Options

Option	Purpose
--preserve-keys	The keys of the two DB should be distinct, and they will be preserved in the concatenation
--take-larger-entry	Only keep the larger entry (dataSize >) in the concatenation, both databases need the same keys in the index
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs concatdbs <i:DB> <i:DB> <o:DB> [options]
```

By Clovis Galiez, Eli Levy Karin & Martin Steinegger
(martin.steinegger@snu.ac.kr)

options: misc:

```
--preserve-keys BOOL      The keys of the two DB should be distinct, and they
will be preserved in the concatenation [0]
```

```
--take-larger-entry BOOL  Only keep the larger entry (dataSize >) in the
concatenation, both databases need the same keys in the index [0]
```

common:

```
--compressed INT          Write compressed output [0]
```

```
--threads INT             Number of CPU-cores used (all by default) [1]
```

```
-v INT                    Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
```

```
3: +info [3]
```

examples:

```
# Download two sequences databases and concat them
```

```
mmseqs databases PDB pdbDB tmp
```

```
mmseqs UniProtKB/Swiss-Prot swissprotDB tmp
```

```
# Works only single threaded since seq. and header DB need the same ordering
```

```
mmseqs concatdbs pdbDB swissprotDB pdbAndSwissprotDB --threads 1
mmseqs concatdbs pdbDB_h swissprotDB_h pdbAndSwissprotDB_h --threads 1
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

6.2.17. **convert2fasta**

Convert sequence DB to FASTA format.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: `result_handling`.

Usage

usage: `mmseqs convert2fasta <DB> [args] [options]`

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs convert2fasta` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.18. **convertalis**

Convert alignment DB to BLAST-tab, SAM or custom format.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 4 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result_handling.

Usage

```
usage: mmseqs convertalis <i:queryDb> <i:targetDb> <i:alignmentDB>
<o:alignmentFile> [options]
```

Key Options

Option	Purpose
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--format-mode	Output format:
--format-output	Choose comma separated list of output columns from: query,target,value,gapopen,pident,fident,nident,qs-tart,qend,qlen
--translation-table	1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3) YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILLATE
--search-type	Search type 0: auto 1: amino acid, 2: translated, 3: nucleotide, 4: translated nucleotide alignment
--sub-mat	Substitution matrix file
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--db-output	Return a result DB instead of a text file

Full CLI Help Snapshot

```
usage: mmseqs convertalis <i:queryDb> <i:targetDb> <i:alignmentDB>
<o:alignmentFile> [options]
By Martin Steinegger <martin.steinegger@snu.ac.kr>
```

```

options: align:
  --gap-open TWIN          Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN        Gap extension cost [aa:1,nucl:2]
misc:
  --format-mode INT         Output format:
                           0: BLAST-TAB
                           1: SAM
                           2: BLAST-TAB + query/db length
                           3: Pretty HTML
                           4: BLAST-TAB + column headers
                           BLAST-TAB (0) and BLAST-TAB + column headers (4)
support custom output formats (--format-output) [0]
  --format-output STR       Choose comma separated list of output columns from:
query,target,evalue,gapopen,pident,fident,nident,qstart,qend,qlen
tstart,tend,tlen,alnlen,raw,bits,cigar,qseq,tseq,qheader,theader,qaln,taln,qframe,tframe,mismatch,
qset,qsetid,tset,tsetid,taxid,taxname,taxlineage,qorfststart,qorfend,torfstart,torfend,ppos
[query,target,fident,alnlen,mismatch,gapopen,qstart,qend,tstart,tend,evalue,bits]
  --translation-table INT   1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
                           9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
                           15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL, 21)
TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
                           23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
                           29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA [1]
  --search-type INT         Search type 0: auto 1: amino acid, 2: translated, 3:
nucleotide, 4: translated nucleotide alignment [0]
common:
  --sub-mat TWIN            Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
  --db-load-mode INT        Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]
  --threads INT             Number of CPU-cores used (all by default) [10]
  --compressed INT          Write compressed output [0]
  -v INT                    Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
3: +info [3]
expert:
  --db-output B00L          Return a result DB instead of a text file [0]

examples:
# Create output in BLAST M8 format (12 columns):
# (1,2) identifiers for query and target sequences/profiles,
# (3) sequence identity, (4) alignment length, (5) number of mismatches,
# (6) number of gap openings, (7-8, 9-10) alignment start and end-position in
query and in target,

```

```
# (11) E-value, and (12) bit score
mmseqs convertalis queryDB targetDB result.m8

# Create a TSV containing pairwise alignments
mmseqs convertalis queryDB targetDB result.tsv --format-output
query,target,qaln,taln

# Annotate a alignment result with taxonomy information from targetDB
mmseqs convertalis queryDB targetDB result.tsv --format-output
query,target,taxid,taxname,taxlineage

# Create SAM output
mmseqs convertalis queryDB targetDB result.sam --format-mode 1

# Create a TSV containing which query file a result comes from
mmseqs createdb euk_queries.fasta bac_queries.fasta queryDB
mmseqs convertalis queryDB targetDB result.tsv --format-output qset,query,target
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.19. convertca3m

Convert a cA3M DB to a result DB.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

```
usage: mmseqs convertca3m <inputDB(s)> <outputDB> [options]
```

The syntax line above is source-derived from command layer/category metadata. Run mmseqs convertca3m locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.20. **convertkb**

Convert UniProtKB data to a DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

usage: mmseqs convertkb <DB> [args] [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs convertkb locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.21. **convertmsa**

Convert Stockholm/PFAM MSA file to a MSA DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

```
usage: mmseqs convertmsa <i:msaFile.sto[.gz]> <o:msaDB> [options]
```

Key Options

Option	Purpose
<code>--identifier-field</code>	Field from STOCKHOLM comments for choosing the MSA identifier: 0: ID, 1: AC. If the respective comment does not exist, the name of the first sequence will become the identifier
<code>--compressed</code>	Write compressed output
<code>-v</code>	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs convertmsa <i:msaFile.sto[.gz]> <o:msaDB> [options]
```

By Milot Mirdita <milot@mirdita.de>

options: common:

```
--identifier-field INT  Field from STOCKHOLM comments for choosing the MSA
identifier: 0: ID, 1: AC. If the respective comment does not exist, the name of
the first sequence will become the identifier [1]
--compressed INT       Write compressed output [0]
-v INT                 Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Mirdita M, Steinegger M, Soding J: MMseqs2 desktop and local web server app for fast, interactive sequence searches. *Bioinformatics*, 35(16), 2856-2858 (2019)

6.2.22. convertprofiledb

Convert a HH-suite HHM DB to a profile DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

```
usage: mmseqs convertprofiledb <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs convertprofiledb` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.23. countkmer

Count k-mers.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family is the main acceleration gate that prunes candidate pairs before expensive alignment. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command in custom pipelines that must expose candidate-generation behavior before alignment stages.

Dependency entry: Open in map; functional module: prefiltering.

Usage

```
usage: mmseqs countkmer <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs countkmer` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.24. cpdb

Copy a DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

```
usage: mmseqs cpdb <i:srcDB> <o:dstDB> [options]
```

Key Options

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs cpdb <i:srcDB> <o:dstDB> [options]
```

By Milot Mirdita <milot@mirdita.de>

options: common:

```
-v INT    Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.25. createbintaxmapping

Create binary taxonomy mapping from tabular taxonomy mapping.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

```
usage: mmseqs createbintaxmapping <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs createbintaxmapping` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.26. createbintaxonomy

Create binary taxonomy from NCBI input.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

```
usage: mmseqs createbintaxonomy <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs createbintaxonomy` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.27. **createdb**

Convert FASTA/Q file(s) to a sequence DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 8 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

```
usage: mmseqs createdb <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>|
<i:stdin> <o:sequenceDB> [options]
```

Key Options

Option	Purpose
--dbtype	Database type 0: auto, 1: amino acid 2: nucleotides
--shuffle	Shuffle input database
--createdb-mode	Createdb mode 0: copy data, 1: soft link data and write new index (works only with single line fasta/q)
--id-offset	Numeric ids in index file are offset by this value
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Option	Purpose
<code>--write-lookup</code>	write <code>.lookup</code> file containing mapping from internal id, fasta id and file number

Full CLI Help Snapshot

```
usage: mmseqs createdb <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>|
<i:stdin> <o:sequenceDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --dbtype INT          Database type 0: auto, 1: amino acid 2: nucleotides [0]
  --shuffle BOOL        Shuffle input database [1]
  --createdb-mode INT    Createdb mode 0: copy data, 1: soft link data and write
new index (works only with single line fasta/q) [0]
  --id-offset INT       Numeric ids in index file are offset by this value [0]
common:
  --compressed INT      Write compressed output [0]
  -v INT                Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
expert:
  --write-lookup INT     write .lookup file containing mapping from internal id,
fasta id and file number [1]

examples:
# Create a sequence database from multiple FASTA files
mmseqs createdb file1.fa file2.fa.gz file3.fa sequenceDB

# Create a seqDB from stdin
cat seq.fasta | mmseqs createdb stdin sequenceDB

# Create a seqDB by indexing existing FASTA/Q (for single line fasta entries
only)
mmseqs createdb seq.fasta sequenceDB --createdb-mode 1

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

6.2.28. createdmptaxonomy

Create dmp files from binary taxonomy.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

```
usage: mmseqs createdmptaxonomy <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs createdmptaxonomy` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.29. createindex

Store precomputed index on disk to reduce search overhead.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 0 upstream caller(s) and 4 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

```
usage: mmseqs createindex <i:sequenceDB> <tmpDir> [options]
```

Key Options

Option	Purpose
<code>--seed-sub-mat</code>	Substitution matrix file for k-mer generation
<code>-k</code>	k-mer length (0: automatically set to optimum)
<code>--alph-size</code>	Alphabet size (range 2-21)

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probablity is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occure > threshold in a rwo
--spaced-kmer-mode	0: use consecutive positions in k-mers; 1: use spaced k-mers
--spaced-kmer-pattern	User-specified spaced k-mer pattern

Full CLI Help Snapshot

```
usage: mmseqs createindex <i:sequenceDB> <tmpDir> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -k INT                      k-mer length (0: automatically set to optimum) [0]
  --alph-size TWIN            Alphabet size (range 2-21) [aa:21,nucl:5]
  --comp-bias-corr INT        Correct for locally biased amino acid composition
(range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]
  --max-seqs INT              Maximum results per query sequence allowed to pass
the prefilter (affects sensitivity) [300]
  --mask INT                  Mask sequences in prefilter stage with tantan: 0:
w/o low complexity masking, 1: with low complexity masking [1]
  --mask-prob FLOAT           Mask sequences is probablity is above threshold
[0.900]
  --mask-lower-case INT       Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
  --mask-n-repeat INT         Repeat letters that occure > threshold in a rwo [0]
  --spaced-kmer-mode INT      0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
```

```

--spaced-kmer-pattern STR    User-specified spaced k-mer pattern []
-s FLOAT                    Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
[7.500]
--k-score TWIN              k-mer threshold for generating similar k-mer lists
[seq:0,prof:0]
--split INT                 Split input into N equally distributed chunks. 0:
set the best split automatically [0]
--split-memory-limit BYTE   Set max memory per split. E.g. 800B, 5K, 10M, 1G.
Default (0) to all available system memory [0]
misc:
--check-compatible INT      0: Always recreate index, 1: Check if recreating
index is needed, 2: Fail if index is incompatible [0]
--search-type INT          Search type 0: auto 1: amino acid, 2: translated,
3: nucleotide, 4: translated nucleotide alignment [0]
--min-length INT           Minimum codon number in open reading frames [30]
--max-length INT           Maximum codon number in open reading frames [32734]
--max-gaps INT             Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT    Contig start can be 0: incomplete, 1: complete, 2:
both [2]
--contig-end-mode INT      Contig end can be 0: incomplete, 1: complete, 2:
both [2]
--orf-start-mode INT       Orf fragment can be 0: from start to stop, 1: from
any to stop, 2: from last encountered start to stop (no start in the middle) [1]
--forward-frames STR       Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR       Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT    1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL,
21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA
[1]
--translate INT            Translate ORF to amino acid [0]
--use-all-table-starts BOOL Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT           Numeric ids in index file are offset by this value
[0]
--sequence-overlap INT     Overlap between sequences [0]
--sequence-split-mode INT  Sequence split mode 0: copy data, 1: soft link data
and write new index, [1]
--headers-split-mode INT   Header split mode: 0: split position, 1: original

```

```

header [0]
  --translation-mode INT      Translation AA seq from nucleotide by 0: ORFs, 1:
full reading frames [0]
common:
  --max-seq-len INT           Maximum sequence length [65535]
  -v INT                       Verboesity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
  --threads INT               Number of CPU-cores used (all by default) [10]
  --compressed INT            Write compressed output [0]
  --remove-tmp-files BOOL      Delete temporary files [0]
expert:
  --index-subset INT           Create specialized index with subset of entries
                                0: normal index
                                1: index without headers
                                2: index without prefiltering data
                                4: index without aln (for cluster db)
                                Flags can be combined bit wise [0]
  --create-lookup INT          Create database lookup file (can be very large) [0]
  --strand INT                 Strand selection only works for DNA/DNA search 0:
reverse, 1: forward, 2: both [1]

```

examples:

```

# Create protein sequence index
mmseqs createindex sequenceDB tmp

# Create TBLASTX/N index from nucleotide sequences
mmseqs createindex sequenceDB tmp --search-type 2

# Create BLASTN index from nucleotide sequences
mmseqs createindex sequenceDB tmp --search-type 3

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Mirdita M, Steinegger M, Soding J: MMseqs2 desktop and local web server app for fast, interactive sequence searches. *Bioinformatics*, 35(16), 2856-2858 (2019)

6.2.30. createlinindex

Create linsearch index.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 2 upstream caller(s) and 4 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

usage: mmseqs createlinindex <i:sequenceDB> <tmpDir> [options]

Key Options

Option	Purpose
--seed-sub-mat	Substitution matrix file for k-mer generation
-k	k-mer length (0: automatically set to optimum)
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
--alph-size	Alphabet size (range 2-21)
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probability is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occur > threshold in a row
--spaced-kmer-mode	0: use consecutive positions in k-mers; 1: use spaced k-mers
--spaced-kmer-pattern	User-specified spaced k-mer pattern
--min-seq-id	List matches above this sequence identity (for clustering) (range 0.0-1.0)
--hash-shift	Shift k-mer hash initialization

Full CLI Help Snapshot

usage: mmseqs createlinindex <i:sequenceDB> <tmpDir> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

```
--seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:blosom62.out,nucl:nucleotide.out]
-k INT                       k-mer length (0: automatically set to optimum) [0]
--split-memory-limit BYTE    Set max memory per split. E.g. 800B, 5K, 10M, 1G.
Default (0) to all available system memory [0]
--alph-size TWIN            Alphabet size (range 2-21) [aa:21,nucl:5]
--mask INT                  Mask sequences in prefilter stage with tantan: 0:
w/o low complexity masking, 1: with low complexity masking [0]
```

```

--mask-prob FLOAT          Mask sequences is probablity is above threshold
[0.900]
--mask-lower-case INT      Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT        Repeat letters that occure > threshold in a rwo [0]
--spaced-kmer-mode INT      0: use consecutive positions in k-mers; 1: use
spaced k-mers [0]
--spaced-kmer-pattern STR   User-specified spaced k-mer pattern []
align:
--min-seq-id FLOAT         List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
kmermatcher:
--hash-shift INT           Shift k-mer hash initialization [67]
--kmer-per-seq INT          k-mers per sequence [21]
--kmer-per-seq-scale TWIN   Scale k-mer per sequence based on sequence length
as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
--adjust-kmer-len BOOL      Adjust k-mer length based on specificity (only for
nucleotides) [0]
--ignore-multi-kmer BOOL    Skip k-mers occurring multiple times (>=2) [0]
misc:
--check-compatible INT      0: Always recreate index, 1: Check if recreating
index is needed, 2: Fail if index is incompatible [0]
--search-type INT           Search type 0: auto 1: amino acid, 2: translated,
3: nucleotide, 4: translated nucleotide alignment [0]
--min-length INT            Minimum codon number in open reading frames [30]
--max-length INT            Maximum codon number in open reading frames [32734]
--max-gaps INT              Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT     Contig start can be 0: incomplete, 1: complete, 2:
both [2]
--contig-end-mode INT       Contig end can be 0: incomplete, 1: complete, 2:
both [2]
--orf-start-mode INT        Orf fragment can be 0: from start to stop, 1: from
any to stop, 2: from last encountered start to stop (no start in the middle) [1]
--forward-frames STR        Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR        Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT     1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL,
21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA

```



```

[1]
--translate INT           Translate ORF to amino acid [0]
--use-all-table-starts BOOL Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT           Numeric ids in index file are offset by this value
[0]
common:
--max-seq-len INT         Maximum sequence length [65535]
-v INT                   Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--threads INT             Number of CPU-cores used (all by default) [10]
--compressed INT          Write compressed output [0]
--remove-tmp-files BOOL   Delete temporary files [0]
expert:
--create-lookup INT       Create database lookup file (can be very large) [0]

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Mirdita M, Steinegger M, Soding J: MMseqs2 desktop and local web server app for fast, interactive sequence searches. *Bioinformatics*, 35(16), 2856-2858 (2019)

6.2.31. createseqfiledb

Create a DB of unaligned FASTA entries.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result_handling.

Usage

usage: mmseqs createseqfiledb <i:sequenceDB> <i:resultDB> <o:fastaDB> [options]

Key Options

Option	Purpose
--min-sequences	Minimum number of sequences a cluster may contain
--max-sequences	Maximum number of sequences a cluster may contain

Option	Purpose
--hh-format	Format entries to use with hhsuite (for singleton clusters)
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: mmseqs createseqfiledb <i:sequenceDB> <i:resultDB> <o:fastaDB> [options]

By Milot Mirdita <milot@mirdita.de>

options: misc:

--min-sequences INT Minimum number of sequences a cluster may contain [1]
--max-sequences INT Maximum number of sequences a cluster may contain
[2147483647]

--hh-format BOOL Format entries to use with hhsuite (for singleton
clusters) [0]

common:

--db-load-mode INT Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]

--threads INT Number of CPU-cores used (all by default) [10]

--compressed INT Write compressed output [0]

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]

examples:

Gather all sequences from a cluster DB
mmseqs createseqfiledb sequenceDB clusterDB unalignedDB --min-sequences 2
Build MSAs with Clustal-Omega
mmseqs apply unalignedDB msadb -- clustalo -i - -o stdout --threads=1

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

6.2.32. createsubdb

Create a subset of a DB from list of DB keys.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 5 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

```
usage: mmseqs createsubdb <i:subsetFile|DB> <i:DB> <o:DB> [options]
```

Key Options

Option	Purpose
--subdb-mode	Subdb mode 0: copy data 1: soft link data and write index
--id-mode	Select DB entries based on 0: database keys, 1: FASTA identifiers (.lookup)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs createsubdb <i:subsetFile|DB> <i:DB> <o:DB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: misc:
  --subdb-mode INT    Subdb mode 0: copy data 1: soft link data and write index [0]
  --id-mode INT       Select DB entries based on 0: database keys, 1: FASTA
identifiers (.lookup) [0]
common:
  -v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

examples:

```
# Create a new sequenceDB from sequenceDB entries with keys 1, 2 and 3
mmseqs createsubdb <(printf '1
2
3
') sequenceDB oneTwoThreeDB
```

```
# Create a new sequence database with representatives of clusterDB
mmseqs cluster sequenceDB clusterDB tmp
mmseqs createsubdb clusterDB sequenceDB representativesDB
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching

for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.33. createtaxdb

Add taxonomic labels to sequence DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 1 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

```
usage: mmseqs createtaxdb <i:sequenceDB> <tmpDir> [options]
```

Key Options

Option	Purpose
--ncbi-tax-dump	tax dump directory. The tax dump can be downloaded here "ftp://ftp.ncbi.nlm.nih.gov/pub/taxonomy/taxdump.tar.gz"
--tax-mapping-file	File to map sequence identifier to taxonomical identifier
--tax-mapping-mode	Map taxonomy based on sequence database 0: .lookup file 1: .source file
--tax-db-mode	Create taxonomy database as: 0: .dmp flat files (human readable) 1: binary dump (faster readin)
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs createtaxdb <i:sequenceDB> <tmpDir> [options]
```

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: misc:

```
--ncbi-tax-dump STR      NCBI tax dump directory. The tax dump can be downloaded here "ftp://ftp.ncbi.nlm.nih.gov/pub/taxonomy/taxdump.tar.gz" []
```

```
--tax-mapping-file STR   File to map sequence identifier to taxonomical
```

```

identifier []
--tax-mapping-mode INT    Map taxonomy based on sequence database 0: .lookup file
1: .source file [0]
--tax-db-mode INT         Create taxonomy database as: 0: .dmp flat files (human
readable) 1: binary dump (faster readin) [1]
common:
--threads INT             Number of CPU-cores used (all by default) [10]
-v INT                   Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]

```

references:

- Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and sensitive taxonomic assignment to metagenomic contigs. Bioinformatics, btabl84 (2021)

6.2.34. createtsv

Convert result DB to tab-separated flat file.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result_handling.

Usage

```
usage: mmseqs createtsv <i:queryDB> [<i:targetDB>] <i:resultDB> <o:tsvFile>
[options]
```

Key Options

Option	Purpose
--first-seq-as-repr	Use the first sequence of the clustering result as representative sequence
--target-column	Select a target column (default 1), 0 if no target id exists
--full-header	Replace DB ID by its corresponding Full Header
--idx-seq-src	0: auto, 1: split/translated sequences, 2: input sequences
--threads	Number of CPU-cores used (all by default)

Option	Purpose
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--db-output	Return a result DB instead of a text file

Full CLI Help Snapshot

```
usage: mmseqs createtsv <i:queryDB> [<i:targetDB>] <i:resultDB> <o:tsvFile>
[options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --first-seq-as-repr B00L  Use the first sequence of the clustering result as
representative sequence [0]
  --target-column INT       Select a target column (default 1), 0 if no target id
exists [1]
  --full-header B00L        Replace DB ID by its corresponding Full Header [0]
  --idx-seq-src INT         0: auto, 1: split/translated sequences, 2: input
sequences [0]
common:
  --threads INT             Number of CPU-cores used (all by default) [10]
  --compressed INT         Write compressed output [0]
  -v INT                   Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
3: +info [3]
expert:
  --db-output B00L         Return a result DB instead of a text file [0]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

6.2.35. databases

List and download databases.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 0 upstream caller(s) and 8 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

usage: mmseqs databases <DB> [args] [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs databases locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.36. db2tar

Archive contents of a DB to a tar archive.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

usage: mmseqs db2tar <i:DB> <o:tar[.gz]> [options]

Key Options

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: mmseqs db2tar <i:DB> <o:tar[.gz]> [options]

By Milot Mirdita <milot@mirdita.de>

options: common:

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

examples:

```
# Create a tar from a MSA DB
mmseqs db2tar msaDB archive.tar.gz
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.37. decompress

Decompress DB entries.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

usage: mmseqs decompress <i:DB> <o:DB> [options]

Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: mmseqs decompress <i:DB> <o:DB> [options]

By Milot Mirdita <milot@mirdita.de>

options: common:

--threads INT Number of CPU-cores used (all by default) [10]
-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.38. diffseqdbs

Compute diff of two sequence DBs.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

```
usage: mmseqs diffseqdbs <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs diffseqdbs` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.39. easy-cluster

Slower, sensitive clustering.

Execution role: workflow entrypoint that coordinates lower-level modules rather than acting as a single compute kernel.

This command family favors fast adoption and default-safe orchestration over maximal low-level control. The current dependency map records 0 upstream caller(s) and 7 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: `easy_workflows`.

Usage

```
usage: mmseqs easy-cluster <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>  
<o:clusterPrefix> <tmpDir> [options]
```

Key Options

Option	Purpose
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists
--alph-size	Alphabet size (range 2-21)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--split	Split input into N equally distributed chunks. 0: set the best split automatically
--split-mode	0: split target db; 1: split query db; 2: auto, depending on main memory
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)

Full CLI Help Snapshot

usage: mmseqs easy-cluster <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>
<o:clusterPrefix> <tmpDir> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

--seed-sub-mat TWIN	Substitution matrix file for k-mer generation [aa:VTML80.out,nucl:nucleotide.out]
-s FLOAT	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive [4.000]
-k INT	k-mer length (0: automatically set to optimum) [0]
--target-search-mode INT	target search mode (0: regular k-mer, 1: similar k-mer) [0]
--k-score TWIN	k-mer threshold for generating similar k-mer lists [seq:2147483647,prof:2147483647]
--alph-size TWIN	Alphabet size (range 2-21) [aa:21,nucl:5]
--max-seqs INT	Maximum results per query sequence allowed to

pass the prefilter (affects sensitivity) [20]

--split INT Split input into N equally distributed chunks.
0: set the best split automatically [0]

--split-mode INT 0: split target db; 1: split query db; 2: auto,
depending on main memory [2]

--split-memory-limit BYTE Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]

--comp-bias-corr INT Correct for locally biased amino acid
composition (range 0-1) [1]

--comp-bias-corr-scale FLOAT Correct for locally biased amino acid
composition (range 0-1) [1.000]

--diag-score BOOL Use ungapped diagonal scoring during prefilter
[1]

--exact-kmer-matching INT Extract only exact k-mers for matching (range
0-1) [0]

--mask INT Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [1]

--mask-prob FLOAT Mask sequences is probability is above threshold
[0.900]

--mask-lower-case INT Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]

--mask-n-repeat INT Repeat letters that occur > threshold in a row
[0]

--min-ungapped-score INT Accept only matches with ungapped alignment
score above threshold [15]

--add-self-matches BOOL Artificially add entries of queries with
themselves (for clustering) [0]

--spaced-kmer-mode INT 0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]

--spaced-kmer-pattern STR User-specified spaced k-mer pattern []

--local-tmp STR Path where some of the temporary files will be
created []

align:

-c FLOAT List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.800]

--cov-mode INT 0: coverage of query and target
1: coverage of target
2: coverage of query
3: target seq. length has to be at least x% of
query length
4: query seq. length has to be at least x% of
target length
5: short seq. needs to be at least x% of the
other seq. length [0]

-a BOOL Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]

--alignment-mode INT How to compute the alignment:
0: automatic
1: only score and end_pos

2: also start_pos and cov
 3: also seq.id
 4: only ungapped alignment [3]
 --alignment-output-mode INT How to compute the alignment:
 0: automatic
 1: only score and end_pos
 2: also start_pos and cov
 3: also seq.id
 4: only ungapped alignment
 5: score only (output) cluster format [0]
 --wrapped-scoring BOOL Double the (nucleotide) query sequence during
 the scoring process to allow wrapped diagonal scoring around end and start [0]
 -e DOUBLE List matches below this E-value (range 0.0-inf)
 [1.000E-03]
 --min-seq-id FLOAT List matches above this sequence identity (for
 clustering) (range 0.0-1.0) [0.000]
 --min-aln-len INT Minimum alignment length (range 0-INT_MAX) [0]
 --seq-id-mode INT 0: alignment length 1: shorter, 2: longer
 sequence [0]
 --alt-ali INT Show up to this many alternative alignments [0]
 --max-rejected INT Maximum rejected alignments before alignment
 calculation for a query is stopped [2147483647]
 --max-accept INT Maximum accepted alignments before alignment
 calculation for a query is stopped [2147483647]
 --score-bias FLOAT Score bias when computing SW alignment (in
 bits) [0.000]
 --realign BOOL Compute more conservative, shorter alignments
 (scores and E-values not changed) [0]
 --realign-score-bias FLOAT Additional bias when computing realignment
 [-0.200]
 --realign-max-seqs INT Maximum number of results to return in
 realignment [2147483647]
 --corr-score-weight FLOAT Weight of backtrace correlation score that is
 added to the alignment score [0.000]
 --gap-open TWIN Gap open cost [aa:11,nucl:5]
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]
 --zdrop INT Maximal allowed difference between score values
 before alignment is truncated (nucleotide alignment only) [40]
 clust:
 --cluster-mode INT 0: Set-Cover (greedy)
 1: Connected component (BLASTclust)
 2,3: Greedy clustering by sequence length
 (CDHIT) [0]
 --max-iterations INT Maximum depth of breadth first search in
 connected component clustering [1000]
 --similarity-type INT Type of score used for clustering. 1: alignment
 score 2: sequence identity [2]
 --single-step-clustering BOOL Switch from cascaded to simple clustering
 workflow [0]

```

--cluster-steps INT          Cascaded clustering steps from 1 to -s [3]
--cluster-reassign BOOL      Cascaded clustering can cluster sequence that
do not fulfill the clustering criteria.
                             Cluster reassignment corrects these errors [0]

kmermatcher:
--weights STR                Weights used for cluster prioritization []
--cluster-weight-threshold FLOAT Weight threshold used for cluster prioritization
[0.900]
--kmer-per-seq INT           k-mers per sequence [21]
--kmer-per-seq-scale TWIN     Scale k-mer per sequence based on sequence
length as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
--adjust-kmer-len BOOL       Adjust k-mer length based on specificity (only
for nucleotides) [0]
--hash-shift INT             Shift k-mer hash initialization [67]
--include-only-extendable BOOL Include only extendable [0]
--ignore-multi-kmer BOOL     Skip k-mers occurring multiple times (>=2) [0]

profile:
--pca                        Pseudo count admixture strength []
--pcb                        Pseudo counts: Neff at half of maximum

admixture (range 0.0-inf) []

misc:
--taxon-list STR             Taxonomy ID, possibly multiple values separated
by ',' []
--rescore-mode INT           Rescore diagonals with:
                             0: Hamming distance
                             1: local alignment (score only)
                             2: local alignment
                             3: global alignment
                             4: longest alignment fulfilling window quality

criterion [0]
--dbtype INT                 Database type 0: auto, 1: amino acid 2:
nucleotides [0]
--shuffle BOOL               Shuffle input database [1]
--createdb-mode INT          Createdb mode 0: copy data, 1: soft link data
and write new index (works only with single line fasta/q) [1]
--id-offset INT              Numeric ids in index file are offset by this
value [0]

common:
--sub-mat TWIN               Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--max-seq-len INT            Maximum sequence length [65535]
--db-load-mode INT           Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
--threads INT                Number of CPU-cores used (all by default) [10]
--compressed INT             Write compressed output [0]
--v INT                       Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--remove-tmp-files BOOL      Delete temporary files [1]
--force-reuse BOOL           Reuse tmp filse in tmp/latest folder ignoring

```

parameters and version changes [0]

```
--mpi-runner STR          Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
expert:
--filter-hits BOOL          Filter hits by seq.id. and coverage [0]
--sort-results INT          Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--write-lookup INT          write .lookup file containing mapping from
internal id, fasta id and file number [0]
```

examples:

```
mmseqs easy-cluster examples/DB.fasta result tmp
# Cluster output
# - result_rep_seq.fasta: Representatives
# - result_all_seqs.fasta: FASTA-like per cluster
# - result_cluster.tsv: Adjacency list

# Important parameter: --min-seq-id, --cov-mode and -c
#
#           --cov-mode
#           0      1      2
# Q: MAVGTACRPA  60%  IGN  60%
# T: -AVGTAC--   60% 100%  IGN
#           -c 0.7   -    +    -
#           -c 0.6   +    +    +

# Cascaded clustering with reassignment
# - Corrects criteria-violations of cascaded merging
# - Produces more clusters and is a bit slower
mmseqs easy-cluster examples/DB.fasta result tmp --cluster-reassign
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Steinegger M, Soding J: Clustering huge protein sequence sets in linear time. *Nature Communications*, 9(1), 2542 (2018)

6.2.40. easy-linclud

Fast linear time cluster, less sensitive clustering.

Execution role: workflow entrypoint that coordinates lower-level modules rather than acting as a single compute kernel.

This command family favors fast adoption and default-safe orchestration over maximal low-level control. The current dependency map records 0 upstream caller(s) and 7 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: easy_workflows.

Usage

```
usage: mmseqs easy-linlust <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>
<o:clusterPrefix> <tmpDir> [options]
```

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--alph-size	Alphabet size (range 2-21)
--spaced-kmer-mode	0: use consecutive positions in k-mers; 1: use spaced k-mers
--spaced-kmer-pattern	User-specified spaced k-mer pattern
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probability is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occur > threshold in a row
-k	k-mer length (0: automatically set to optimum)
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory

Full CLI Help Snapshot

```
usage: mmseqs easy-linlust <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>
<o:clusterPrefix> <tmpDir> [options]
```

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

```
--comp-bias-corr INT          Correct for locally biased amino acid
composition (range 0-1) [1]
--comp-bias-corr-scale FLOAT  Correct for locally biased amino acid
```

```

composition (range 0-1) [1.000]
--add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
--alph-size TWIN             Alphabet size (range 2-21) [aa:21,nucl:5]
--spaced-kmer-mode INT        0: use consecutive positions in k-mers; 1: use
spaced k-mers [0]
--spaced-kmer-pattern STR     User-specified spaced k-mer pattern []
--mask INT                   Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [1]
--mask-prob FLOAT            Mask sequences is probablity is above threshold
[0.900]
--mask-lower-case INT        Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT          Repeat letters that occure > threshold in a rwo
[0]
-k INT                       k-mer length (0: automatically set to optimum)
[0]
--split-memory-limit BYTE     Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
align:
-a BOOL                      Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]
--alignment-mode INT          How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment [0]
--alignment-output-mode INT   How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment
5: score only (output) cluster format [0]
--wrapped-scoring BOOL       Double the (nucleotide) query sequence during
the scoring process to allow wrapped diagonal scoring around end and start [0]
-e DOUBLE                    List matches below this E-value (range 0.0-inf)
[1.000E-03]
--min-seq-id FLOAT           List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
--min-aln-len INT            Minimum alignment length (range 0-INT_MAX) [0]
--seq-id-mode INT            0: alignment length 1: shorter, 2: longer
sequence [0]
--alt-ali INT                Show up to this many alternative alignments [0]
-c FLOAT                     List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.800]
--cov-mode INT               0: coverage of query and target
1: coverage of target

```


2: coverage of query
 3: target seq. length has to be at least x% of
 query length
 4: query seq. length has to be at least x% of
 target length
 5: short seq. needs to be at least x% of the
 other seq. length [0]
 --max-rejected INT Maximum rejected alignments before alignment
 calculation for a query is stopped [2147483647]
 --max-accept INT Maximum accepted alignments before alignment
 calculation for a query is stopped [2147483647]
 --score-bias FLOAT Score bias when computing SW alignment (in
 bits) [0.000]
 --realign BOOL Compute more conservative, shorter alignments
 (scores and E-values not changed) [0]
 --realign-score-bias FLOAT Additional bias when computing realignment
 [-0.200]
 --realign-max-seqs INT Maximum number of results to return in
 realignment [2147483647]
 --corr-score-weight FLOAT Weight of backtrace correlation score that is
 added to the alignment score [0.000]
 --gap-open TWIN Gap open cost [aa:11,nucl:5]
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]
 --zdrop INT Maximal allowed difference between score values
 before alignment is truncated (nucleotide alignment only) [40]
 clust:
 --cluster-mode INT 0: Set-Cover (greedy)
 1: Connected component (BLASTclust)
 2,3: Greedy clustering by sequence length
 (CDHIT) [0]
 --max-iterations INT Maximum depth of breadth first search in
 connected component clustering [1000]
 --similarity-type INT Type of score used for clustering. 1: alignment
 score 2: sequence identity [2]
 kmermatcher:
 --weights STR Weights used for cluster prioritization []
 --cluster-weight-threshold FLOAT Weight threshold used for cluster prioritization
 [0.900]
 --kmer-per-seq INT k-mers per sequence [21]
 --kmer-per-seq-scale TWIN Scale k-mer per sequence based on sequence
 length as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
 --adjust-kmer-len BOOL Adjust k-mer length based on specificity (only
 for nucleotides) [0]
 --hash-shift INT Shift k-mer hash initialization [67]
 --include-only-extendable BOOL Include only extendable [0]
 --ignore-multi-kmer BOOL Skip k-mers occurring multiple times (>=2) [0]
 profile:
 --pca Pseudo count admixture strength []
 --pcb Pseudo counts: Neff at half of maximum

```

admixture (range 0.0-inf) []
misc:
  --rescore-mode INT          Rescore diagonals with:
                              0: Hamming distance
                              1: local alignment (score only)
                              2: local alignment
                              3: global alignment
                              4: longest alignment fulfilling window quality

criterion [0]
  --dbtype INT                Database type 0: auto, 1: amino acid 2:
nucleotides [0]
  --shuffle BOOL              Shuffle input database [1]
  --createdb-mode INT         Createdb mode 0: copy data, 1: soft link data
and write new index (works only with single line fasta/q) [1]
  --id-offset INT             Numeric ids in index file are offset by this
value [0]
common:
  --threads INT               Number of CPU-cores used (all by default) [10]
  --compressed INT            Write compressed output [0]
  -v INT                      Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
  --sub-mat TWIN              Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
  --max-seq-len INT           Maximum sequence length [65535]
  --db-load-mode INT          Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
  --remove-tmp-files BOOL     Delete temporary files [1]
  --force-reuse BOOL          Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
  --mpi-runner STR            Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
expert:
  --filter-hits BOOL          Filter hits by seq.id. and coverage [0]
  --sort-results INT          Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
  --write-lookup INT          write .lookup file containing mapping from
internal id, fasta id and file number [0]

examples:
mmseqs easy-linclust examples/DB.fasta result tmp

# Linclust output
# - result_rep_seq.fasta: Representatives
# - result_all_seqs.fasta: FASTA-like per cluster
# - result_cluster.tsv: Adjacency list

# Important parameter: --min-seq-id, --cov-mode and -c
# --cov-mode
# 0 1 2

```

```
# Q: MAVGTACRPA 60% IGN 60%
# T: -AVGTAC--- 60% 100% IGN
#      -c 0.7    -    +    -
#      -c 0.6    +    +    +
```

```
# Cluster nucleotide sequences
mmseqs easy-lincludst examples/DB.fasta result tmp --kmer-per-seq-scale 0.3
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Steinegger M, Soding J: Clustering huge protein sequence sets in linear time. *Nature Communications*, 9(1), 2542 (2018)

6.2.41. easy-linsearch

Fast, less sensitive homology search.

Execution role: workflow entrypoint that coordinates lower-level modules rather than acting as a single compute kernel.

This command family favors fast adoption and default-safe orchestration over maximal low-level control. The current dependency map records 0 upstream caller(s) and 7 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: easy_workflows.

Usage

```
usage:                                     mmseqs
easy-linsearch <i:queryFastaFile1[.gz|.bz2]> ... <i:queryFastaFileN[.gz|.bz2]>
<i:targetFastaFile[.gz|.bz2]>|<i:targetDB> <o:alignmentFile> <tmpDir> [options]
```

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation

Option	Purpose
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probability is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occure > threshold in a rwo
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
-a	Add backtrace string (convert to alignments with mmseqs convertalis module)
--alignment-mode	How to compute the alignment:
--alignment-output-mode	How to compute the alignment:

Full CLI Help Snapshot

```
usage: mmseqs easy-linsearch <i:queryFastaFile1[.gz|.bz2]> ...
<i:queryFastaFileN[.gz|.bz2]> <i:targetFastaFile[.gz|.bz2]>|<i:targetDB>
<o:alignmentFile> <tmpDir> [options]
  By Milot Mirdita <milot@mirdita.de> & Martin Steinegger
<martin.steinegger@snu.ac.kr>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
(range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  --mask INT                   Mask sequences in prefilter stage with tantan: 0:
w/o low complexity masking, 1: with low complexity masking [1]
  --mask-prob FLOAT            Mask sequences is probability is above threshold
[0.900]
  --mask-lower-case INT        Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
  --mask-n-repeat INT          Repeat letters that occure > threshold in a rwo
[0]
  --split-memory-limit BYTE     Set max memory per split. E.g. 800B, 5K, 10M, 1G.
Default (0) to all available system memory [0]
align:
  -a BOOL                      Add backtrace string (convert to alignments with
```

```

mmseqs convertalis module) [0]
--alignment-mode INT          How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment [3]
--alignment-output-mode INT   How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment
                                5: score only (output) cluster format [0]
--wrapped-scoring BOOL        Double the (nucleotide) query sequence during the
scoring process to allow wrapped diagonal scoring around end and start [0]
-e DOUBLE                     List matches below this E-value (range 0.0-inf)
[1.000E-03]
--min-seq-id FLOAT            List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
--min-aln-len INT             Minimum alignment length (range 0-INT_MAX) [0]
--seq-id-mode INT             0: alignment length 1: shorter, 2: longer sequence
[0]
--alt-ali INT                 Show up to this many alternative alignments [0]
-c FLOAT                      List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.000]
--cov-mode INT                0: coverage of query and target
                                1: coverage of target
                                2: coverage of query
                                3: target seq. length has to be at least x% of
query length
                                4: query seq. length has to be at least x% of
target length
                                5: short seq. needs to be at least x% of the other
seq. length [0]
--max-rejected INT            Maximum rejected alignments before alignment
calculation for a query is stopped [2147483647]
--max-accept INT              Maximum accepted alignments before alignment
calculation for a query is stopped [2147483647]
--score-bias FLOAT            Score bias when computing SW alignment (in bits)
[0.000]
--realign BOOL                Compute more conservative, shorter alignments
(scores and E-values not changed) [0]
--realign-score-bias FLOAT     Additional bias when computing realignment
[-0.200]
--realign-max-seqs INT         Maximum number of results to return in realignment
[2147483647]
--corr-score-weight FLOAT      Weight of backtrace correlation score that is
added to the alignment score [0.000]

```

```

--gap-open TWIN          Gap open cost [aa:11,nucl:5]
--gap-extend TWIN        Gap extension cost [aa:1,nucl:2]
--zdrop INT              Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
kmermatcher:
--kmer-per-seq INT        k-mers per sequence [21]
--kmer-per-seq-scale TWIN Scale k-mer per sequence based on sequence length
as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
--pick-n-sim-kmer INT     Add N similar k-mers to search [1]
--result-direction INT    result is 0: query, 1: target centric [1]
profile:
--pca                    Pseudo count admixture strength []
--pcb                    Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
misc:
--min-length INT          Minimum codon number in open reading frames [30]
--max-length INT          Maximum codon number in open reading frames
[32734]
--max-gaps INT            Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT   Contig start can be 0: incomplete, 1: complete, 2:
both [2]
--contig-end-mode INT     Contig end can be 0: incomplete, 1: complete, 2:
both [2]
--orf-start-mode INT      Orf fragment can be 0: from start to stop, 1: from
any to stop, 2: from last encountered start to stop (no start in the middle) [1]
--forward-frames STR      Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR      Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT    1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL,
21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA
[1]
--translate INT           Translate ORF to amino acid [0]
--use-all-table-starts BOOL Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT           Numeric ids in index file are offset by this value
[0]
--search-type INT         Search type 0: auto 1: amino acid, 2: translated,
3: nucleotide, 4: translated nucleotide alignment [0]

```

```

--format-mode INT          Output format:
                           0: BLAST-TAB
                           1: SAM
                           2: BLAST-TAB + query/db length
                           3: Pretty HTML
                           4: BLAST-TAB + column headers
                           BLAST-TAB (0) and BLAST-TAB + column headers (4)
support custom output formats (--format-output) [0]
--format-output STR        Choose comma separated list of output columns
from: query,target,value,gapopen,pident,fident,nident,qstart,qend,qlen
tstart,tend,tlen,alnlen,raw,bits,cigar,qseq,tseq,qheader,theader,qaln,taln,qframe,tframe,mismatch,
qset,qsetid,tset,tsetid,taxid,taxname,taxlineage,qorfststart,qorfend,torfstart,torfend,ppos
[query,target,fident,alnlen,mismatch,gapopen,qstart,qend,tstart,tend,value,bits]
common:
--sub-mat TWIN             Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--max-seq-len INT          Maximum sequence length [65535]
--db-load-mode INT         Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--threads INT              Number of CPU-cores used (all by default) [10]
--compressed INT           Write compressed output [0]
-v INT                     Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--mpi-runner STR           Use MPI on compute cluster with this MPI command
(e.g. "mpirun -np 42") []
--force-reuse BOOL         Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL    Delete temporary files [1]
expert:
--create-lookup INT        Create database lookup file (can be very large)
[0]
--chain-alignments INT     Chain overlapping alignments [0]
--merge-query INT          Combine ORFs/split sequences to a single entry [1]
--db-output BOOL           Return a result DB instead of a text file [0]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
- Steinegger M, Soding J: Clustering huge protein sequence sets in linear time.
Nature Communications, 9(1), 2542 (2018)

```

6.2.42. easy-rbh

Find reciprocal best hit.

Execution role: workflow entrypoint that coordinates lower-level modules rather than acting as a single compute kernel.

This command family favors fast adoption and default-safe orchestration over maximal low-level control. The current dependency map records 0 upstream caller(s) and 4 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: easy_workflows.

Usage

```
usage: mmseqs easy-rbh <i:queryFastaFile1[.gz|.bz2]>
<i:targetFastaFile[.gz|.bz2]>|<i:targetDB> <o:alignmentFile> <tmpDir> [options]
```

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists
--alph-size	Alphabet size (range 2-21)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--split	Split input into N equally distributed chunks. 0: set the best split automatically
--split-mode	0: split target db; 1: split query db; 2: auto, depending on main memory

Full CLI Help Snapshot

```
usage: mmseqs easy-rbh <i:queryFastaFile1[.gz|.bz2]>
<i:targetFastaFile[.gz|.bz2]>|<i:targetDB> <o:alignmentFile> <tmpDir> [options]
```


By Eli Levy Karin & Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

--comp-bias-corr INT	Correct for locally biased amino acid composition (range 0-1) [1]
--comp-bias-corr-scale FLOAT	Correct for locally biased amino acid composition (range 0-1) [1.000]
--add-self-matches BOOL	Artificially add entries of queries with themselves (for clustering) [0]
--seed-sub-mat TWIN	Substitution matrix file for k-mer generation [aa:VTML80.out,nucl:nucleotide.out]
-s FLOAT	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive [5.700]
-k INT	k-mer length (0: automatically set to optimum) [0]
--target-search-mode INT	target search mode (0: regular k-mer, 1: similar k-mer) [0]
--k-score TWIN	k-mer threshold for generating similar k-mer lists [seq:2147483647,prof:2147483647]
--alph-size TWIN	Alphabet size (range 2-21) [aa:21,nucl:5]
--max-seqs INT	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) [300]
--split INT	Split input into N equally distributed chunks. 0: set the best split automatically [0]
--split-mode INT	0: split target db; 1: split query db; 2: auto, depending on main memory [2]
--split-memory-limit BYTE	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory [0]
--diag-score BOOL	Use ungapped diagonal scoring during prefilter [1]
--exact-kmer-matching INT	Extract only exact k-mers for matching (range 0-1) [0]
--mask INT	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking [1]
--mask-prob FLOAT	Mask sequences is probablity is above threshold [0.900]
--mask-lower-case INT	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region [0]
--mask-n-repeat INT	Repeat letters that occure > threshold in a rwo [0]
--min-ungapped-score INT	Accept only matches with ungapped alignment score above threshold [15]
--spaced-kmer-mode INT	0: use consecutive positions in k-mers; 1: use spaced k-mers [1]
--spaced-kmer-pattern STR	User-specified spaced k-mer pattern []
--local-tmp STR	Path where some of the temporary files will be created []
--disk-space-limit BYTE	Set max disk space to use for reverse profile searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the temp folder [0]

```

align:
  -a BOOL          Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]
  --alignment-mode INT    How to compute the alignment:
                        0: automatic
                        1: only score and end_pos
                        2: also start_pos and cov
                        3: also seq.id
                        4: only ungapped alignment [3]
  --alignment-output-mode INT    How to compute the alignment:
                        0: automatic
                        1: only score and end_pos
                        2: also start_pos and cov
                        3: also seq.id
                        4: only ungapped alignment
                        5: score only (output) cluster format [0]
  --wrapped-scoring BOOL    Double the (nucleotide) query sequence during
the scoring process to allow wrapped diagonal scoring around end and start [0]
  -e DOUBLE              List matches below this E-value (range 0.0-inf)
[1.000E-03]
  --min-seq-id FLOAT      List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
  --min-aln-len INT       Minimum alignment length (range 0-INT_MAX) [0]
  --seq-id-mode INT       0: alignment length 1: shorter, 2: longer
sequence [0]
  --alt-ali INT           Show up to this many alternative alignments [0]
  -c FLOAT               List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.000]
  --cov-mode INT          0: coverage of query and target
                        1: coverage of target
                        2: coverage of query
                        3: target seq. length has to be at least x% of
query length
                        4: query seq. length has to be at least x% of
target length
                        5: short seq. needs to be at least x% of the
other seq. length [0]
  --max-rejected INT      Maximum rejected alignments before alignment
calculation for a query is stopped [2147483647]
  --max-accept INT        Maximum accepted alignments before alignment
calculation for a query is stopped [2147483647]
  --score-bias FLOAT      Score bias when computing SW alignment (in
bits) [0.000]
  --realign BOOL          Compute more conservative, shorter alignments
(scores and E-values not changed) [0]
  --realign-score-bias FLOAT    Additional bias when computing realignment
[-0.200]
  --realign-max-seqs INT   Maximum number of results to return in
realignment [2147483647]

```

```

--corr-score-weight FLOAT      Weight of backtrace correlation score that is
added to the alignment score [0.000]
--gap-open TWIN                Gap open cost [aa:11,nucl:5]
--gap-extend TWIN              Gap extension cost [aa:1,nucl:2]
--zdrop INT                    Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
--exhaustive-search-filter INT  Filter result during search: 0: do not filter,
1: filter [0]
profile:
--pca                          Pseudo count admixture strength []
--pcb                          Pseudo counts: Neff at half of maximum
admixture (range 0.0-inf) []
--mask-profile INT             Mask query sequence of profile using tantan
[0,1] [1]
--e-profile DOUBLE             Include sequences matches with < E-value thr.
into the profile (>=0.0) [1.000E-03]
--wg B00L                     Use global sequence weighting for profile
calculation [0]
--filter-msa INT               Filter msa: 0: do not filter, 1: filter [1]
--filter-min-enable INT        Only filter MSAs with more than N sequences, 0
always filters [0]
--max-seq-id FLOAT             Reduce redundancy of output MSA using max.
pairwise sequence identity [0.0,1.0] [0.900]
--qid STR                      Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
Alternatively, can be a list of multiple
thresholds:
E.g.: 0.15,0.30,0.50 to defines filter buckets
of ]0.15-0.30] and ]0.30-0.50] [0.0]
--qsc FLOAT                    Reduce diversity of output MSAs using min.
score per aligned residue with query sequences [-50.0,100.0] [-20.000]
--cov FLOAT                    Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
--diff INT                     Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
--pseudo-cnt-mode INT          use 0: substitution-matrix or 1: context-
specific pseudocounts [0]
--profile-output-mode INT       Profile output mode: 0: binary log-odds 1:
human-readable frequencies [0]
--num-iterations INT           Number of iterative profile search iterations
[1]
--exhaustive-search B00L        For bigger profile DB, run iteratively the
search by greedily swapping the search results [0]
--lca-search B00L              Efficient search for LCA candidates [0]
misc:
--taxon-list STR                Taxonomy ID, possibly multiple values separated
by ',' []
--prefilter-mode INT            prefilter mode: 0: kmer/ungapped 1: ungapped,
2: nofilter, 3: ungapped&gapped [0]

```

```

--rescore-mode INT          Rescore diagonals with:
                             0: Hamming distance
                             1: local alignment (score only)
                             2: local alignment
                             3: global alignment
                             4: longest alignment fulfilling window quality

criterion [0]
--allow-deletion BOOL       Allow deletions in a MSA [0]
--min-length INT            Minimum codon number in open reading frames
[30]
--max-length INT            Maximum codon number in open reading frames
[32734]
--max-gaps INT              Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT     Contig start can be 0: incomplete, 1: complete,
2: both [2]
--contig-end-mode INT       Contig end can be 0: incomplete, 1: complete,
2: both [2]
--orf-start-mode INT         Orf fragment can be 0: from start to stop, 1:
from any to stop, 2: from last encountered start to stop (no start in the middle)
[1]
--forward-frames STR        Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR        Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT      1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16)
CHLOROPHYCEAN_MITOCHONDRIAL, 21) TREMATODE_MITOCHONDRIAL, 22)
SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31)

BLASTOCRITHIDIA [1]
--translate INT             Translate ORF to amino acid [0]
--use-all-table-starts BOOL Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT             Numeric ids in index file are offset by this
value [0]
--sequence-overlap INT      Overlap between sequences [0]
--sequence-split-mode INT    Sequence split mode 0: copy data, 1: soft link
data and write new index, [1]
--headers-split-mode INT     Header split mode: 0: split position, 1:
original header [0]
--search-type INT           Search type 0: auto 1: amino acid, 2:

```

translated, 3: nucleotide, 4: translated nucleotide alignment [0]
 --start-sens FLOAT Start sensitivity [4.000]
 --sens-steps INT Number of search steps performed from --start-sens to -s [1]
 --translation-mode INT Translation AA seq from nucleotide by 0: ORFs, 1: full reading frames [0]
 --format-mode INT Output format:
 0: BLAST-TAB
 1: SAM
 2: BLAST-TAB + query/db length
 3: Pretty HTML
 4: BLAST-TAB + column headers
 BLAST-TAB (0) and BLAST-TAB + column headers
 (4) support custom output formats (--format-output) [0]
 --format-output STR Choose comma separated list of output columns from: query,target,evalue,gapopen,pident,fident,nident,qstart,qend,qlen,tstart,tend,tlen,alnlen,raw,bits,cigar,qseq,tseq,qheader,theader,qaln,taln,qframe,tframe,mismatch,qset,qsetid,tset,tsetid,taxid,taxname,taxlineage,qorfstart,qorfend,torfstart,torfend,ppos [query,target,fident,alnlen,mismatch,gapopen,qstart,qend,tstart,tend,evalue,bits]
 --overlap FLOAT Maximum overlap of covered regions [0.000]
 --dbtype INT Database type 0: auto, 1: amino acid 2: nucleotides [0]
 --shuffle BOOL Shuffle input database [1]
 --createdb-mode INT Createdb mode 0: copy data, 1: soft link data and write new index (works only with single line fasta/q) [1]
 --greedy-best-hits BOOL Choose the best hits greedily to cover the query [0]
 common:
 --sub-mat TWIN Substitution matrix file [aa:blosum62.out,nucl:nucleotide.out]
 --max-seq-len INT Maximum sequence length [65535]
 --db-load-mode INT Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch [0]
 --threads INT Number of CPU-cores used (all by default) [10]
 --compressed INT Write compressed output [0]
 -v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]
 --gpu INT Use GPU (CUDA) if possible [0]
 --gpu-server INT Use GPU server [0]
 --gpu-server-wait-timeout INT Wait for GPU server for 0: don't wait -1: no wait limit: >0 this many seconds [600]
 --mpi-runner STR Use MPI on compute cluster with this MPI command (e.g. "mpirun -np 42") []
 --force-reuse BOOL Reuse tmp filse in tmp/latest folder ignoring parameters and version changes [0]
 --remove-tmp-files BOOL Delete temporary files [1]
 expert:
 --filter-hits BOOL Filter hits by seq.id. and coverage [0]

```

--sort-results INT          Sort results: 0: no sorting, 1: sort by E-value
                             (Alignment) or seq.id. (Hamming) [0]
--create-lookup INT         Create database lookup file (can be very large)
                             [0]
--chain-alignments INT      Chain overlapping alignments [0]
--merge-query INT           Combine ORFs/split sequences to a single entry
                             [1]
--strand INT                Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]
--db-output B00L           Return a result DB instead of a text file [0]
--write-lookup INT          write .lookup file containing mapping from
                             internal id, fasta id and file number [0]

```

examples:

```

# Assign reciprocal best hit
mmseqs easy-rbh examples/QUERY.fasta examples/DB.fasta result tmp

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.43. easy-search

Sensitive homology search.

Execution role: workflow entrypoint that coordinates lower-level modules rather than acting as a single compute kernel.

This command family favors fast adoption and default-safe orchestration over maximal low-level control. The current dependency map records 0 upstream caller(s) and 7 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: easy_workflows.

Usage

```

usage:      mmseqs      easy-search      <i:queryFastaFile1[.gz|.bz2]>      ...
<i:queryFastaFileN[.gz|.bz2]>|<i:stdin>      <i:targetFastaFile[.gz]>|<i:targetDB>
<o:alignmentFile> <tmpDir> [options]

```

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)

Option	Purpose
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists
--alph-size	Alphabet size (range 2-21)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--split	Split input into N equally distributed chunks. 0: set the best split automatically
--split-mode	0: split target db; 1: split query db; 2: auto, depending on main memory

Full CLI Help Snapshot

```
usage: mmseqs easy-search <i:queryFastaFile1[.gz|.bz2]> ...
<i:queryFastaFileN[.gz|.bz2]>|<i:stdin> <i:targetFastaFile[.gz]>|<i:targetDB>
<o:alignmentFile> <tmpDir> [options]
  By Milot Mirdita <milot@mirdita.de> & Martin Steinegger
<martin.steinegger@snu.ac.kr>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid
composition (range 0-1) [1]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid
composition (range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nuc:nucleotide.out]
  -s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [5.700]
  -k INT                       k-mer length (0: automatically set to optimum)
[0]
  --target-search-mode INT      target search mode (0: regular k-mer, 1:
```

```

similar k-mer) [0]
--k-score TWIN          k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
--alph-size TWIN        Alphabet size (range 2-21) [aa:21,nucl:5]
--max-seqs INT          Maximum results per query sequence allowed to
pass the prefilter (affects sensitivity) [300]
--split INT             Split input into N equally distributed chunks.
0: set the best split automatically [0]
--split-mode INT        0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
--split-memory-limit BYTE Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
--diag-score BOOL       Use ungapped diagonal scoring during prefilter
[1]
--exact-kmer-matching INT Extract only exact k-mers for matching (range
0-1) [0]
--mask INT              Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [1]
--mask-prob FLOAT       Mask sequences is probablity is above threshold
[0.900]
--mask-lower-case INT   Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT     Repeat letters that occure > threshold in a rwo
[0]
--min-ungapped-score INT Accept only matches with ungapped alignment
score above threshold [15]
--spaced-kmer-mode INT   0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
--spaced-kmer-pattern STR User-specified spaced k-mer pattern []
--local-tmp STR         Path where some of the temporary files will be
created []
--disk-space-limit BYTE Set max disk space to use for reverse profile
searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the
temp folder [0]
align:
-a BOOL                Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]
--alignment-mode INT     How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment [3]
--alignment-output-mode INT How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment

```


5: score only (output) cluster format [0]
 --wrapped-scoring BOOL Double the (nucleotide) query sequence during
 the scoring process to allow wrapped diagonal scoring around end and start [0]
 -e DOUBLE List matches below this E-value (range 0.0-inf)
 [1.000E-03]
 --min-seq-id FLOAT List matches above this sequence identity (for
 clustering) (range 0.0-1.0) [0.000]
 --min-aln-len INT Minimum alignment length (range 0-INT_MAX) [0]
 --seq-id-mode INT 0: alignment length 1: shorter, 2: longer
 sequence [0]
 --alt-ali INT Show up to this many alternative alignments [0]
 -c FLOAT List matches above this fraction of aligned
 (covered) residues (see --cov-mode) [0.000]
 --cov-mode INT 0: coverage of query and target
 1: coverage of target
 2: coverage of query
 3: target seq. length has to be at least x% of
 query length
 4: query seq. length has to be at least x% of
 target length
 5: short seq. needs to be at least x% of the
 other seq. length [0]
 --max-rejected INT Maximum rejected alignments before alignment
 calculation for a query is stopped [2147483647]
 --max-accept INT Maximum accepted alignments before alignment
 calculation for a query is stopped [2147483647]
 --score-bias FLOAT Score bias when computing SW alignment (in
 bits) [0.000]
 --realign BOOL Compute more conservative, shorter alignments
 (scores and E-values not changed) [0]
 --realign-score-bias FLOAT Additional bias when computing realignment
 [-0.200]
 --realign-max-seqs INT Maximum number of results to return in
 realignment [2147483647]
 --corr-score-weight FLOAT Weight of backtrace correlation score that is
 added to the alignment score [0.000]
 --gap-open TWIN Gap open cost [aa:11,nucl:5]
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]
 --zdrop INT Maximal allowed difference between score values
 before alignment is truncated (nucleotide alignment only) [40]
 --exhaustive-search-filter INT Filter result during search: 0: do not filter,
 1: filter [0]
 profile:
 --pca Pseudo count admixture strength []
 --pcb Pseudo counts: Neff at half of maximum
 admixture (range 0.0-inf) []
 --mask-profile INT Mask query sequence of profile using tantan
 [0,1] [1]
 --e-profile DOUBLE Include sequences matches with < E-value thr.

into the profile (≥ 0.0) [1.000E-03]

--wg BOOL Use global sequence weighting for profile calculation [0]

--filter-msa INT Filter msa: 0: do not filter, 1: filter [1]

--filter-min-enable INT Only filter MSAs with more than N sequences, 0 always filters [0]

--max-seq-id FLOAT Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0] [0.900]

--qid STR Reduce diversity of output MSAs using min.seq. identity with query sequences [0.0,1.0]

Alternatively, can be a list of multiple thresholds:

E.g.: 0.15,0.30,0.50 to defines filter buckets of]0.15-0.30] and]0.30-0.50] [0.0]

--qsc FLOAT Reduce diversity of output MSAs using min. score per aligned residue with query sequences [-50.0,100.0] [-20.000]

--cov FLOAT Filter output MSAs using min. fraction of query residues covered by matched sequences [0.0,1.0] [0.000]

--diff INT Filter MSAs by selecting most diverse set of sequences, keeping at least this many seqs in each MSA block of length 50 [1000]

--pseudo-cnt-mode INT use 0: substitution-matrix or 1: context-specific pseudocounts [0]

--profile-output-mode INT Profile output mode: 0: binary log-odds 1: human-readable frequencies [0]

--num-iterations INT Number of iterative profile search iterations [1]

--exhaustive-search BOOL For bigger profile DB, run iteratively the search by greedily swapping the search results [0]

--lca-search BOOL Efficient search for LCA candidates [0]

misc:

--taxon-list STR Taxonomy ID, possibly multiple values separated by ',' []

--prefilter-mode INT prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped [0]

--rescore-mode INT Rescore diagonals with:
0: Hamming distance
1: local alignment (score only)
2: local alignment
3: global alignment
4: longest alignment fulfilling window quality criterion [0]

--allow-deletion BOOL Allow deletions in a MSA [0]

--min-length INT Minimum codon number in open reading frames [30]

--max-length INT Maximum codon number in open reading frames [32734]

--max-gaps INT Maximum number of codons with gaps or unknown residues before an open reading frame is rejected [2147483647]

--contig-start-mode INT Contig start can be 0: incomplete, 1: complete,

2: both [2]
 --contig-end-mode INT Contig end can be 0: incomplete, 1: complete,
 2: both [2]
 --orf-start-mode INT Orf fragment can be 0: from start to stop, 1:
 from any to stop, 2: from last encountered start to stop (no start in the middle)
 [1]
 --forward-frames STR Comma-separated list of frames on the forward
 strand to be extracted [1,2,3]
 --reverse-frames STR Comma-separated list of frames on the reverse
 strand to be extracted [1,2,3]
 --translation-table INT 1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
 YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
 9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
 PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
 ALT_FLATWORM_MITOCHONDRIAL
 15) BLEPHARISMA, 16)
 CHLOROPHYCEAN_MITOCHONDRIAL, 21) TREMATODE_MITOCHONDRIAL, 22)
 SCENEDESMUS_MITOCHONDRIAL
 23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
 PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
 KARYORELICT, 28) CONDYLOSTOMA
 29) MESODINIUM, 30) PERTRICH, 31)
 BLASTOCRITHIDIA [1]
 --translate INT Translate ORF to amino acid [0]
 --use-all-table-starts BOOL Use all alternatives for a start codon in the
 genetic table, if false - only ATG (AUG) [0]
 --id-offset INT Numeric ids in index file are offset by this
 value [0]
 --sequence-overlap INT Overlap between sequences [0]
 --sequence-split-mode INT Sequence split mode 0: copy data, 1: soft link
 data and write new index, [1]
 --headers-split-mode INT Header split mode: 0: split position, 1:
 original header [0]
 --search-type INT Search type 0: auto 1: amino acid, 2:
 translated, 3: nucleotide, 4: translated nucleotide alignment [0]
 --start-sens FLOAT Start sensitivity [4.000]
 --sens-steps INT Number of search steps performed from --start-
 sens to -s [1]
 --translation-mode INT Translation AA seq from nucleotide by 0: ORFs,
 1: full reading frames [0]
 --format-mode INT Output format:
 0: BLAST-TAB
 1: SAM
 2: BLAST-TAB + query/db length
 3: Pretty HTML
 4: BLAST-TAB + column headers
 BLAST-TAB (0) and BLAST-TAB + column headers
 (4) support custom output formats (--format-output) [0]
 --format-output STR Choose comma separated list of output columns

```

from: query,target,evalue,gapopen,pident,fident,nident,qstart,qend,qlen
tstart,tend,tlen,alnlen,raw,bits,cigar,qseq,tseq,qheader,theader,qaln,taln,qframe,tframe,mismatch,c
qset,qsetid,tset,tsetid,taxid,taxname,taxlineage,qorfstart,qorfend,torfstart,torfend,ppos
[query,target,fident,alnlen,mismatch,gapopen,qstart,qend,tstart,tend,evalue,bits]
--overlap FLOAT                Maximum overlap of covered regions [0.000]
--dbtype INT                    Database type 0: auto, 1: amino acid 2:
nucleotides [0]
--shuffle BOOL                  Shuffle input database [1]
--createdb-mode INT             Createdb mode 0: copy data, 1: soft link data
and write new index (works only with single line fasta/q) [0]
--greedy-best-hits BOOL         Choose the best hits greedily to cover the
query [0]
common:
--sub-mat TWIN                  Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--max-seq-len INT               Maximum sequence length [65535]
--db-load-mode INT              Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
--threads INT                   Number of CPU-cores used (all by default) [10]
--compressed INT                Write compressed output [0]
-v INT                          Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--gpu INT                       Use GPU (CUDA) if possible [0]
--gpu-server INT                Use GPU server [0]
--gpu-server-wait-timeout INT    Wait for GPU server for 0: don't wait -1: no
wait limit: >0 this many seconds [600]
--mpi-runner STR                Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
--force-reuse BOOL              Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL         Delete temporary files [1]
expert:
--filter-hits BOOL               Filter hits by seq.id. and coverage [0]
--sort-results INT               Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--create-lookup INT              Create database lookup file (can be very large)
[0]
--chain-alignments INT           Chain overlapping alignments [0]
--merge-query INT                Combine ORFs/split sequences to a single entry
[1]
--strand INT                     Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]
--db-output BOOL                 Return a result DB instead of a text file [0]
--write-lookup INT                write .lookup file containing mapping from
internal id, fasta id and file number [0]

examples:
# Search multiple FASTA against FASTA (like BLASTP, TBLASTN, BLASTX, BLASTN --

```

```

search-type 3, TBLASTX --search-type 2)
mmseqs easy-search examples/QUERY.fasta examples/QUERY.fasta examples/DB.fasta
result.m8 tmp

# Iterative profile search from stdin (like PSI-BLAST)
cat examples/QUERY.fasta | mmseqs easy-search stdin examples/DB.fasta result.m8
tmp --num-iterations 2

# Profile search against small databases (e.g. PFAM, eggNOG)
mmseqs databases PFAM pfam_db tmp
mmseqs easy-search examples/QUERY.fasta pfam_db res.m8 tmp

# Exhaustive search against sequences or profiles (works for large DBs)
mmseqs easy-search examples/QUERY.fasta targetProfiles res.m8 tmp --exhaustive-
search

# Increasing sensitivity search (from 2 to 7 in 3 steps)
mmseqs easy-search examples/QUERY.fasta examples/DB.fasta result.m8 tmp --start-
sens 2 -s 7 --sens-steps 3

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Mirdita M, Steinegger M, Soding J: MMseqs2 desktop and local web server app for fast, interactive sequence searches. *Bioinformatics*, 35(16), 2856-2858 (2019)
- Kallenborn F, Chacon A, Hundt C, Sirelkhatim H, Didi K, Dallago C, Mirdita M, Schmidt B, Steinegger M: GPU-accelerated homology search with MMseqs2. *bioRxiv*, 2024.11.13.623350 (2024)

6.2.44. easy-taxonomy

Taxonomic classification.

Execution role: workflow entrypoint that coordinates lower-level modules rather than acting as a single compute kernel.

This command family favors fast adoption and default-safe orchestration over maximal low-level control. The current dependency map records 0 upstream caller(s) and 11 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: easy_workflows.

Usage

```
usage: mmseqs easy-taxonomy <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>
<i:targetDB> <o:taxReports> <tmpDir> [options]
```

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists
--alph-size	Alphabet size (range 2-21)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--split	Split input into N equally distributed chunks. 0: set the best split automatically
--split-mode	0: split target db; 1: split query db; 2: auto, depending on main memory

Full CLI Help Snapshot

usage: mmseqs easy-taxonomy <i:fastaFile1[.gz|.bz2]> ... <i:fastaFileN[.gz|.bz2]>
<i:targetDB> <o:taxReports> <tmpDir> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

--comp-bias-corr INT	Correct for locally biased amino acid composition (range 0-1) [1]
--comp-bias-corr-scale FLOAT	Correct for locally biased amino acid composition (range 0-1) [1.000]
--add-self-matches BOOL	Artificially add entries of queries with themselves (for clustering) [0]
--seed-sub-mat TWIN	Substitution matrix file for k-mer generation [aa:VTML80.out,nucl:nucleotide.out]
-s FLOAT	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive [4.000]
-k INT	k-mer length (0: automatically set to optimum) [0]

```

--target-search-mode INT          target search mode (0: regular k-mer, 1:
similar k-mer) [0]
--k-score TWIN                    k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
--alph-size TWIN                  Alphabet size (range 2-21) [aa:21,nucl:5]
--max-seqs INT                    Maximum results per query sequence allowed to
pass the prefilter (affects sensitivity) [300]
--split INT                       Split input into N equally distributed chunks.
0: set the best split automatically [0]
--split-mode INT                  0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
--split-memory-limit BYTE         Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
--diag-score BOOL                 Use ungapped diagonal scoring during prefilter
[1]
--exact-kmer-matching INT         Extract only exact k-mers for matching (range
0-1) [0]
--mask INT                        Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [1]
--mask-prob FLOAT                 Mask sequences is probability is above threshold
[0.900]
--mask-lower-case INT             Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT              Repeat letters that occur > threshold in a row
[0]
--min-ungapped-score INT          Accept only matches with ungapped alignment
score above threshold [15]
--spaced-kmer-mode INT            0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
--spaced-kmer-pattern STR         User-specified spaced k-mer pattern []
--local-tmp STR                   Path where some of the temporary files will be
created []
--disk-space-limit BYTE           Set max disk space to use for reverse profile
searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the
temp folder [0]
align:
-a BOOL                           Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]
--alignment-mode INT              How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment [0]
--alignment-output-mode INT       How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id

```

4: only ungapped alignment
 5: score only (output) cluster format [0]
 --wrapped-scoring BOOL Double the (nucleotide) query sequence during
 the scoring process to allow wrapped diagonal scoring around end and start [0]
 -e DOUBLE List matches below this E-value (range 0.0-inf)
 [1.000E-03]
 --min-seq-id FLOAT List matches above this sequence identity (for
 clustering) (range 0.0-1.0) [0.000]
 --min-aln-len INT Minimum alignment length (range 0-INT_MAX) [0]
 --seq-id-mode INT 0: alignment length 1: shorter, 2: longer
 sequence [0]
 --alt-ali INT Show up to this many alternative alignments [0]
 -c FLOAT List matches above this fraction of aligned
 (covered) residues (see --cov-mode) [0.000]
 --cov-mode INT 0: coverage of query and target
 1: coverage of target
 2: coverage of query
 3: target seq. length has to be at least x% of
 query length
 4: query seq. length has to be at least x% of
 target length
 5: short seq. needs to be at least x% of the
 other seq. length [0]
 --max-rejected INT Maximum rejected alignments before alignment
 calculation for a query is stopped [2147483647]
 --max-accept INT Maximum accepted alignments before alignment
 calculation for a query is stopped [2147483647]
 --score-bias FLOAT Score bias when computing SW alignment (in
 bits) [0.000]
 --realign BOOL Compute more conservative, shorter alignments
 (scores and E-values not changed) [0]
 --realign-score-bias FLOAT Additional bias when computing realignment
 [-0.200]
 --realign-max-seqs INT Maximum number of results to return in
 realignment [2147483647]
 --corr-score-weight FLOAT Weight of backtrace correlation score that is
 added to the alignment score [0.000]
 --gap-open TWIN Gap open cost [aa:11,nucl:5]
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]
 --zdrop INT Maximal allowed difference between score values
 before alignment is truncated (nucleotide alignment only) [40]
 --exhaustive-search-filter INT Filter result during search: 0: do not filter,
 1: filter [0]
 profile:
 --pca Pseudo count admixture strength []
 --pcb Pseudo counts: Neff at half of maximum
 admixture (range 0.0-inf) []
 --mask-profile INT Mask query sequence of profile using tantan
 [0,1] [1]


```

--e-profile DOUBLE          Include sequences matches with < E-value thr.
into the profile (>=0.0) [1.000E-03]
--wg BOOL                   Use global sequence weighting for profile
calculation [0]
--filter-msa INT            Filter msa: 0: do not filter, 1: filter [1]
--filter-min-enable INT     Only filter MSAs with more than N sequences, 0
always filters [0]
--max-seq-id FLOAT         Reduce redundancy of output MSA using max.
pairwise sequence identity [0.0,1.0] [0.900]
--qid STR                  Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
                             Alternatively, can be a list of multiple
thresholds:
                             E.g.: 0.15,0.30,0.50 to defines filter buckets
of [0.15-0.30] and [0.30-0.50] [0.0]
--qsc FLOAT                Reduce diversity of output MSAs using min.
score per aligned residue with query sequences [-50.0,100.0] [-20.000]
--cov FLOAT                Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
--diff INT                 Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
--pseudo-cnt-mode INT      use 0: substitution-matrix or 1: context-
specific pseudocounts [0]
--profile-output-mode INT   Profile output mode: 0: binary log-odds 1:
human-readable frequencies [0]
--exhaustive-search BOOL    For bigger profile DB, run iteratively the
search by greedily swapping the search results [0]
--lca-search BOOL           Efficient search for LCA candidates [0]
misc:
--orf-filter INT            Prefilter query ORFs with non-selective search
                             Only used during nucleotide-vs-protein
classification
                             NOTE: Consider disabling when classifying short
reads [0]
--orf-filter-e DOUBLE       E-value threshold used for query ORF
prefiltering [1.000E+02]
--orf-filter-s FLOAT        Sensitivity used for query ORF prefiltering
[2.000]
--lca-mode INT              LCA Mode 1: single search LCA , 2/3:
approximate 2bLCA, 4: top hit [3]
--majority FLOAT            minimal fraction of agreement among
taxonomically assigned sequences of a set [0.500]
--vote-mode INT             Mode of assigning weights to compute majority.
0: uniform, 1: minus log E-value, 2: score [1]
--lca-ranks STR             Add column with specified ranks (',' separated)
[]
--tax-lineage INT           0: don't show, 1: add all lineage names, 2: add
all lineage taxids [0]
--blacklist STR             Comma separated list of ignored taxa in LCA

```

```

computation [12908:unclassified sequences,28384:other sequences]
--taxon-list STR          Taxonomy ID, possibly multiple values separated
by ',' []
--prefilter-mode INT      prefilter mode: 0: kmer/ungapped 1: ungapped,
2: nofilter, 3: ungapped&gapped [0]
--rescore-mode INT        Rescore diagonals with:
                          0: Hamming distance
                          1: local alignment (score only)
                          2: local alignment
                          3: global alignment
                          4: longest alignment fulfilling window quality

criterion [0]
--allow-deletion BOOL      Allow deletions in a MSA [0]
--min-length INT           Minimum codon number in open reading frames
[30]
--max-length INT           Maximum codon number in open reading frames
[32734]
--max-gaps INT             Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT    Contig start can be 0: incomplete, 1: complete,
2: both [2]
--contig-end-mode INT      Contig end can be 0: incomplete, 1: complete,
2: both [2]
--orf-start-mode INT       Orf fragment can be 0: from start to stop, 1:
from any to stop, 2: from last encountered start to stop (no start in the middle)
[1]
--forward-frames STR       Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR       Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT     1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16)
CHLOROPHYCEAN_MITOCHONDRIAL, 21) TREMATODE_MITOCHONDRIAL, 22)
SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31)

BLASTOCRITHIDIA [1]
--translate INT            Translate ORF to amino acid [0]
--use-all-table-starts BOOL Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT            Numeric ids in index file are offset by this
value [0]
--sequence-overlap INT     Overlap between sequences [0]

```

```

--sequence-split-mode INT      Sequence split mode 0: copy data, 1: soft link
data and write new index, [1]
--headers-split-mode INT      Header split mode: 0: split position, 1:
original header [0]
--search-type INT             Search type 0: auto 1: amino acid, 2:
translated, 3: nucleotide, 4: translated nucleotide alignment [0]
--translation-mode INT        Translation AA seq from nucleotide by 0: ORFs,
1: full reading frames [0]
--report-mode INT             Taxonomy report mode 0: Kraken 1: Krona [0]
--format-mode INT             Output format:
                                0: BLAST-TAB
                                1: SAM
                                2: BLAST-TAB + query/db length
                                3: Pretty HTML
                                4: BLAST-TAB + column headers
                                BLAST-TAB (0) and BLAST-TAB + column headers
(4) support custom output formats (--format-output) [0]
--format-output STR           Choose comma separated list of output columns
from: query,target,evalue,gapopen,pident,fident,nident,qstart,qend,qlen
tstart,tend,tlen,alnlen,raw,bits,cigar,qseq,tseq,qheader,theader,qaln,taln,qframe,tframe,mismatch,c
qset,qsetid,tset,tsetid,taxid,taxname,taxlineage,qorfstart,qorfend,torfstart,torfend,ppos
[query,target,fident,alnlen,mismatch,gapopen,qstart,qend,tstart,tend,evalue,bits]
--first-seq-as-repr BOOL      Use the first sequence of the clustering result
as representative sequence [0]
--target-column INT           Select a target column (default 1), 0 if no
target id exists [1]
--full-header BOOL            Replace DB ID by its corresponding Full Header
[0]
--idx-seq-src INT             0: auto, 1: split/translated sequences, 2:
input sequences [0]
--dbtype INT                  Database type 0: auto, 1: amino acid 2:
nucleotides [0]
--shuffle BOOL                Shuffle input database [1]
--createdb-mode INT           Createdb mode 0: copy data, 1: soft link data
and write new index (works only with single line fasta/q) [1]
common:
--compressed INT              Write compressed output [0]
--threads INT                 Number of CPU-cores used (all by default) [10]
-v INT                        Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--sub-mat TWIN                Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--max-seq-len INT             Maximum sequence length [65535]
--db-load-mode INT            Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
--gpu INT                     Use GPU (CUDA) if possible [0]
--gpu-server INT              Use GPU server [0]
--gpu-server-wait-timeout INT Wait for GPU server for 0: don't wait -1: no

```

```

wait limit: >0 this many seconds [600]
--mpi-runner STR          Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
--force-reuse BOOL        Reuse tmp file in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL   Delete temporary files [1]
expert:
--filter-hits BOOL        Filter hits by seq.id. and coverage [0]
--sort-results INT        Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--create-lookup INT       Create database lookup file (can be very large)
[0]
--chain-alignments INT     Chain overlapping alignments [0]
--merge-query INT         Combine ORFs/split sequences to a single entry
[1]
--strand INT              Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]
--db-output BOOL          Return a result DB instead of a text file [0]
--write-lookup INT        write .lookup file containing mapping from
internal id, fasta id and file number [0]

```

examples:

```

# Assign taxonomic labels to FASTA sequences
- result_tophit_aln: top hits
- result_tophit_report: coverage profiles per database entry
- result_report: kraken style report
# Download a sequence database with taxonomy information
mmseqs databases UniProtKB/Swiss-Prot swissprotDB tmp

# Assign taxonomy based on 2bLCA hit
mmseqs easy-taxonomy examples/DB.fasta swissprotDB result tmp

# Assign taxonomy based on top hit
mmseqs easy-taxonomy examples/DB.fasta swissprotDB result tmp --lca-mode 4

# Assign taxonomy without ORF prefilter
# Classifies higher percentage for short nucleotide input (e.g. short reads) at
the cost of speed
mmseqs easy-taxonomy queryNuclDB swissprotDB result tmp --orf-filter 0

```

references:

```

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
- Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and
sensitive taxonomic assignment to metagenomic contigs. Bioinformatics, btab184
(2021)

```

6.2.45. **expand2profile**

Expand an alignment result based on another and create a profile.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

```
usage: mmseqs expand2profile <inputDB(s)> <outputDB> [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs expand2profile` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.46. **expandaln**

Expand an alignment result based on another.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

Usage

usage: mmseqs expandaln <i:queryDB> <i:targetDB> <i:resultDB> <i:resultDB|ca3mDB>
<o:alignmentDB> [options]

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--score-bias	Score bias when computing SW alignment (in bits)
-e	List matches below this E-value (range 0.0-inf)
--min-seq-id	List matches above this sequence identity (for clustering) (range 0.0-1.0)
-c	List matches above this fraction of aligned (covered) residues (see --cov-mode)
--cov-mode	0: coverage of query and target
--pseudo-cnt-mode	use 0: substitution-matrix or 1: context-specific pseudocounts
--pca	Pseudo count admixture strength
--pcb	Pseudo counts: Neff at half of maximum admixture (range 0.0-inf)

Full CLI Help Snapshot

usage: mmseqs expandaln <i:queryDB> <i:targetDB> <i:resultDB> <i:resultDB|ca3mDB>
<o:alignmentDB> [options]

By Milot Mirdita <milot@mirdita.de>

options: prefilter:

--comp-bias-corr INT Correct for locally biased amino acid composition
(range 0-1) [1]

--comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]

align:

--gap-open TWIN Gap open cost [aa:11,nucl:5]
--gap-extend TWIN Gap extension cost [aa:1,nucl:2]
--score-bias FLOAT Score bias when computing SW alignment (in bits)
[0.000]

-e DOUBLE List matches below this E-value (range 0.0-inf)

```

[1.000E-03]
--min-seq-id FLOAT          List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
-c FLOAT                    List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.000]
--cov-mode INT              0: coverage of query and target
                           1: coverage of target
                           2: coverage of query
                           3: target seq. length has to be at least x% of
query length
                           4: query seq. length has to be at least x% of
target length
                           5: short seq. needs to be at least x% of the
other seq. length [0]
profile:
--pseudo-cnt-mode INT       use 0: substitution-matrix or 1: context-specific
pseudocounts [0]
--pca                       Pseudo count admixture strength []
--pcb                       Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
--filter-min-enable INT     Only filter MSAs with more than N sequences, 0
always filters [0]
--max-seq-id FLOAT          Reduce redundancy of output MSA using max.
pairwise sequence identity [0.0,1.0] [0.900]
--qid STR                   Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
                           Alternatively, can be a list of multiple
thresholds:
                           E.g.: 0.15,0.30,0.50 to defines filter buckets
of [0.15-0.30] and [0.30-0.50] [0.0]
--qsc FLOAT                 Reduce diversity of output MSAs using min. score
per aligned residue with query sequences [-50.0,100.0] [-20.000]
--cov FLOAT                 Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
--diff INT                  Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
misc:
--expansion-mode INT         Update score, E-value, and sequence identity by
0: input alignment 1: rescoring the inferred backtrace [0]
--expand-filter-clusters INT Filter each target cluster during expansion 0: no
filter 1: filter [0]
common:
--sub-mat TWIN               Substitution matrix file
[aa:blosum62.out,nuc1:nucleotide.out]
--max-seq-len INT            Maximum sequence length [65535]
--db-load-mode INT           Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--compressed INT             Write compressed output [0]
--threads INT                Number of CPU-cores used (all by default) [10]

```

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.47. extractalignedregion

Extract aligned sequence region from query.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence_manipulation.

Usage

usage: mmseqs extractalignedregion <i:queryDB> <i:targetDB> <i:resultDB> <o:sequenceDB> [options]

Key Options

Option	Purpose
--extract-mode	Extract from 1: Query, 2: Target
--compressed	Write compressed output
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: mmseqs extractalignedregion <i:queryDB> <i:targetDB> <i:resultDB> <o:sequenceDB> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: misc:

--extract-mode INT Extract from 1: Query, 2: Target [2]

common:

--compressed INT Write compressed output [0]


```
--db-load-mode INT    Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]
--threads INT         Number of CPU-cores used (all by default) [10]
-v INT               Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

6.2.48. extractdomains

Extract highest scoring alignment regions for each sequence from BLAST-tab file.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result_handling.

Usage

usage: mmseqs extractdomains <DB> [args] [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs extractdomains locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.49. extractframes

Extract frames from a nucleotide sequence DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 4 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence_manipulation.

Usage

usage: mmseqs extractframes <i:sequenceDB> <o:sequenceDB> [options]

Key Options

Option	Purpose
--forward-frames	Comma-separated list of frames on the forward strand to be extracted
--reverse-frames	Comma-separated list of frames on the reverse strand to be extracted
--translation-table	1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3) YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
--translate	Translate ORF to amino acid
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--create-lookup	Create database lookup file (can be very large)

Full CLI Help Snapshot

usage: mmseqs extractframes <i:sequenceDB> <o:sequenceDB> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: misc:

--forward-frames STR Comma-separated list of frames on the forward strand to be extracted [1,2,3]

--reverse-frames STR Comma-separated list of frames on the reverse strand to be extracted [1,2,3]

--translation-table INT 1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3) YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE 9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11) PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)

ALT_FLATWORM_MITOCHONDRIAL 15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL, 21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL 23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24) PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27) KARYORELICT, 28) CONDYLOSTOMA 29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA [1]

--translate INT Translate ORF to amino acid [0]

common:

--threads INT Number of CPU-cores used (all by default) [10]

--compressed INT Write compressed output [0]

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

expert:

--create-lookup INT Create database lookup file (can be very large) [0]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.50. extractorfs

Six-frame extraction of open reading frames.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 6 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence_manipulation.

Usage

usage: mmseqs extractorfs <i:sequenceDB> <o:sequenceDB> [options]

Key Options

Option	Purpose
--min-length	Minimum codon number in open reading frames
--max-length	Maximum codon number in open reading frames

Option	Purpose
--max-gaps	Maximum number of codons with gaps or unknown residues before an open reading frame is rejected
--contig-start-mode	Contig start can be 0: incomplete, 1: complete, 2: both
--contig-end-mode	Contig end can be 0: incomplete, 1: complete, 2: both
--orf-start-mode	Orf fragment can be 0: from start to stop, 1: from any to stop, 2: from last encountered start to stop (no start in the middle)
--forward-frames	Comma-separated list of frames on the forward strand to be extracted
--reverse-frames	Comma-separated list of frames on the reverse strand to be extracted
--translation-table	1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3) YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILLATE
--translate	Translate ORF to amino acid
--use-all-table-starts	Use all alternatives for a start codon in the genetic table, if false - only ATG (AUG)
--id-offset	Numeric ids in index file are offset by this value

Full CLI Help Snapshot

```
usage: mmseqs extractorfs <i:sequenceDB> <o:sequenceDB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: misc:
  --min-length INT           Minimum codon number in open reading frames [30]
  --max-length INT           Maximum codon number in open reading frames [32734]
  --max-gaps INT             Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
  --contig-start-mode INT     Contig start can be 0: incomplete, 1: complete, 2:
both [2]
  --contig-end-mode INT       Contig end can be 0: incomplete, 1: complete, 2:
both [2]
  --orf-start-mode INT        Orf fragment can be 0: from start to stop, 1: from
any to stop, 2: from last encountered start to stop (no start in the middle) [1]
```

```

--forward-frames STR          Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR          Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT        1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL,
21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA
[1]
--translate INT                Translate ORF to amino acid [0]
--use-all-table-starts BOOL   Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT                Numeric ids in index file are offset by this value
[0]
common:
--threads INT                  Number of CPU-cores used (all by default) [10]
--compressed INT               Write compressed output [0]
-v INT                         Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
expert:
--create-lookup INT            Create database lookup file (can be very large) [0]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

```

6.2.51. filterdb

DB filtering by given conditions.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 8 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

usage: mmseqs filterdb <i:resultDB> <o:resultDB> [options]

Key Options

Option	Purpose
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--filter-expression	Specify a mathematical expression to filter lines
--filter-column	column
--column-to-take	column to take in join mode. If -1, the whole line is taken
--filter-regex	Regex to select column (example float: [0-9]*([0-9]+)? int:[1-9]{1}[0-9])
--positive-filter	Used in conjunction with --filter-file. If true, out = in filter ; if false, out = in - filter
--filter-file	Specify a file that contains the filtering elements
--beats-first	Filter by comparing each entry to the first entry
--mapping-file	Specify a file that translates the keys of a DB to new keys, TSV format
--weights	Weights used for cluster prioritization
--trim-to-one-column	Output only the column specified by --filter-column
--extract-lines	Extract n lines of each entry

Full CLI Help Snapshot

usage: mmseqs filterdb <i:resultDB> <o:resultDB> [options]

By Clovis Galiez & Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

--add-self-matches BOOL Artificially add entries of queries with themselves (for clustering) [0]

misc:

--filter-expression STR Specify a mathematical expression to filter lines []
--filter-column INT column [1]

--column-to-take INT column to take in join mode. If -1, the whole line is taken [-1]

--filter-regex STR Regex to select column (example float: [0-9]*([0-9]+)? int:[1-9]{1}[0-9]) [^.*\$]

--positive-filter BOOL Used in conjunction with --filter-file. If true, out

```

= in \intersect filter ; if false, out = in - filter [1]
--filter-file STR          Specify a file that contains the filtering elements
[]
--beats-first BOOL         Filter by comparing each entry to the first entry
[0]
--mapping-file STR         Specify a file that translates the keys of a DB to
new keys, TSV format []
--weights STR             Weights used for cluster prioritization []
--trim-to-one-column BOOL  Output only the column specified by --filter-column
[0]
--extract-lines INT        Extract n lines of each entry [0]
--comparison-operator STR  Filter by comparing each entry row numerically by
using the le) less-than-equal, ge) greater-than-equal or e) equal operator []
--comparison-value DOUBLE  Filter by comparing each entry to this value
[0.000E+00]
--sort-entries INT         Sort column set by --filter-column, by 0: no
sorting, 1: increasing, 2: decreasing, 3: random shuffle, 4: priority [0]
--join-db STR             Join another database entry with respect to the
database identifier in the chosen column []
common:
--threads INT             Number of CPU-cores used (all by default) [10]
--compressed INT          Write compressed output [0]
-v INT                   Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
3: +info [3]

```

examples:

```

# Retain top alignment for each query (alignment DBs are sorted by E-value)
mmseqs filterdb alignmentDB topHitAlignmentDB --extract-lines 1

# Extract alignments with Seq.id. greater than 90%
mmseqs filterdb alignmentDB scoreGreater35AlignmentDB --comparison-operator ge
--comparison-value 0.9 --filter-column 2

# Retain all hits matching a regular expression
mmseqs filterdb alignmentDB regexFilteredDB --filter-regex '^[1-9].$' --filter-
column 2

# Remove all hits to target keys contained in file db.index
mmseqs filterdb --filter-file db.index --positive-filter false

# Retain all hits matching any boolean expression
mmseqs filterdb --filter-expression '$1 * $2 >= 200'

```

references:

```

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

```

6.2.52. filterresult

Pairwise alignment result filter.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result_handling.

Usage

```
usage: mmseqs filterresult <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB>
[options]
```

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	keep the query (representative) sequence
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--filter-min-enable	Only filter MSAs with more than N sequences, 0 always filters
--max-seq-id	Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0]
--qid	Reduce diversity of output MSAs using min.seq. identity with query sequences
--qsc	Reduce diversity of output MSAs using min. score per aligned residue with query sequences [-50.0,100.0]
--cov	Filter output MSAs using min. fraction of query residues covered by matched sequences [0.0,1.0]

Option	Purpose
--diff	Filter MSAs by selecting most diverse set of sequences, keeping at least this many seqs in each MSA block of length 50
--allow-deletion	Allow deletions in a MSA

Full CLI Help Snapshot

```
usage: mmseqs filterresult <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB>
[options]
  By Milot Mirdita <milot@mirdita.de>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
                                (range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
                                (range 0-1) [1.000]
  --add-self-matches BOOL      keep the query (representative) sequence [0]
align:
  --gap-open TWIN              Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN           Gap extension cost [aa:1,nucl:2]
profile:
  --filter-min-enable INT      Only filter MSAs with more than N sequences, 0
                                always filters [0]
  --max-seq-id FLOAT          Reduce redundancy of output MSA using max. pairwise
                                sequence identity [0.0,1.0] [0.900]
  --qid STR                   Reduce diversity of output MSAs using min.seq.
                                identity with query sequences [0.0,1.0]
                                Alternatively, can be a list of multiple
thresholds:
                                E.g.: 0.15,0.30,0.50 to defines filter buckets
                                of ]0.15-0.30] and ]0.30-0.50] [0.0]
  --qsc FLOAT                 Reduce diversity of output MSAs using min. score
                                per aligned residue with query sequences [-50.0,100.0] [-20.000]
  --cov FLOAT                 Filter output MSAs using min. fraction of query
                                residues covered by matched sequences [0.0,1.0] [0.000]
  --diff INT                  Filter MSAs by selecting most diverse set of
                                sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
misc:
  --allow-deletion BOOL       Allow deletions in a MSA [0]
common:
  --sub-mat TWIN              Substitution matrix file
                                [aa:blosum62.out,nucl:nucleotide.out]
  --db-load-mode INT          Database preload mode 0: auto, 1: fread, 2: mmap,
                                3: mmap+touch [0]
  --threads INT               Number of CPU-cores used (all by default) [10]
  --compressed INT            Write compressed output [0]
  -v INT                      Verbosity level: 0: quiet, 1: +errors, 2:
                                +warnings, 3: +info [3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.53. **filtertaxdb**

Filter taxonomy result database.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

usage: mmseqs filtertaxdb <i:targetDB> <i:taxDB> <o:taxDB> [options]

Key Options

Option	Purpose
--taxon-list	Taxonomy ID, possibly multiple values separated by ‘,’
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: mmseqs filtertaxdb <i:targetDB> <i:taxDB> <o:taxDB> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: misc:

--taxon-list STR Taxonomy ID, possibly multiple values separated by ',' []

common:

--compressed INT Write compressed output [0]

--threads INT Number of CPU-cores used (all by default) [10]

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

examples:

Download a sequence database with taxonomy information

```

mmseqs databases UniProtKB/Swiss-Prot swissprotDB tmp
# Annotate a queryDB with taxonomy information
mmseqs taxonomy queryDB swissprotDB taxDB tmp

# Retain all unclassified hits
mmseqs filtertaxdb swissprotDB taxDB filteredTaxDB --taxon-list 0
mmseqs createsubdb <(awk '$3 == 1' filteredTaxDB.index) queryDB
queryUnclassifiedDB

# Retain all eukaryotic hits except fungi
mmseqs filtertaxdb swissprotDB taxDB filteredTaxDB --taxon-list '2759&&!4751'

# Retain all human and chlamydia hits
mmseqs filtertaxdb swissprotDB taxDB filteredTaxDB --taxon-list '9606||810'

```

references:

- Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and sensitive taxonomic assignment to metagenomic contigs. Bioinformatics, btabl84 (2021)

6.2.54. filtertaxseqdb

Filter taxonomy sequence database.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

usage: mmseqs filtertaxseqdb <i:taxSeqDB> <o:taxSeqDB> [options]

Key Options

Option	Purpose
--taxon-list	Taxonomy ID, possibly multiple values separated by ‘,’
--subdb-mode	Subdb mode 0: copy data 1: soft link data and write index
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs filtertaxseqdb <i:taxSeqDB> <o:taxSeqDB> [options]
  By Eli Levy Karin <eli.levy.karin@gmail.com> & Martin Steinegger
<martin.steinegger@snu.ac.kr>
options: misc:
  --taxon-list STR    Taxonomy ID, possibly multiple values separated by ',' []
  --subdb-mode INT    Subdb mode 0: copy data 1: soft link data and write index [0]
common:
  --compressed INT    Write compressed output [0]
  --threads INT       Number of CPU-cores used (all by default) [10]
  -v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

examples:
# Download a sequence database with taxonomy information
mmseqs databases UniProtKB/Swiss-Prot swissprotDB tmp

# Retain all bacterial sequences
mmseqs filtertaxseqdb swissprotDB swissprotDB_only_bac --taxon-list 2

# Retain all eukaryotic sequences except fungi
mmseqs filtertaxseqdb swissprotDB swissprotDB_euk_wo_fungi --taxon-list '2759&&!
4751'

# Retain all human and chlamydia sequences
mmseqs filtertaxseqdb swissprotDB swissprotDB_human_and_chlamydia --taxon-list
'9606||810'
```

references:

- Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and sensitive taxonomic assignment to metagenomic contigs. *Bioinformatics*, btabl84 (2021)

6.2.55. fwbw

Forward Backward Alignment.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

Usage

```
usage: mmseqs fwbw <inputDB(s)> <outputDB> [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs fwbw` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.56. gappedprefilter

Optimal Smith-Waterman-based prefiltering (slow).

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family is the main acceleration gate that prunes candidate pairs before expensive alignment. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command in custom pipelines that must expose candidate-generation behavior before alignment stages.

Dependency entry: Open in map; functional module: prefiltering.

Usage

```
usage: mmseqs gappedprefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]
```

Key Options

Option	Purpose
<code>--comp-bias-corr</code>	Correct for locally biased amino acid composition (range 0-1)
<code>--comp-bias-corr-scale</code>	Correct for locally biased amino acid composition (range 0-1)

Option	Purpose
--min-ungapped-score	Accept only matches with ungapped alignment score above threshold
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--gap-open	Gap open cost
--gap-extend	Gap extension cost
-e	List matches below this E-value (range 0.0-inf)
-c	List matches above this fraction of aligned (covered) residues (see --cov-mode)
--cov-mode	0: coverage of query and target
--taxon-list	Taxonomy ID, possibly multiple values separated by ‘,’
--sub-mat	Substitution matrix file
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch

Full CLI Help Snapshot

usage: mmseqs gappedprefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]
 By Milot Mirdita <milot@mirdita.de> & Martin Steinegger
 <martin.steinegger@snu.ac.kr>

options: prefilter:

--comp-bias-corr INT Correct for locally biased amino acid composition
 (range 0-1) [1]
 --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
 (range 0-1) [1.000]
 --min-ungapped-score INT Accept only matches with ungapped alignment score
 above threshold [15]
 --max-seqs INT Maximum results per query sequence allowed to pass
 the prefilter (affects sensitivity) [300]

align:

--gap-open TWIN Gap open cost [aa:11,nucl:5]
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]
 -e DOUBLE List matches below this E-value (range 0.0-inf)
 [1.000E-03]
 -c FLOAT List matches above this fraction of aligned
 (covered) residues (see --cov-mode) [0.000]
 --cov-mode INT 0: coverage of query and target
 1: coverage of target
 2: coverage of query
 3: target seq. length has to be at least x% of
 query length

target length 4: query seq. length has to be at least x% of
seq. length [0] 5: short seq. needs to be at least x% of the other
misc:
--taxon-list STR Taxonomy ID, possibly multiple values separated by
',' []
common:
--sub-mat TWIN Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--db-load-mode INT Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--threads INT Number of CPU-cores used (all by default) [10]
--compressed INT Write compressed output [0]
-v INT Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Kallenborn F, Chacon A, Hundt C, Sirelkhatim H, Didi K, Dallago C, Mirdita M, Schmidt B, Steinegger M: GPU-accelerated homology search with MMseqs2. *bioRxiv*, 2024.11.13.623350 (2024)

6.2.57. gff2db

Extract regions from a sequence database based on a GFF3 file.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

usage: mmseqs gff2db <DB> [args] [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs gff2db locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.58. gpuserver

Start a GPU server.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

```
usage: mmseqs gpuserver <i:DB> [options]
```

Key Options

Option	Purpose
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--prefilter-mode	prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped
--gpu	Use GPU (CUDA) if possible
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch

Full CLI Help Snapshot

```
usage: mmseqs gpuserver <i:DB> [options]
```

By Milot Mirdita <milot@mirdita.de> & Martin Steinegger
<martin.steinegger@snu.ac.kr>

options: prefilter:

```
--max-seqs INT          Maximum results per query sequence allowed to pass the  
prefilter (affects sensitivity) [300]
```

misc:

```
--prefilter-mode INT    prefilter mode: 0: kmer/ungapped 1: ungapped, 2:
```


nofilter, 3: ungapped&gapped [0]

common:

--gpu INT Use GPU (CUDA) if possible [0]
--db-load-mode INT Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]

references:

- Kallenborn F, Chacon A, Hundt C, Sirelkhatim H, Didi K, Dallago C, Mirdita M, Schmidt B, Steinegger M: GPU-accelerated homology search with MMseqs2. bioRxiv, 2024.11.13.623350 (2024)

6.2.59. kmermatcher

Find bottom-m-hashed k-mer matches within sequence DB.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family is the main acceleration gate that prunes candidate pairs before expensive alignment. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command in custom pipelines that must expose candidate-generation behavior before alignment stages.

Dependency entry: Open in map; functional module: prefiltering.

Usage

usage: mmseqs kmermatcher <i:sequenceDB> <o:prefilterDB> [options]

Key Options

Option	Purpose
--alph-size	Alphabet size (range 2-21)
--spaced-kmer-mode	0: use consecutive positions in k-mers; 1: use spaced k-mers
--spaced-kmer-pattern	User-specified spaced k-mer pattern
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probability is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occur > threshold in a row

Option	Purpose
-k	k-mer length (0: automatically set to optimum)
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
--min-seq-id	List matches above this sequence identity (for clustering) (range 0.0-1.0)
--cov-mode	0: coverage of query and target
-c	List matches above this fraction of aligned (covered) residues (see --cov-mode)

Full CLI Help Snapshot

```
usage: mmseqs kmermatcher <i:sequenceDB> <o:prefilterDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --alph-size TWIN           Alphabet size (range 2-21) [aa:13,nucl:5]
  --spaced-kmer-mode INT     0: use consecutive positions in k-mers; 1: use
spaced k-mers [0]
  --spaced-kmer-pattern STR  User-specified spaced k-mer pattern []
  --mask INT                 Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [0]
  --mask-prob FLOAT         Mask sequences is probablity is above threshold
[0.900]
  --mask-lower-case INT     Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
  --mask-n-repeat INT       Repeat letters that occure > threshold in a rwo
[0]
  -k INT                     k-mer length (0: automatically set to optimum)
[0]
  --split-memory-limit BYTE  Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
align:
  --min-seq-id FLOAT        List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
  --cov-mode INT            0: coverage of query and target
                           1: coverage of target
                           2: coverage of query
                           3: target seq. length has to be at least x% of
query length
                           4: query seq. length has to be at least x% of
target length
                           5: short seq. needs to be at least x% of the
other seq. length [0]
  -c FLOAT                 List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.800]
```

```

kmermatcher:
  --kmer-per-seq INT          k-mers per sequence [0]
  --kmer-per-seq-scale TWIN    Scale k-mer per sequence based on sequence
length as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
  --adjust-kmer-len BOOL      Adjust k-mer length based on specificity (only
for nucleotides) [0]
  --hash-shift INT            Shift k-mer hash initialization [67]
  --include-only-extendable BOOL Include only extendable [0]
  --ignore-multi-kmer BOOL    Skip k-mers occurring multiple times (>=2) [0]
  --weights STR               Weights used for cluster prioritization []
  --cluster-weight-threshold FLOAT Weight threshold used for cluster prioritization
[0.900]
common:
  --sub-mat TWIN              Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
  --max-seq-len INT           Maximum sequence length [65535]
  --threads INT               Number of CPU-cores used (all by default) [10]
  --compressed INT            Write compressed output [0]
  -v INT                      Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.60. kmersearch

Find bottom-m-hashed k-mer matches between target and query DB.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family is the main acceleration gate that prunes candidate pairs before expensive alignment. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command in custom pipelines that must expose candidate-generation behavior before alignment stages.

Dependency entry: Open in map; functional module: prefiltering.

Usage

usage: mmseqs kmersearch <i:queryDB> <i:kmerIndexDB> <o:prefilterDB> [options]

Key Options

Option	Purpose
--seed-sub-mat	Substitution matrix file for k-mer generation

Option	Purpose
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probability is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occur > threshold in a row
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
--cov-mode	0: coverage of query and target
-c	List matches above this fraction of aligned (covered) residues (see --cov-mode)
--kmer-per-seq	k-mers per sequence
--kmer-per-seq-scale	Scale k-mer per sequence based on sequence length as kmer-per-seq val + scale x seq len
--pick-n-sim-kmer	Add N similar k-mers to search
--result-direction	result is 0: query, 1: target centric

Full CLI Help Snapshot

```
usage: mmseqs kmersearch <i:queryDB> <i:kmerIndexDB> <o:prefilterDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
                                [aa:VTML80.out,nucl:nucleotide.out]
  --mask INT                   Mask sequences in prefilter stage with tantan: 0: w/o
                                low complexity masking, 1: with low complexity masking [0]
  --mask-prob FLOAT            Mask sequences is probability is above threshold
                                [0.900]
  --mask-lower-case INT        Lowercase letters will be excluded from k-mer search
                                0: include region, 1: exclude region [0]
  --mask-n-repeat INT          Repeat letters that occur > threshold in a row [0]
  --split-memory-limit BYTE    Set max memory per split. E.g. 800B, 5K, 10M, 1G.
                                Default (0) to all available system memory [0]
align:
  --cov-mode INT                0: coverage of query and target
                                1: coverage of target
                                2: coverage of query
                                3: target seq. length has to be at least x% of query
                                length
```

length 4: query seq. length has to be at least x% of target

5: short seq. needs to be at least x% of the other

seq. length [0]

-c FLOAT List matches above this fraction of aligned (covered) residues (see --cov-mode) [0.800]

kmermatcher:

--kmer-per-seq INT k-mers per sequence [0]

--kmer-per-seq-scale TWIN Scale k-mer per sequence based on sequence length as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]

--pick-n-sim-kmer INT Add N similar k-mers to search [1]

--result-direction INT result is 0: query, 1: target centric [1]

common:

--max-seq-len INT Maximum sequence length [65535]

--threads INT Number of CPU-cores used (all by default) [10]

--compressed INT Write compressed output [0]

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.61. lcaalign

Efficient gapped alignment for lca computation.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

usage: mmseqs lcaalign <i:queryDB> <i:targetDB> <i:resultDB> <o:alignmentDB> [options]

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)

Option	Purpose
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
-a	Add backtrace string (convert to alignments with mmseqs convertalis module)
--alignment-mode	How to compute the alignment:
--alignment-output-mode	How to compute the alignment:
--wrapped-scoring	Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start
-e	List matches below this E-value (range 0.0-inf)
--min-seq-id	List matches above this sequence identity (for clustering) (range 0.0-1.0)
--min-aln-len	Minimum alignment length (range 0-INT_MAX)
--seq-id-mode	0: alignment length 1: shorter, 2: longer sequence
--alt-ali	Show up to this many alternative alignments

Full CLI Help Snapshot

```
usage: mmseqs lcaalign <i:queryDB> <i:targetDB> <i:resultDB> <o:alignmentDB>
[options]
  By Milot Mirdita <milot@mirdita.de>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
(range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
align:
  -a BOOL                      Add backtrace string (convert to alignments with
mmseqs convertalis module) [0]
  --alignment-mode INT          How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id [0]
  --alignment-output-mode INT   How to compute the alignment:
```

0: automatic
 1: only score and end_pos
 2: also start_pos and cov
 3: also seq.id
 4: only ungapped alignment
 5: score only (output) cluster format [0]

--wrapped-scoring BOOL Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start [0]

-e DOUBLE List matches below this E-value (range 0.0-inf) [1.000E-03]

--min-seq-id FLOAT List matches above this sequence identity (for clustering) (range 0.0-1.0) [0.000]

--min-aln-len INT Minimum alignment length (range 0-INT_MAX) [0]

--seq-id-mode INT 0: alignment length 1: shorter, 2: longer sequence [0]

--alt-ali INT Show up to this many alternative alignments [0]

-c FLOAT List matches above this fraction of aligned (covered) residues (see --cov-mode) [0.000]

--cov-mode INT 0: coverage of query and target
 1: coverage of target
 2: coverage of query
 3: target seq. length has to be at least x% of query length
 4: query seq. length has to be at least x% of target length
 5: short seq. needs to be at least x% of the other seq. length [0]

--max-rejected INT Maximum rejected alignments before alignment calculation for a query is stopped [2147483647]

--max-accept INT Maximum accepted alignments before alignment calculation for a query is stopped [2147483647]

--score-bias FLOAT Score bias when computing SW alignment (in bits) [0.000]

--realign BOOL Compute more conservative, shorter alignments (scores and E-values not changed) [0]

--realign-score-bias FLOAT Additional bias when computing realignment [-0.200]

--realign-max-seqs INT Maximum number of results to return in realignment [2147483647]

--corr-score-weight FLOAT Weight of backtrace correlation score that is added to the alignment score [0.000]

--gap-open TWIN Gap open cost [aa:11,nucl:5]

--gap-extend TWIN Gap extension cost [aa:1,nucl:2]

--zdrop INT Maximal allowed difference between score values before alignment is truncated (nucleotide alignment only) [40]

profile:

--pca Pseudo count admixture strength []

--pcb Pseudo counts: Neff at half of maximum admixture (range 0.0-inf) []

```

common:
  --sub-mat TWIN          Substitution matrix file
                           [aa:blosom62.out,nucl:nucleotide.out]
  --max-seq-len INT      Maximum sequence length [65535]
  --db-load-mode INT      Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
  --threads INT          Number of CPU-cores used (all by default) [10]
  --compressed INT       Write compressed output [0]
  -v INT                 Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]

```

references:

- Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and sensitive taxonomic assignment to metagenomic contigs. *Bioinformatics*, btab184 (2021)

6.2.62. lca

Compute the lowest common ancestor.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

usage: mmseqs lca <i:targetDB> <i:resultDB> <o:taxaDB> [options]

Key Options

Option	Purpose
--lca-ranks	Add column with specified ranks (',' separated)
--blacklist	Comma separated list of ignored taxa in LCA computation
--tax-lineage	0: don't show, 1: add all lineage names, 2: add all lineage taxids
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs lca <i:targetDB> <i:resultDB> <o:taxaDB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: misc:
  --lca-ranks STR      Add column with specified ranks (',' separated) []
  --blacklist STR      Comma separated list of ignored taxa in LCA computation
[12908:unclassified sequences,28384:other sequences]
  --tax-lineage INT    0: don't show, 1: add all lineage names, 2: add all lineage
taxids [0]
common:
  --compressed INT    Write compressed output [0]
  --threads INT       Number of CPU-cores used (all by default) [10]
  -v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
  - Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and
sensitive taxonomic assignment to metagenomic contigs. Bioinformatics, btab184
(2021)
```

6.2.63. linclust

Fast, less sensitive clustering.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 2 upstream caller(s) and 8 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

Usage

```
usage: mmseqs linclust <i:sequenceDB> <o:clusterDB> <tmpDir> [options]
```

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--alph-size	Alphabet size (range 2-21)
--spaced-kmer-mode	0: use consecutive positions in k-mers; 1: use spaced k-mers
--spaced-kmer-pattern	User-specified spaced k-mer pattern
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probability is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occur > threshold in a row
-k	k-mer length (0: automatically set to optimum)
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory

Full CLI Help Snapshot

```
usage: mmseqs linclust <i:sequenceDB> <o:clusterDB> <tmpDir> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid
composition (range 0-1) [1]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid
composition (range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --alph-size TWIN             Alphabet size (range 2-21) [aa:21,nucl:5]
  --spaced-kmer-mode INT       0: use consecutive positions in k-mers; 1: use
spaced k-mers [0]
  --spaced-kmer-pattern STR     User-specified spaced k-mer pattern []
  --mask INT                   Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [0]
```

```

--mask-prob FLOAT           Mask sequences is probablity is above threshold
[0.900]
--mask-lower-case INT       Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT         Repeat letters that occure > threshold in a rwo
[0]
-k INT                       k-mer length (0: automatically set to optimum)
[0]
--split-memory-limit BYTE   Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
align:
-a BOOL                      Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]
--alignment-mode INT         How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment [2]
--alignment-output-mode INT  How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment
                                5: score only (output) cluster format [0]
--wrapped-scoring BOOL      Double the (nucleotide) query sequence during
the scoring process to allow wrapped diagonal scoring around end and start [0]
-e DOUBLE                   List matches below this E-value (range 0.0-inf)
[1.000E-03]
--min-seq-id FLOAT          List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.900]
--min-aln-len INT           Minimum alignment length (range 0-INT_MAX) [0]
--seq-id-mode INT           0: alignment length 1: shorter, 2: longer
sequence [0]
--alt-ali INT               Show up to this many alternative alignments [0]
-c FLOAT                    List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.800]
--cov-mode INT              0: coverage of query and target
                                1: coverage of target
                                2: coverage of query
                                3: target seq. length has to be at least x% of
query length
                                4: query seq. length has to be at least x% of
target length
                                5: short seq. needs to be at least x% of the
other seq. length [0]
--max-rejected INT          Maximum rejected alignments before alignment
calculation for a query is stopped [2147483647]

```

```

--max-accept INT           Maximum accepted alignments before alignment
calculation for a query is stopped [2147483647]
--score-bias FLOAT         Score bias when computing SW alignment (in
bits) [0.000]
--realign BOOL             Compute more conservative, shorter alignments
(scores and E-values not changed) [0]
--realign-score-bias FLOAT Additional bias when computing realignment
[-0.200]
--realign-max-seqs INT     Maximum number of results to return in
realignment [2147483647]
--corr-score-weight FLOAT  Weight of backtrace correlation score that is
added to the alignment score [0.000]
--gap-open TWIN            Gap open cost [aa:11,nucl:5]
--gap-extend TWIN          Gap extension cost [aa:1,nucl:2]
--zdrop INT                Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
clust:
--cluster-mode INT         0: Set-Cover (greedy)
                           1: Connected component (BLASTclust)
                           2,3: Greedy clustering by sequence length
(CDHIT) [0]
--max-iterations INT       Maximum depth of breadth first search in
connected component clustering [1000]
--similarity-type INT      Type of score used for clustering. 1: alignment
score 2: sequence identity [2]
kmermatcher:
--weights STR              Weights used for cluster prioritization []
--cluster-weight-threshold FLOAT Weight threshold used for cluster prioritization
[0.900]
--kmer-per-seq INT         k-mers per sequence [21]
--kmer-per-seq-scale TWIN  Scale k-mer per sequence based on sequence
length as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
--adjust-kmer-len BOOL     Adjust k-mer length based on specificity (only
for nucleotides) [0]
--hash-shift INT           Shift k-mer hash initialization [67]
--include-only-extendable BOOL Include only extendable [0]
--ignore-multi-kmer BOOL   Skip k-mers occurring multiple times (>=2) [0]
profile:
--pca                      Pseudo count admixture strength []
--pcb                      Pseudo counts: Neff at half of maximum
admixture (range 0.0-inf) []
misc:
--rescore-mode INT         Rescore diagonals with:
                           0: Hamming distance
                           1: local alignment (score only)
                           2: local alignment
                           3: global alignment
                           4: longest alignment fulfilling window quality
criterion [0]

```

```

common:
  --threads INT                Number of CPU-cores used (all by default) [10]
  --compressed INT             Write compressed output [0]
  -v INT                        Verbosity level: 0: quiet, 1: +errors, 2:
                               +warnings, 3: +info [3]
  --sub-mat TWIN                Substitution matrix file
                               [aa:blosum62.out,nucl:nucleotide.out]
  --max-seq-len INT            Maximum sequence length [65535]
  --db-load-mode INT           Database preload mode 0: auto, 1: fread, 2:
                               mmap, 3: mmap+touch [0]
  --remove-tmp-files BOOL      Delete temporary files [0]
  --force-reuse BOOL           Reuse tmp filse in tmp/latest folder ignoring
                               parameters and version changes [0]
  --mpi-runner STR             Use MPI on compute cluster with this MPI
                               command (e.g. "mpirun -np 42") []
expert:
  --filter-hits BOOL           Filter hits by seq.id. and coverage [0]
  --sort-results INT           Sort results: 0: no sorting, 1: sort by E-value
                               (Alignment) or seq.id. (Hamming) [0]

```

examples:

```

# Linear-time clustering of FASTA file
mmseqs linclust sequenceDB clusterDB tmp

```

```

                                --cov-mode
# Sequence                    0    1    2
# Q: MAVGTACRPA              60%  IGN  60%
# T: -AVGTAC---              60% 100%  IGN
# Cutoff -c 0.7              -    +    -
#                               -c 0.6    +    +    +

```

```

# Cluster nucleotide sequences
mmseqs easy-linclust nucl.fasta result tmp --kmer-per-seq-scale 0.3

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Steinegger M, Soding J: Clustering huge protein sequence sets in linear time. *Nature Communications*, 9(1), 2542 (2018)

6.2.64. linsearch

Fast, less sensitive homology search.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family balances sensitivity against runtime by controlling candidate generation, alignment workload, and split policy. The current dependency map records 2 upstream caller(s) and 9 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control over search sensitivity, filtering, and alignment behavior for DB-to-DB runs.

Dependency entry: Open in map; functional module: search_workflows.

Usage

usage: mmseqs linsearch <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probability is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occur > threshold in a row
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
-a	Add backtrace string (convert to alignments with mmseqs convertalis module)
--alignment-mode	How to compute the alignment:
--alignment-output-mode	How to compute the alignment:

Full CLI Help Snapshot

usage: mmseqs linsearch <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

```

options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
                                (range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
                                (range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
                                themselves (for clustering) [0]
  --seed-sub-mat TWIN           Substitution matrix file for k-mer generation
                                [aa:blosum62.out,nucl:nucleotide.out]
  --mask INT                    Mask sequences in prefilter stage with tantan: 0:
                                w/o low complexity masking, 1: with low complexity masking [0]
  --mask-prob FLOAT            Mask sequences is probability is above threshold
                                [0.900]
  --mask-lower-case INT        Lowercase letters will be excluded from k-mer
                                search 0: include region, 1: exclude region [0]
  --mask-n-repeat INT          Repeat letters that occure > threshold in a rwo
                                [0]
  --split-memory-limit BYTE     Set max memory per split. E.g. 800B, 5K, 10M, 1G.
                                Default (0) to all available system memory [0]
align:
  -a BOOL                      Add backtrace string (convert to alignments with
                                mmseqs convertalis module) [0]
  --alignment-mode INT          How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment [2]
  --alignment-output-mode INT   How to compute the alignment:
                                0: automatic
                                1: only score and end_pos
                                2: also start_pos and cov
                                3: also seq.id
                                4: only ungapped alignment
                                5: score only (output) cluster format [0]
  --wrapped-scoring BOOL       Double the (nucleotide) query sequence during the
                                scoring process to allow wrapped diagonal scoring around end and start [0]
  -e DOUBLE                    List matches below this E-value (range 0.0-inf)
                                [1.000E-03]
  --min-seq-id FLOAT           List matches above this sequence identity (for
                                clustering) (range 0.0-1.0) [0.000]
  --min-aln-len INT            Minimum alignment length (range 0-INT_MAX) [0]
  --seq-id-mode INT            0: alignment length 1: shorter, 2: longer sequence
                                [0]
  --alt-ali INT                Show up to this many alternative alignments [0]
  -c FLOAT                     List matches above this fraction of aligned
                                (covered) residues (see --cov-mode) [0.000]
  --cov-mode INT               0: coverage of query and target
                                1: coverage of target

```

2: coverage of query
 3: target seq. length has to be at least x% of
 query length
 4: query seq. length has to be at least x% of
 target length
 5: short seq. needs to be at least x% of the other
 seq. length [0]
 --max-rejected INT Maximum rejected alignments before alignment
 calculation for a query is stopped [2147483647]
 --max-accept INT Maximum accepted alignments before alignment
 calculation for a query is stopped [2147483647]
 --score-bias FLOAT Score bias when computing SW alignment (in bits)
 [0.000]
 --realign BOOL Compute more conservative, shorter alignments
 (scores and E-values not changed) [0]
 --realign-score-bias FLOAT Additional bias when computing realignment
 [-0.200]
 --realign-max-seqs INT Maximum number of results to return in realignment
 [2147483647]
 --corr-score-weight FLOAT Weight of backtrace correlation score that is
 added to the alignment score [0.000]
 --gap-open TWIN Gap open cost [aa:11,nucl:5]
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]
 --zdrop INT Maximal allowed difference between score values
 before alignment is truncated (nucleotide alignment only) [40]
 kmermatcher:
 --kmer-per-seq INT k-mers per sequence [21]
 --kmer-per-seq-scale TWIN Scale k-mer per sequence based on sequence length
 as kmer-per-seq val + scale x seqlen [aa:0.000,nucl:0.200]
 --pick-n-sim-kmer INT Add N similar k-mers to search [1]
 --result-direction INT result is 0: query, 1: target centric [1]
 profile:
 --pca Pseudo count admixture strength []
 --pcb Pseudo counts: Neff at half of maximum admixture
 (range 0.0-inf) []
 misc:
 --min-length INT Minimum codon number in open reading frames [30]
 --max-length INT Maximum codon number in open reading frames
 [32734]
 --max-gaps INT Maximum number of codons with gaps or unknown
 residues before an open reading frame is rejected [2147483647]
 --contig-start-mode INT Contig start can be 0: incomplete, 1: complete, 2:
 both [2]
 --contig-end-mode INT Contig end can be 0: incomplete, 1: complete, 2:
 both [2]
 --orf-start-mode INT Orf fragment can be 0: from start to stop, 1: from
 any to stop, 2: from last encountered start to stop (no start in the middle) [1]
 --forward-frames STR Comma-separated list of frames on the forward
 strand to be extracted [1,2,3]


```

--reverse-frames STR          Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT       1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL,
21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA
[1]
--translate INT              Translate ORF to amino acid [0]
--use-all-table-starts BOOL Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT              Numeric ids in index file are offset by this value
[0]
--search-type INT            Search type 0: auto 1: amino acid, 2: translated,
3: nucleotide, 4: translated nucleotide alignment [0]
common:
--sub-mat TWIN                Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--max-seq-len INT             Maximum sequence length [65535]
--db-load-mode INT            Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--threads INT                 Number of CPU-cores used (all by default) [10]
--compressed INT              Write compressed output [0]
-v INT                        Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--mpi-runner STR              Use MPI on compute cluster with this MPI command
(e.g. "mpirun -np 42") []
--force-reuse BOOL            Reuse tmp file in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL       Delete temporary files [0]
expert:
--create-lookup INT           Create database lookup file (can be very large)
[0]
--chain-alignments INT         Chain overlapping alignments [0]
--merge-query INT             Combine ORFs/split sequences to a single entry [1]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

```

6.2.65. lndb

Symlink a DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

```
usage: mmseqs lndb <i:srcDB> <o:dstDB> [options]
```

Key Options

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs lndb <i:srcDB> <o:dstDB> [options]
```

```
By Milot Mirdita <milot@mirdita.de>
```

```
options: common:
```

```
-v INT    Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]
```

```
references:
```

```
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)
```

6.2.66. majoritylca

Compute the lowest common ancestor using majority voting.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

```
usage: mmseqs majoritylca <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs majoritylca` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.67. map

Map nearly identical sequences.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family balances sensitivity against runtime by controlling candidate generation, alignment workload, and split policy. The current dependency map records 0 upstream caller(s) and 1 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control over search sensitivity, filtering, and alignment behavior for DB-to-DB runs.

Dependency entry: Open in map; functional module: `search_workflows`.

Usage

usage: `mmseqs map <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]`

Key Options

Option	Purpose
<code>--seed-sub-mat</code>	Substitution matrix file for k-mer generation
<code>-s</code>	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
<code>-k</code>	k-mer length (0: automatically set to optimum)
<code>--target-search-mode</code>	target search mode (0: regular k-mer, 1: similar k-mer)
<code>--k-score</code>	k-mer threshold for generating similar k-mer lists
<code>--alph-size</code>	Alphabet size (range 2-21)

Option	Purpose
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--split	Split input into N equally distributed chunks. 0: set the best split automatically
--split-mode	0: split target db; 1: split query db; 2: auto, depending on main memory
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)

Full CLI Help Snapshot

```
usage: mmseqs map <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]
  By Milot Mirdita <milot@mirdita.de> & Martin Steinegger
<martin.steinegger@snu.ac.kr>
options: prefilter:
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
[2.000]
  -k INT                       k-mer length (0: automatically set to optimum) [0]
  --target-search-mode INT     target search mode (0: regular k-mer, 1: similar k-
mer) [0]
  --k-score TWIN               k-mer threshold for generating similar k-mer lists
[seq:2147483647,prof:2147483647]
  --alph-size TWIN             Alphabet size (range 2-21) [aa:21,nucl:5]
  --max-seqs INT               Maximum results per query sequence allowed to pass
the prefilter (affects sensitivity) [300]
  --split INT                   Split input into N equally distributed chunks. 0:
set the best split automatically [0]
  --split-mode INT             0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
  --split-memory-limit BYTE     Set max memory per split. E.g. 800B, 5K, 10M, 1G.
Default (0) to all available system memory [0]
  --comp-bias-corr INT          Correct for locally biased amino acid composition
(range 0-1) [0]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]
  --diag-score BOOL            Use ungapped diagonal scoring during prefilter [1]
  --exact-kmer-matching INT     Extract only exact k-mers for matching (range 0-1)
[0]
```

--mask INT Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking [0]
 --mask-prob FLOAT Mask sequences is probability is above threshold [0.900]
 --mask-lower-case INT Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region [0]
 --mask-n-repeat INT Repeat letters that occur > threshold in a row [0]
 --min-ungapped-score INT Accept only matches with ungapped alignment score above threshold [15]
 --add-self-matches BOOL Artificially add entries of queries with themselves (for clustering) [0]
 --spaced-kmer-mode INT 0: use consecutive positions in k-mers; 1: use spaced k-mers [1]
 --spaced-kmer-pattern STR User-specified spaced k-mer pattern []
 --local-tmp STR Path where some of the temporary files will be created []
 align:
 -c FLOAT List matches above this fraction of aligned (covered) residues (see --cov-mode) [0.950]
 --cov-mode INT 0: coverage of query and target
 1: coverage of target
 2: coverage of query
 3: target seq. length has to be at least x% of query length
 4: query seq. length has to be at least x% of target length
 5: short seq. needs to be at least x% of the other seq. length [2]
 --wrapped-scoring BOOL Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start [0]
 -e DOUBLE List matches below this E-value (range 0.0-inf) [1.000E-03]
 -a BOOL Add backtrace string (convert to alignments with mmseqs convertalis module) [0]
 --min-seq-id FLOAT List matches above this sequence identity (for clustering) (range 0.0-1.0) [0.900]
 --min-aln-len INT Minimum alignment length (range 0-INT_MAX) [0]
 --seq-id-mode INT 0: alignment length 1: shorter, 2: longer sequence [0]
 profile:
 --pca Pseudo count admixture strength []
 --pcb Pseudo counts: Neff at half of maximum admixture (range 0.0-inf) []
 misc:
 --taxon-list STR Taxonomy ID, possibly multiple values separated by ',' []
 --rescore-mode INT Rescore diagonals with:
 0: Hamming distance
 1: local alignment (score only)

```

2: local alignment
3: global alignment
4: longest alignment fulfilling window quality

criterion [2]
--min-length INT           Minimum codon number in open reading frames [10]
--max-length INT           Maximum codon number in open reading frames [32734]
--max-gaps INT             Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT    Contig start can be 0: incomplete, 1: complete, 2:
both [2]
--contig-end-mode INT      Contig end can be 0: incomplete, 1: complete, 2:
both [2]
--orf-start-mode INT       Orf fragment can be 0: from start to stop, 1: from
any to stop, 2: from last encountered start to stop (no start in the middle) [1]
--forward-frames STR       Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR       Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT    1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL,
21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA
[1]
--translate INT            Translate ORF to amino acid [0]
--use-all-table-starts BOOL Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT            Numeric ids in index file are offset by this value
[0]
--start-sens FLOAT         Start sensitivity [4.000]
--sens-steps INT           Number of search steps performed from --start-sens
to -s [1]
common:
--sub-mat TWIN             Substitution matrix file
[aa:blosom62.out,nucl:nucleotide.out]
--max-seq-len INT          Maximum sequence length [65535]
--db-load-mode INT         Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--threads INT              Number of CPU-cores used (all by default) [10]
--compressed INT           Write compressed output [0]
-v INT                     Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--mpi-runner STR           Use MPI on compute cluster with this MPI command

```

```
(e.g. "mpirun -np 42") []
--force-reuse BOOL          Reuse tmp files in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL     Delete temporary files [0]
expert:
--filter-hits BOOL          Filter hits by seq.id. and coverage [0]
--sort-results INT          Sort results: 0: no sorting, 1: sort by E-value
                             (Alignment) or seq.id. (Hamming) [1]
--create-lookup INT         Create database lookup file (can be very large) [0]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)
- Steinegger M, Mirdita M, Soding J: Protein-level assembly increases protein sequence recovery from metagenomic samples manyfold. *Nature Methods*, 16(7), 603-606 (2019)

6.2.68. maskbygff

Mask out sequence regions in a sequence DB by features selected from a GFF3 file.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

```
usage: mmseqs maskbygff <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs maskbygff` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.69. masksequence

Soft mask sequence DB using tantan.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence_manipulation.

Usage

```
usage: mmseqs masksequence <i:sequenceDB> <o:sequenceDB> [options]
```

Key Options

Option	Purpose
--mask	Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking
--mask-prob	Mask sequences is probablity is above threshold
--mask-lower-case	Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region
--mask-n-repeat	Repeat letters that occure > threshold in a rwo
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs masksequence <i:sequenceDB> <o:sequenceDB> [options]
```

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: prefilter:

```
--mask INT          Mask sequences in prefilter stage with tantan: 0: w/o
low complexity masking, 1: with low complexity masking [1]
--mask-prob FLOAT    Mask sequences is probablity is above threshold [0.900]
--mask-lower-case INT Lowercase letters will be excluded from k-mer search 0:
include region, 1: exclude region [0]
--mask-n-repeat INT  Repeat letters that occure > threshold in a rwo [0]
```



```

common:
  --threads INT          Number of CPU-cores used (all by default) [10]
  --compressed INT       Write compressed output [0]
  -v INT                 Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

6.2.70. mergeclusters

Merge multiple cascaded clustering steps.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

Usage

```

usage: mmseqs mergeclusters <i:sequenceDB> <o:clusterDB> <i:clusterDB1> ...
<i:clusterDBn> [options]

```

Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```

usage: mmseqs mergeclusters <i:sequenceDB> <o:clusterDB> <i:clusterDB1> ...
<i:clusterDBn> [options]
  By Maria Hauser & Martin Steinegger <martin.steinegger@snu.ac.kr>
options: common:
  --threads INT          Number of CPU-cores used (all by default) [10]
  --compressed INT       Write compressed output [0]
  -v INT                 Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.71. mergedbs

Merge entries from multiple DBs.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 4 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

usage: mmseqs mergedbs <i:DB> <o:DB> <i:DB1> ... <i:DBn> [options]

Key Options

Option	Purpose
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--prefixes	Comma separated list of prefixes for each entry
--merge-stop-empty	Don't continue merging entries after an empty entry

Full CLI Help Snapshot

usage: mmseqs mergedbs <i:DB> <o:DB> <i:DB1> ... <i:DBn> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: common:

--compressed INT Write compressed output [0]
 -v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

expert:

--prefixes STR Comma separated list of prefixes for each entry []
 --merge-stop-empty BOOL Don't continue merging entries after an empty entry [0]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching

for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.72. **mergeresultsbyset**

Merge results from multiple ORFs back to their respective contig.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family aggregates sequence-level hits into set-level statistics and decisions. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: multi_hit.

Usage

usage: mmseqs mergeresultsbyset <i:setDB> <i:DB> <o:DB> [options]

Key Options

Option	Purpose
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: mmseqs mergeresultsbyset <i:setDB> <i:DB> <o:DB> [options]

By Ruoshi Zhang, Clovis Norroy & Milot Mirdita <milot@mirdita.de>

options: common:

--db-load-mode INT Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch [0]
--threads INT Number of CPU-cores used (all by default) [10]
--compressed INT Write compressed output [0]
-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.73. msa2profile

Convert a MSA DB to a profile DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

```
usage: mmseqs msa2profile <i:msaDB> <o:profileDB> [options]
```

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--match-mode	0: Columns that have a residue in the first sequence are kept, 1: columns that have a residue in --match-ratio of all sequences are kept
--match-ratio	Columns that have a residue in this ratio of all sequences are kept
--pseudo-cnt-mode	use 0: substitution-matrix or 1: context-specific pseudocounts
--pca	Pseudo count admixture strength
--pcb	Pseudo counts: Neff at half of maximum admixture (range 0.0-inf)
--wg	Use global sequence weighting for profile calculation
--filter-msa	Filter msa: 0: do not filter, 1: filter

Option	Purpose
--filter-min-enable	Only filter MSAs with more than N sequences, 0 always filters

Full CLI Help Snapshot

usage: mmseqs msa2profile <i:msaDB> <o:profileDB> [options]

By Milot Mirdita <milot@mirdita.de>

options: prefilter:

--comp-bias-corr INT Correct for locally biased amino acid composition
(range 0-1) [1]

--comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]

align:

--gap-open TWIN Gap open cost [aa:11,nucl:5]

--gap-extend TWIN Gap extension cost [aa:1,nucl:2]

profile:

--match-mode INT 0: Columns that have a residue in the first
sequence are kept, 1: columns that have a residue in --match-ratio of all
sequences are kept [0]

--match-ratio FLOAT Columns that have a residue in this ratio of all
sequences are kept [0.500]

--pseudo-cnt-mode INT use 0: substitution-matrix or 1: context-specific
pseudocounts [0]

--pca Pseudo count admixture strength []

--pcb Pseudo counts: Neff at half of maximum admixture

(range 0.0-inf) []

--wg BOOL Use global sequence weighting for profile

calculation [0]

--filter-msa INT Filter msa: 0: do not filter, 1: filter [1]

--filter-min-enable INT Only filter MSAs with more than N sequences, 0
always filters [0]

--cov FLOAT Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]

--qid STR Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]

Alternatively, can be a list of multiple

thresholds:

E.g.: 0.15,0.30,0.50 to defines filter buckets

of]0.15-0.30] and]0.30-0.50] [0.0]

--qsc FLOAT Reduce diversity of output MSAs using min. score
per aligned residue with query sequences [-50.0,100.0] [-20.000]

--max-seq-id FLOAT Reduce redundancy of output MSA using max. pairwise
sequence identity [0.0,1.0] [0.900]

--diff INT Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]

misc:

--msa-type INT MSA Type: 0: cA3M, 1: A3M, 2: FASTA [2]

```

common:
  --sub-mat TWIN          Substitution matrix file
                           [aa:blosum62.out,nucl:nucleotide.out]
  --threads INT           Number of CPU-cores used (all by default) [10]
  --compressed INT        Write compressed output [0]
  -v INT                  Verbosity level: 0: quiet, 1: +errors, 2:
                           +warnings, 3: +info [3]
expert:
  --skip-query B00L       Skip the query sequence [0]

examples:
# Convert globally aligned MSAs to profiles
# Defines columns as match columns if more than 50% of residues are not gaps
# Non-match columns are discarded
mmseqs msa2profile msaDB profileDB --match-mode 1 --match-ratio 0.5

# Assign match-columns through the first sequence
# Gaps in query sequence define non-match columns and are discarded
mmseqs msa2profile msaDB profileDB --match-mode 0

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
  for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
  (2017)
- Mirdita M, Steinegger M, Soding J: MMseqs2 desktop and local web server app
  for fast, interactive sequence searches. Bioinformatics, 35(16), 2856-2858 (2019)

```

6.2.74. msa2result

Convert a MSA DB to a profile DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

```
usage: mmseqs msa2result <i:msaDB> <o:seqDB> <o:profileDB> [options]
```

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--match-mode	0: Columns that have a residue in the first sequence are kept, 1: columns that have a residue in --match-ratio of all sequences are kept
--match-ratio	Columns that have a residue in this ratio of all sequences are kept
--pseudo-cnt-mode	use 0: substitution-matrix or 1: context-specific pseudocounts
--pca	Pseudo count admixture strength
--pcb	Pseudo counts: Neff at half of maximum admixture (range 0.0-inf)
--wg	Use global sequence weighting for profile calculation
--filter-msa	Filter msa: 0: do not filter, 1: filter
--filter-min-enable	Only filter MSAs with more than N sequences, 0 always filters

Full CLI Help Snapshot

```
usage: mmseqs msa2result <i:msaDB> <o:seqDB> <o:profileDB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
  (range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
  (range 0-1) [1.000]
align:
  --gap-open TWIN              Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN            Gap extension cost [aa:1,nucl:2]
profile:
  --match-mode INT             0: Columns that have a residue in the first
  sequence are kept, 1: columns that have a residue in --match-ratio of all
  sequences are kept [0]
  --match-ratio FLOAT          Columns that have a residue in this ratio of all
```

```

sequences are kept [0.500]
--pseudo-cnt-mode INT          use 0: substitution-matrix or 1: context-specific
pseudocounts [0]
--pca                          Pseudo count admixture strength []
--pcb                          Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
--wg BOOL                      Use global sequence weighting for profile
calculation [0]
--filter-msa INT               Filter msa: 0: do not filter, 1: filter [1]
--filter-min-enable INT        Only filter MSAs with more than N sequences, 0
always filters [0]
--cov FLOAT                   Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
--qid STR                     Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
                                Alternatively, can be a list of multiple
thresholds:
                                E.g.: 0.15,0.30,0.50 to defines filter buckets
of [0.15-0.30] and [0.30-0.50] [0.0]
--qsc FLOAT                   Reduce diversity of output MSAs using min. score
per aligned residue with query sequences [-50.0,100.0] [-20.000]
--max-seq-id FLOAT            Reduce redundancy of output MSA using max. pairwise
sequence identity [0.0,1.0] [0.900]
--diff INT                    Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
misc:
--msa-type INT                MSA Type: 0: cA3M, 1: A3M, 2: FASTA [2]
common:
--sub-mat TWIN                Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--threads INT                 Number of CPU-cores used (all by default) [10]
--compressed INT              Write compressed output [0]
-v INT                        Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
expert:
--skip-query BOOL             Skip the query sequence [0]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
- Mirdita M, Steinegger M, Soding J: MMseqs2 desktop and local web server app
for fast, interactive sequence searches. Bioinformatics, 35(16), 2856-2858 (2019)

```

6.2.75. multihitdb

Create sequence DB for multi hit searches.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family aggregates sequence-level hits into set-level statistics and decisions. The current dependency map records 0 upstream caller(s) and 8 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: multi_hit.

Usage

```
usage: mmseqs multihitdb <i:fastaFile1.gz|bz2> ... <i:fastaFileN.gz|bz2>
<o:setDB> <tmpDir> [options]
```

Key Options

Option	Purpose
--dbtype	Database type 0: auto, 1: amino acid 2: nucleotides
--shuffle	Shuffle input database
--createdb-mode	Createdb mode 0: copy data, 1: soft link data and write new index (works only with single line fasta/q)
--id-offset	Numeric ids in index file are offset by this value
--min-length	Minimum codon number in open reading frames
--max-length	Maximum codon number in open reading frames
--max-gaps	Maximum number of codons with gaps or unknown residues before an open reading frame is rejected
--contig-start-mode	Contig start can be 0: incomplete, 1: complete, 2: both
--contig-end-mode	Contig end can be 0: incomplete, 1: complete, 2: both
--orf-start-mode	Orf fragment can be 0: from start to stop, 1: from any to stop, 2: from last encountered start to stop (no start in the middle)
--forward-frames	Comma-separated list of frames on the forward strand to be extracted

Option	Purpose
--reverse-frames	Comma-separated list of frames on the reverse strand to be extracted

Full CLI Help Snapshot

```
usage: mmseqs multihitdb <i:fastaFile1[.gz|bz2]> ... <i:fastaFileN[.gz|bz2]>
<o:setDB> <tmpDir> [options]
  By Ruoshi Zhang, Clovis Norroy & Milot Mirdita <milot@mirdita.de>
options: misc:
  --dbtype INT                Database type 0: auto, 1: amino acid 2: nucleotides
[0]
  --shuffle BOOL              Shuffle input database [1]
  --createdb-mode INT         Createdb mode 0: copy data, 1: soft link data and
write new index (works only with single line fasta/q) [0]
  --id-offset INT             Numeric ids in index file are offset by this value
[0]
  --min-length INT            Minimum codon number in open reading frames [30]
  --max-length INT            Maximum codon number in open reading frames [32734]
  --max-gaps INT              Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
  --contig-start-mode INT     Contig start can be 0: incomplete, 1: complete, 2:
both [2]
  --contig-end-mode INT       Contig end can be 0: incomplete, 1: complete, 2:
both [2]
  --orf-start-mode INT        Orf fragment can be 0: from start to stop, 1: from
any to stop, 2: from last encountered start to stop (no start in the middle) [1]
  --forward-frames STR        Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
  --reverse-frames STR        Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
  --translation-table INT     1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL,
21) TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA
[1]
  --translate INT             Translate ORF to amino acid [0]
  --use-all-table-starts BOOL Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
  --add-orf-stop BOOL         Add stop codon '*' at complete start and end [0]
  --stat STR                  One of: linecount, mean, min, max, doolittle,
```

```

charges, seqlen, firstline []
--tsv BOOL                Return output in TSV format [0]
common:
--compressed INT          Write compressed output [0]
-v INT                    Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--threads INT             Number of CPU-cores used (all by default) [10]
expert:
--write-lookup INT        write .lookup file containing mapping from internal
id, fasta id and file number [1]
--create-lookup INT       Create database lookup file (can be very large) [0]

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

6.2.76. multihitsearch

Search with a grouped set of sequences against another grouped set.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family aggregates sequence-level hits into set-level statistics and decisions. The current dependency map records 0 upstream caller(s) and 4 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: multi_hit.

Usage

```
usage: mmseqs multihitsearch <i:querySetDB> <i:targetSetDB> <o:resultDB> <tmpDir>
[options]
```

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation

Option	Purpose
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists
--alph-size	Alphabet size (range 2-21)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--split	Split input into N equally distributed chunks. 0: set the best split automatically
--split-mode	0: split target db; 1: split query db; 2: auto, depending on main memory

Full CLI Help Snapshot

usage: mmseqs multihitsearch <i:querySetDB> <i:targetSetDB> <o:resultDB> <tmpDir> [options]

By Ruoshi Zhang, Clovis Norroy & Milot Mirdita <milot@mirdita.de>

options: prefilter:

--comp-bias-corr INT	Correct for locally biased amino acid composition (range 0-1) [1]
--comp-bias-corr-scale FLOAT	Correct for locally biased amino acid composition (range 0-1) [1.000]
--add-self-matches BOOL	Artificially add entries of queries with themselves (for clustering) [0]
--seed-sub-mat TWIN	Substitution matrix file for k-mer generation [aa:VTML80.out,nucl:nucleotide.out]
-s FLOAT	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive [5.700]
-k INT	k-mer length (0: automatically set to optimum) [0]
--target-search-mode INT	target search mode (0: regular k-mer, 1: similar k-mer) [0]
--k-score TWIN	k-mer threshold for generating similar k-mer lists [seq:2147483647,prof:2147483647]
--alph-size TWIN	Alphabet size (range 2-21) [aa:21,nucl:5]
--max-seqs INT	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) [300]
--split INT	Split input into N equally distributed chunks. 0: set the best split automatically [0]
--split-mode INT	0: split target db; 1: split query db; 2: auto,

depending on main memory [2]

--split-memory-limit BYTE Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory [0]

--diag-score BOOL Use ungapped diagonal scoring during prefilter [1]

--exact-kmer-matching INT Extract only exact k-mers for matching (range 0-1) [0]

--mask INT Mask sequences in prefilter stage with tantan: 0: w/o low complexity masking, 1: with low complexity masking [1]

--mask-prob FLOAT Mask sequences is probablity is above threshold [0.900]

--mask-lower-case INT Lowercase letters will be excluded from k-mer search 0: include region, 1: exclude region [0]

--mask-n-repeat INT Repeat letters that occure > threshold in a two [0]

--min-ungapped-score INT Accept only matches with ungapped alignment score above threshold [15]

--spaced-kmer-mode INT 0: use consecutive positions in k-mers; 1: use spaced k-mers [1]

--spaced-kmer-pattern STR User-specified spaced k-mer pattern []

--local-tmp STR Path where some of the temporary files will be created []

--disk-space-limit BYTE Set max disk space to use for reverse profile searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the temp folder [0]

align:

-a BOOL Add backtrace string (convert to alignments with mmseqs convertalis module) [0]

--alignment-mode INT How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment [2]

--alignment-output-mode INT How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment
5: score only (output) cluster format [0]

--wrapped-scoring BOOL Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start [0]

-e DOUBLE List matches below this E-value (range 0.0-inf) [1.000E+02]

--min-seq-id FLOAT List matches above this sequence identity (for clustering) (range 0.0-1.0) [0.000]

--min-aln-len INT Minimum alignment length (range 0-INT_MAX) [0]

--seq-id-mode INT 0: alignment length 1: shorter, 2: longer

```

sequence [0]
  --alt-ali INT          Show up to this many alternative alignments [0]
  -c FLOAT              List matches above this fraction of aligned
                        (covered) residues (see --cov-mode) [0.000]
  --cov-mode INT         0: coverage of query and target
                        1: coverage of target
                        2: coverage of query
                        3: target seq. length has to be at least x% of
query length
                        4: query seq. length has to be at least x% of
target length
                        5: short seq. needs to be at least x% of the
other seq. length [0]
  --max-rejected INT     Maximum rejected alignments before alignment
calculation for a query is stopped [2147483647]
  --max-accept INT       Maximum accepted alignments before alignment
calculation for a query is stopped [2147483647]
  --score-bias FLOAT     Score bias when computing SW alignment (in
bits) [0.000]
  --realign BOOL         Compute more conservative, shorter alignments
(scores and E-values not changed) [0]
  --realign-score-bias FLOAT Additional bias when computing realignment
[-0.200]
  --realign-max-seqs INT Maximum number of results to return in
realignment [2147483647]
  --corr-score-weight FLOAT Weight of backtrace correlation score that is
added to the alignment score [0.000]
  --gap-open TWIN        Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN      Gap extension cost [aa:1,nucl:2]
  --zdrop INT            Maximal allowed difference between score values
before alignment is truncated (nucleotide alignment only) [40]
  --exhaustive-search-filter INT Filter result during search: 0: do not filter,
1: filter [0]
profile:
  --pca                  Pseudo count admixture strength []
  --pcb                  Pseudo counts: Neff at half of maximum
admixture (range 0.0-inf) []
  --mask-profile INT     Mask query sequence of profile using tantan
[0,1] [1]
  --e-profile DOUBLE     Include sequences matches with < E-value thr.
into the profile (>=0.0) [1.000E-03]
  --wg BOOL             Use global sequence weighting for profile
calculation [0]
  --filter-msa INT       Filter msa: 0: do not filter, 1: filter [1]
  --filter-min-enable INT Only filter MSAs with more than N sequences, 0
always filters [0]
  --max-seq-id FLOAT     Reduce redundancy of output MSA using max.
pairwise sequence identity [0.0,1.0] [0.900]
  --qid STR              Reduce diversity of output MSAs using min.seq.

```

identity with query sequences [0.0,1.0]
 Alternatively, can be a list of multiple thresholds:
 E.g.: 0.15,0.30,0.50 to defines filter buckets of]0.15-0.30] and]0.30-0.50] [0.0]

--qsc FLOAT Reduce diversity of output MSAs using min. score per aligned residue with query sequences [-50.0,100.0] [-20.000]

--cov FLOAT Filter output MSAs using min. fraction of query residues covered by matched sequences [0.0,1.0] [0.000]

--diff INT Filter MSAs by selecting most diverse set of sequences, keeping at least this many seqs in each MSA block of length 50 [1000]

--pseudo-cnt-mode INT use 0: substitution-matrix or 1: context-specific pseudocounts [0]

--profile-output-mode INT Profile output mode: 0: binary log-odds 1: human-readable frequencies [0]

--num-iterations INT Number of iterative profile search iterations [1]

--exhaustive-search BOOL For bigger profile DB, run iteratively the search by greedily swapping the search results [0]

--lca-search BOOL Efficient search for LCA candidates [0]

misc:

--taxon-list STR Taxonomy ID, possibly multiple values separated by ',' []

--prefilter-mode INT prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped [0]

--rescore-mode INT Rescore diagonals with:
 0: Hamming distance
 1: local alignment (score only)
 2: local alignment
 3: global alignment
 4: longest alignment fulfilling window quality

criterion [0]

--allow-deletion BOOL Allow deletions in a MSA [0]

--min-length INT Minimum codon number in open reading frames [30]

--max-length INT Maximum codon number in open reading frames [32734]

--max-gaps INT Maximum number of codons with gaps or unknown residues before an open reading frame is rejected [2147483647]

--contig-start-mode INT Contig start can be 0: incomplete, 1: complete, 2: both [2]

--contig-end-mode INT Contig end can be 0: incomplete, 1: complete, 2: both [2]

--orf-start-mode INT Orf fragment can be 0: from start to stop, 1: from any to stop, 2: from last encountered start to stop (no start in the middle) [1]

--forward-frames STR Comma-separated list of frames on the forward strand to be extracted [1,2,3]

--reverse-frames STR Comma-separated list of frames on the reverse

```

strand to be extracted [1,2,3]
  --translation-table INT          1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
                                   9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
                                   15) BLEPHARISMA, 16)
CHLOROPHYCEAN_MITOCHONDRIAL, 21) TREMATODE_MITOCHONDRIAL, 22)
SCENEDESMUS_MITOCHONDRIAL
                                   23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
                                   29) MESODINIUM, 30) PERTRICH, 31)

BLASTOCRITHIDIA [1]
  --translate INT                  Translate ORF to amino acid [0]
  --use-all-table-starts BOOL      Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
  --id-offset INT                  Numeric ids in index file are offset by this
value [0]
  --sequence-overlap INT            Overlap between sequences [0]
  --sequence-split-mode INT         Sequence split mode 0: copy data, 1: soft link
data and write new index, [1]
  --headers-split-mode INT          Header split mode: 0: split position, 1:
original header [0]
  --search-type INT                 Search type 0: auto 1: amino acid, 2:
translated, 3: nucleotide, 4: translated nucleotide alignment [0]
  --start-sens FLOAT                Start sensitivity [4.000]
  --sens-steps INT                  Number of search steps performed from --start-
sens to -s [1]
  --translation-mode INT             Translation AA seq from nucleotide by 0: ORFs,
1: full reading frames [0]
  --simple-best-hit BOOL              Update the p-value by a single best hit, or by
best and second best hits [1]
common:
  --sub-mat TWIN                     Substitution matrix file
[aa:blosom62.out,nucl:nucleotide.out]
  --max-seq-len INT                  Maximum sequence length [65535]
  --db-load-mode INT                 Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
  --threads INT                      Number of CPU-cores used (all by default) [10]
  --compressed INT                   Write compressed output [0]
  -v INT                             Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
  --gpu INT                           Use GPU (CUDA) if possible [0]
  --gpu-server INT                     Use GPU server [0]
  --gpu-server-wait-timeout INT        Wait for GPU server for 0: don't wait -1: no
wait limit: >0 this many seconds [600]
  --mpi-runner STR                     Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []

```



```

--force-reuse BOOL          Reuse tmp file in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL    Delete temporary files [0]
expert:
--filter-hits BOOL         Filter hits by seq.id. and coverage [0]
--sort-results INT         Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--create-lookup INT        Create database lookup file (can be very large)
[0]
--chain-alignments INT     Chain overlapping alignments [0]
--merge-query INT          Combine ORFs/split sequences to a single entry
[1]
--strand INT               Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.77. mvdb

Move a DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 5 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

```
usage: mmseqs mvdb <i:srcDB> <o:dstDB> [options]
```

Key Options

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs mvdb <i:srcDB> <o:dstDB> [options]
```

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: common:

```
-v INT  Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

6.2.78. nrtotaxmapping

Create taxonomy mapping for NR database.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

```
usage: mmseqs nrtotaxmapping <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs nrtotaxmapping` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.79. offsetalignment

Offset alignment by ORF start position.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

Usage

```
usage: mmseqs offsetalignment <i:queryDB> <i:queryOrfDB> <i:targetDB>
<i:targetOrfDB> <i:alnDB> <o:alnDB> [options]
```

Key Options

Option	Purpose
--search-type	Search type 0: auto 1: amino acid, 2: translated, 3: nucleotide, 4: translated nucleotide alignment
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--chain-alignments	Chain overlapping alignments
--merge-query	Combine ORFs/split sequences to a single entry

Full CLI Help Snapshot

```
usage: mmseqs offsetalignment <i:queryDB> <i:queryOrfDB> <i:targetDB>
<i:targetOrfDB> <i:alnDB> <o:alnDB> [options]
```

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: misc:

```
--search-type INT      Search type 0: auto 1: amino acid, 2: translated, 3:
nucleotide, 4: translated nucleotide alignment [0]
```

common:

```
--threads INT          Number of CPU-cores used (all by default) [10]
```

```
--compressed INT       Write compressed output [0]
```

```
--db-load-mode INT     Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]
```

```
-v INT                 Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
```

expert:

```
--chain-alignments INT Chain overlapping alignments [0]
```

```
--merge-query INT      Combine ORFs/split sequences to a single entry [1]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.80. orftocontig

Write ORF locations in alignment format.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence_manipulation.

Usage

```
usage: mmseqs orftocontig <i:contigsSequenceDB> <i:extractedOrfsHeadersDB>
<o:orfsAlignedToContigDB> [options]
```

Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs orftocontig <i:contigsSequenceDB> <i:extractedOrfsHeadersDB>
<o:orfsAlignedToContigDB> [options]
  By Eli Levy Karin <eli.levy.karin@gmail.com>
options: common:
  --threads INT      Number of CPU-cores used (all by default) [10]
  --compressed INT   Write compressed output [0]
  -v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
    for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
    (2017)
```

6.2.81. pairaln

Pair sequences to match best protein A and B from a species.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

```
usage: mmseqs pairaln <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs pairaln` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.82. pickconsensusrep

Select new representatives for each cluster based on consensus.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family controls graph construction and cluster assignment behavior, so early filter decisions strongly affect downstream structure. The current dependency map records 0 upstream caller(s) and 7 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when constructing, refining, or updating cluster assignments while preserving explicit coverage/identity criteria.

Dependency entry: Open in map; functional module: clustering.

Usage

```
usage: mmseqs pickconsensusrep <inputDB(s)> <outputDB> [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs pickconsensusrep` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.83. prefilter

Double consecutive diagonal k-mer search.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family is the main acceleration gate that prunes candidate pairs before expensive alignment. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command in custom pipelines that must expose candidate-generation behavior before alignment stages.

Dependency entry: Open in map; functional module: prefiltering.

Usage

```
usage: mmseqs prefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]
```

Key Options

Option	Purpose
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists
--alph-size	Alphabet size (range 2-21)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)

Option	Purpose
--split	Split input into N equally distributed chunks. 0: set the best split automatically
--split-mode	0: split target db; 1: split query db; 2: auto, depending on main memory
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
--comp-bias-corr	Correct for locally biased amino acid compo- sition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid compo- sition (range 0-1)

Full CLI Help Snapshot

```
usage: mmseqs prefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr> & Maria Hauser
options: prefilter:
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
                                [aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
                                [4.000]
  -k INT                       k-mer length (0: automatically set to optimum) [0]
  --target-search-mode INT     target search mode (0: regular k-mer, 1: similar k-
                                mer) [0]
  --k-score TWIN              k-mer threshold for generating similar k-mer lists
                                [seq:2147483647,prof:2147483647]
  --alph-size TWIN            Alphabet size (range 2-21) [aa:21,nucl:5]
  --max-seqs INT              Maximum results per query sequence allowed to pass
                                the prefilter (affects sensitivity) [300]
  --split INT                  Split input into N equally distributed chunks. 0:
                                set the best split automatically [0]
  --split-mode INT            0: split target db; 1: split query db; 2: auto,
                                depending on main memory [2]
  --split-memory-limit BYTE   Set max memory per split. E.g. 800B, 5K, 10M, 1G.
                                Default (0) to all available system memory [0]
  --comp-bias-corr INT        Correct for locally biased amino acid composition
                                (range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
                                (range 0-1) [1.000]
  --diag-score BOOL           Use ungapped diagonal scoring during prefilter [1]
  --exact-kmer-matching INT   Extract only exact k-mers for matching (range 0-1)
                                [0]
  --mask INT                  Mask sequences in prefilter stage with tantan: 0:
                                w/o low complexity masking, 1: with low complexity masking [1]
  --mask-prob FLOAT           Mask sequences is probability is above threshold
```

```

[0.900]
--mask-lower-case INT      Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT        Repeat letters that occur > threshold in a row [0]
--min-ungapped-score INT    Accept only matches with ungapped alignment score
above threshold [15]
--add-self-matches BOOL    Artificially add entries of queries with themselves
(for clustering) [0]
--spaced-kmer-mode INT      0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
--spaced-kmer-pattern STR   User-specified spaced k-mer pattern []
--local-tmp STR             Path where some of the temporary files will be
created []
align:
-c FLOAT                   List matches above this fraction of aligned
(covered) residues (see --cov-mode) [0.000]
--cov-mode INT              0: coverage of query and target
                           1: coverage of target
                           2: coverage of query
                           3: target seq. length has to be at least x% of
query length
                           4: query seq. length has to be at least x% of
target length
                           5: short seq. needs to be at least x% of the other
seq. length [0]
profile:
--pca                      Pseudo count admixture strength []
--pcb                      Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
misc:
--taxon-list STR           Taxonomy ID, possibly multiple values separated by
',' []
common:
--sub-mat TWIN             Substitution matrix file
[aa:blosum62.out,nuc:nucleotide.out]
--max-seq-len INT          Maximum sequence length [65535]
--db-load-mode INT         Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--threads INT              Number of CPU-cores used (all by default) [10]
--compressed INT           Write compressed output [0]
-v INT                    Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

```


6.2.84. prefixid

For each entry in a DB prepend the entry key to the entry itself.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

```
usage: mmseqs prefixid <i:DB> <o:DB> [options]
```

Key Options

Option	Purpose
--prefix	Use this prefix for all entries
--mapping-file	Specify a file that translates the keys of a DB to new keys, TSV format
--tsv	Return output in TSV format
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs prefixid <i:DB> <o:DB> [options]
```

By Milot Mirdita <milot@mirdita.de>

options: misc:

```
--prefix STR      Use this prefix for all entries []
--mapping-file STR Specify a file that translates the keys of a DB to new
keys, TSV format []
--tsv BOOL        Return output in TSV format [0]
```

common:

```
--threads INT     Number of CPU-cores used (all by default) [10]
--compressed INT  Write compressed output [0]
-v INT            Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.85. profile2consensus

Extract consensus sequence DB from a profile DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

```
usage: mmseqs profile2consensus <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs profile2consensus` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.86. profile2neff

Convert a profile DB to a tab-separated list of Neff scores.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

```
usage: mmseqs profile2neff <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs profile2neff` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.87. profile2pssm

Convert a profile DB to a tab-separated PSSM file.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

```
usage: mmseqs profile2pssm <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs profile2pssm` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.88. **profile2repseq**

Extract representative sequence DB from a profile DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

```
usage: mmseqs profile2repseq <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs profile2repseq` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.89. **proteinaln2nucl**

Transform protein alignments to nucleotide alignments.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

Usage

```
usage: mmseqs proteinaln2nucl <i:nuclQueryDB> <i:nuclTargetDB> <i:aaQueryDB>
<i:aaTargetDB> <i:alnDB> <o:alnDB> [options]
```

Key Options

Option	Purpose
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--sub-mat	Substitution matrix file
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs proteinaln2nucl <i:nuclQueryDB> <i:nuclTargetDB> <i:aaQueryDB>
<i:aaTargetDB> <i:alnDB> <o:alnDB> [options]
```

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: align:

```
--gap-open TWIN    Gap open cost [aa:11,nucl:5]
--gap-extend TWIN  Gap extension cost [aa:1,nucl:2]
```

common:

```
--sub-mat TWIN      Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--threads INT        Number of CPU-cores used (all by default) [10]
--compressed INT     Write compressed output [0]
-v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.90. rbh

Reciprocal best hit search.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family balances sensitivity against runtime by controlling candidate generation, alignment workload, and split policy. The current dependency map records 1 upstream caller(s) and 6 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control over search sensitivity, filtering, and alignment behavior for DB-to-DB runs.

Dependency entry: Open in map; functional module: search_workflows.

Usage

usage: mmseqs rbh <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists
--alph-size	Alphabet size (range 2-21)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--split	Split input into N equally distributed chunks. 0: set the best split automatically
--split-mode	0: split target db; 1: split query db; 2: auto, depending on main memory

Full CLI Help Snapshot

usage: mmseqs rbh <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]

By Eli Levy Karin

options: prefilter:

--comp-bias-corr INT	Correct for locally biased amino acid composition (range 0-1) [0]
--comp-bias-corr-scale FLOAT	Correct for locally biased amino acid composition (range 0-1) [1.000]

```

--add-self-matches BOOL          Artificially add entries of queries with
themselves (for clustering) [0]
--seed-sub-mat TWIN              Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
-s FLOAT                          Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [4.000]
-k INT                            k-mer length (0: automatically set to optimum)
[0]
--target-search-mode INT          target search mode (0: regular k-mer, 1:
similar k-mer) [0]
--k-score TWIN                   k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
--alph-size TWIN                 Alphabet size (range 2-21) [aa:21,nucl:5]
--max-seqs INT                   Maximum results per query sequence allowed to
pass the prefilter (affects sensitivity) [300]
--split INT                      Split input into N equally distributed chunks.
0: set the best split automatically [0]
--split-mode INT                 0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
--split-memory-limit BYTE        Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
--diag-score BOOL               Use ungapped diagonal scoring during prefilter
[1]
--exact-kmer-matching INT        Extract only exact k-mers for matching (range
0-1) [0]
--mask INT                       Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [0]
--mask-prob FLOAT               Mask sequences is probablity is above threshold
[0.900]
--mask-lower-case INT            Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT              Repeat letters that occure > threshold in a rwo
[0]
--min-ungapped-score INT         Accept only matches with ungapped alignment
score above threshold [15]
--spaced-kmer-mode INT           0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
--spaced-kmer-pattern STR        User-specified spaced k-mer pattern []
--local-tmp STR                  Path where some of the temporary files will be
created []
--disk-space-limit BYTE          Set max disk space to use for reverse profile
searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the
temp folder [0]
align:
-a BOOL                          Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]
--alignment-mode INT             How to compute the alignment:
0: automatic
1: only score and end_pos

```

2: also start_pos and cov
 3: also seq.id
 4: only ungapped alignment [3]
 --alignment-output-mode INT How to compute the alignment:
 0: automatic
 1: only score and end_pos
 2: also start_pos and cov
 3: also seq.id
 4: only ungapped alignment
 5: score only (output) cluster format [0]
 --wrapped-scoring BOOL Double the (nucleotide) query sequence during
 the scoring process to allow wrapped diagonal scoring around end and start [0]
 -e DOUBLE List matches below this E-value (range 0.0-inf)
 [1.000E-03]
 --min-seq-id FLOAT List matches above this sequence identity (for
 clustering) (range 0.0-1.0) [0.000]
 --min-aln-len INT Minimum alignment length (range 0-INT_MAX) [0]
 --seq-id-mode INT 0: alignment length 1: shorter, 2: longer
 sequence [0]
 --alt-ali INT Show up to this many alternative alignments [0]
 -c FLOAT List matches above this fraction of aligned
 (covered) residues (see --cov-mode) [0.000]
 --cov-mode INT 0: coverage of query and target
 1: coverage of target
 2: coverage of query
 3: target seq. length has to be at least x% of
 query length
 4: query seq. length has to be at least x% of
 target length
 5: short seq. needs to be at least x% of the
 other seq. length [0]
 --max-rejected INT Maximum rejected alignments before alignment
 calculation for a query is stopped [2147483647]
 --max-accept INT Maximum accepted alignments before alignment
 calculation for a query is stopped [2147483647]
 --score-bias FLOAT Score bias when computing SW alignment (in
 bits) [0.000]
 --realign BOOL Compute more conservative, shorter alignments
 (scores and E-values not changed) [0]
 --realign-score-bias FLOAT Additional bias when computing realignment
 [-0.200]
 --realign-max-seqs INT Maximum number of results to return in
 realignment [2147483647]
 --corr-score-weight FLOAT Weight of backtrace correlation score that is
 added to the alignment score [0.000]
 --gap-open TWIN Gap open cost [aa:11,nucl:5]
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]
 --zdrop INT Maximal allowed difference between score values
 before alignment is truncated (nucleotide alignment only) [40]


```

--exhaustive-search-filter INT  Filter result during search: 0: do not filter,
1: filter [0]
profile:
--pca                          Pseudo count admixture strength []
--pcb                          Pseudo counts: Neff at half of maximum
admixture (range 0.0-inf) []
--mask-profile INT             Mask query sequence of profile using tantan
[0,1] [1]
--e-profile DOUBLE             Include sequences matches with < E-value thr.
into the profile (>=0.0) [1.000E-03]
--wg B00L                     Use global sequence weighting for profile
calculation [0]
--filter-msa INT               Filter msa: 0: do not filter, 1: filter [1]
--filter-min-enable INT        Only filter MSAs with more than N sequences, 0
always filters [0]
--max-seq-id FLOAT             Reduce redundancy of output MSA using max.
pairwise sequence identity [0.0,1.0] [0.900]
--qid STR                      Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
                                Alternatively, can be a list of multiple
thresholds:
                                E.g.: 0.15,0.30,0.50 to defines filter buckets
of [0.15-0.30] and [0.30-0.50] [0.0]
--qsc FLOAT                   Reduce diversity of output MSAs using min.
score per aligned residue with query sequences [-50.0,100.0] [-20.000]
--cov FLOAT                   Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
--diff INT                    Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
--pseudo-cnt-mode INT          use 0: substitution-matrix or 1: context-
specific pseudocounts [0]
--profile-output-mode INT      Profile output mode: 0: binary log-odds 1:
human-readable frequencies [0]
--num-iterations INT           Number of iterative profile search iterations
[1]
--exhaustive-search B00L       For bigger profile DB, run iteratively the
search by greedily swapping the search results [0]
--lca-search B00L              Efficient search for LCA candidates [0]
misc:
--taxon-list STR               Taxonomy ID, possibly multiple values separated
by ',' []
--prefilter-mode INT           prefilter mode: 0: kmer/ungapped 1: ungapped,
2: nofilter, 3: ungapped&gapped [0]
--rescore-mode INT             Rescore diagonals with:
                                0: Hamming distance
                                1: local alignment (score only)
                                2: local alignment
                                3: global alignment
                                4: longest alignment fulfilling window quality

```

```

criterion [0]
--allow-deletion BOOL          Allow deletions in a MSA [0]
--min-length INT               Minimum codon number in open reading frames
[10]
--max-length INT               Maximum codon number in open reading frames
[32734]
--max-gaps INT                 Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT        Contig start can be 0: incomplete, 1: complete,
2: both [2]
--contig-end-mode INT          Contig end can be 0: incomplete, 1: complete,
2: both [2]
--orf-start-mode INT           Orf fragment can be 0: from start to stop, 1:
from any to stop, 2: from last encountered start to stop (no start in the middle)
[1]
--forward-frames STR           Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR           Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT         1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16)
CHLOROPHYCEAN_MITOCHONDRIAL, 21) TREMATODE_MITOCHONDRIAL, 22)
SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31)
BLASTOCRITHIDIA [1]
--translate INT                Translate ORF to amino acid [0]
--use-all-table-starts BOOL    Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT                Numeric ids in index file are offset by this
value [0]
--sequence-overlap INT          Overlap between sequences [0]
--sequence-split-mode INT       Sequence split mode 0: copy data, 1: soft link
data and write new index, [1]
--headers-split-mode INT        Header split mode: 0: split position, 1:
original header [0]
--search-type INT               Search type 0: auto 1: amino acid, 2:
translated, 3: nucleotide, 4: translated nucleotide alignment [0]
--start-sens FLOAT              Start sensitivity [4.000]
--sens-steps INT                Number of search steps performed from --start-
sens to -s [1]
--translation-mode INT          Translation AA seq from nucleotide by 0: ORFs,
1: full reading frames [0]

```

```

common:
  --sub-mat TWIN                Substitution matrix file
[aa:blosum62.out,nuc:nucleotide.out]
  --max-seq-len INT             Maximum sequence length [65535]
  --db-load-mode INT            Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
  --threads INT                 Number of CPU-cores used (all by default) [10]
  --compressed INT              Write compressed output [0]
  -v INT                         Verboosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
  --gpu INT                      Use GPU (CUDA) if possible [0]
  --gpu-server INT              Use GPU server [0]
  --gpu-server-wait-timeout INT  Wait for GPU server for 0: don't wait -1: no
wait limit: >0 this many seconds [600]
  --mpi-runner STR              Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
  --force-reuse BOOL             Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
  --remove-tmp-files BOOL        Delete temporary files [0]
expert:
  --filter-hits BOOL             Filter hits by seq.id. and coverage [0]
  --sort-results INT             Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
  --create-lookup INT            Create database lookup file (can be very large)
[0]
  --chain-alignments INT          Chain overlapping alignments [0]
  --merge-query INT              Combine ORFs/split sequences to a single entry
[1]
  --strand INT                   Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

```

6.2.91. recoverlongestorf

Recover longest ORF for taxonomy annotation after elimination.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence_manipulation.

Usage

usage: mmseqs recoverlongestorf <i:orfDB> <i:resultDB> <o:tsvFile> [options]

Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: mmseqs recoverlongestorf <i:orfDB> <i:resultDB> <o:tsvFile> [options]

By Sung-eun Jang

options: common:

--threads INT Number of CPU-cores used (all by default) [10]
-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.92. renamedbkeys

Create a new DB with original keys renamed.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

usage: mmseqs renamedbkeys <i:idMapFile|stdin> <i:DB> <o:DB> [options]

Key Options

Option	Purpose
--subdb-mode	Subdb mode 0: copy data 1: soft link data and write index
--threads	Number of CPU-cores used (all by default)

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs renamedbkeys <i:idMapFile|stdin> <i:DB> <o:DB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: misc:
  --subdb-mode INT    Subdb mode 0: copy data 1: soft link data and write index [0]
common:
  --threads INT       Number of CPU-cores used (all by default) [10]
  -v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

6.2.93. rescorediagonal

Compute sequence identity for diagonal.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 5 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

Usage

```
usage: mmseqs rescorediagonal <i:queryDB> <i:targetDB> <i:prefilterDB> <o:resultDB>
[options]
```

Key Options

Option	Purpose
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--wrapped-scoring	Double the (nucleotide) query sequence during the scoring process to allow wrapped diagonal scoring around end and start

Option	Purpose
-e	List matches below this E-value (range 0.0-inf)
-c	List matches above this fraction of aligned (covered) residues (see --cov-mode)
-a	Add backtrace string (convert to alignments with mmseqs convertalis module)
--cov-mode	0: coverage of query and target
--min-seq-id	List matches above this sequence identity (for clustering) (range 0.0-1.0)
--min-aln-len	Minimum alignment length (range 0-INT_MAX)
--seq-id-mode	0: alignment length 1: shorter, 2: longer sequence
--rescore-mode	Rescore diagonals with:
--sub-mat	Substitution matrix file
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch

Full CLI Help Snapshot

```
usage: mmseqs rescorediagonal <i:queryDB> <i:targetDB> <i:prefilterDB>
<o:resultDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --add-self-matches BOOL  Artificially add entries of queries with themselves
(for clustering) [0]
align:
  --wrapped-scoring BOOL   Double the (nucleotide) query sequence during the
scoring process to allow wrapped diagonal scoring around end and start [0]
  -e DOUBLE                List matches below this E-value (range 0.0-inf)
[1.000E-03]
  -c FLOAT                 List matches above this fraction of aligned (covered)
residues (see --cov-mode) [0.000]
  -a BOOL                  Add backtrace string (convert to alignments with mmseqs
convertalis module) [0]
  --cov-mode INT           0: coverage of query and target
                           1: coverage of target
                           2: coverage of query
                           3: target seq. length has to be at least x% of query
length
                           4: query seq. length has to be at least x% of target
length
                           5: short seq. needs to be at least x% of the other seq.
```

```

length [0]
  --min-seq-id FLOAT      List matches above this sequence identity (for
clustering) (range 0.0-1.0) [0.000]
  --min-aln-len INT       Minimum alignment length (range 0-INT_MAX) [0]
  --seq-id-mode INT        0: alignment length 1: shorter, 2: longer sequence [0]
misc:
  --rescore-mode INT       Rescore diagonals with:
                           0: Hamming distance
                           1: local alignment (score only)
                           2: local alignment
                           3: global alignment
                           4: longest alignment fulfilling window quality

criterion [0]
common:
  --sub-mat TWIN           Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
  --db-load-mode INT       Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]
  --threads INT            Number of CPU-cores used (all by default) [10]
  --compressed INT         Write compressed output [0]
  -v INT                   Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
expert:
  --filter-hits BOOL        Filter hits by seq.id. and coverage [0]
  --sort-results INT        Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

```

6.2.94. result2dnamsa

Compute MSA DB with out insertions in the query for DNA sequences.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result_handling.

Usage

```
usage: mmseqs result2dnamsa <i:queryDB> <i:targetDB> <i:resultDB> <o:msaDB>
[options]
```

Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--skip-query	Skip the query sequence

Full CLI Help Snapshot

```
usage: mmseqs result2dnamsa <i:queryDB> <i:targetDB> <i:resultDB> <o:msaDB>
[options]
  By Martin Steinegger (martin.steinegger@snu.ac.kr)
options: common:
  --threads INT      Number of CPU-cores used (all by default) [10]
  --compressed INT   Write compressed output [0]
  -v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
expert:
  --skip-query B00L  Skip the query sequence [0]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
    for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
    (2017)
```

6.2.95. result2flat

Create flat file by adding FASTA headers to DB entries.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result_handling.

Usage

```
usage: mmseqs result2flat <i:queryDB> <i:targetDB> <i:resultDB> <o:fastaDB>
[options]
```

Key Options

Option	Purpose
<code>--use-fasta-header</code>	Use the id parsed from the fasta header as the index key instead of using incrementing numeric identifiers
<code>-v</code>	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs result2flat <i:queryDB> <i:targetDB> <i:resultDB> <o:fastaDB>
[options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --use-fasta-header BOOL  Use the id parsed from the fasta header as the index
key instead of using incrementing numeric identifiers [0]
common:
  -v INT                    Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

6.2.96. result2msa

Compute MSA DB from a result DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: `result_handling`.

Usage

```
usage: mmseqs result2msa <i:queryDB> <i:targetDB> <i:resultDB> <o:msaDB> [options]
```

Key Options

Option	Purpose
<code>--comp-bias-corr</code>	Correct for locally biased amino acid composition (range 0-1)

Option	Purpose
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--filter-msa	Filter msa: 0: do not filter, 1: filter
--filter-min-enable	Only filter MSAs with more than N sequences, 0 always filters
--max-seq-id	Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0]
--qid	Reduce diversity of output MSAs using min.seq. identity with query sequences
--qsc	Reduce diversity of output MSAs using min. score per aligned residue with query sequences [-50.0,100.0]
--cov	Filter output MSAs using min. fraction of query residues covered by matched sequences [0.0,1.0]
--diff	Filter MSAs by selecting most diverse set of sequences, keeping at least this many seqs in each MSA block of length 50
--allow-deletion	Allow deletions in a MSA

Full CLI Help Snapshot

```
usage: mmseqs result2msa <i:queryDB> <i:targetDB> <i:resultDB> <o:msaDB>
[options]
  By Martin Steinegger (martin.steinegger@snu.ac.kr) & Milot Mirdita
  <milot@mirdita.de> & Clovis Galiez
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
  (range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
  (range 0-1) [1.000]
align:
  --gap-open TWIN              Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN           Gap extension cost [aa:1,nucl:2]
profile:
  --filter-msa INT             Filter msa: 0: do not filter, 1: filter [0]
  --filter-min-enable INT      Only filter MSAs with more than N sequences, 0
  always filters [0]
```

```

--max-seq-id FLOAT          Reduce redundancy of output MSA using max. pairwise
sequence identity [0.0,1.0] [0.900]
--qid STR                   Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
                             Alternatively, can be a list of multiple
thresholds:
                             E.g.: 0.15,0.30,0.50 to defines filter buckets
of ]0.15-0.30] and ]0.30-0.50] [0.0]
--qsc FLOAT                 Reduce diversity of output MSAs using min. score
per aligned residue with query sequences [-50.0,100.0] [-20.000]
--cov FLOAT                 Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
--diff INT                  Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
misc:
--allow-deletion BOOL       Allow deletions in a MSA [0]
--msa-format-mode INT       Format MSA as: 0: binary cA3M DB
                             1: binary ca3m w. consensus DB
                             2: aligned FASTA DB
                             3: aligned FASTA w. header summary
                             4: STOCKHOLM flat file
                             5: A3M format
                             6: A3M format w. alignment info [2]

common:
--sub-mat TWIN              Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--db-load-mode INT          Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
--threads INT               Number of CPU-cores used (all by default) [10]
--compressed INT            Write compressed output [0]
-v INT                      Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
expert:
--summary-prefix STR        Set the cluster summary prefix [cl]
--skip-query BOOL           Skip the query sequence [0]

```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.97. result2profile

Compute profile DB from a result DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

```
usage: mmseqs result2profile <i:queryDB> <i:targetDB> <i:resultDB> <o:profileDB>
[options]
```

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
-e	List matches below this E-value (range 0.0-inf)
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--mask-profile	Mask query sequence of profile using tantan [0,1]
--e-profile	Include sequences matches with < E-value thr. into the profile (>=0.0)
--wg	Use global sequence weighting for profile calculation
--filter-msa	Filter msa: 0: do not filter, 1: filter
--filter-min-enable	Only filter MSAs with more than N sequences, 0 always filters
--max-seq-id	Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0]
--qid	Reduce diversity of output MSAs using min.seq. identity with query sequences

Full CLI Help Snapshot

```
usage: mmseqs result2profile <i:queryDB> <i:targetDB> <i:resultDB> <o:profileDB>
[options]
```

By Martin Steinegger <martin.steinegger@snu.ac.kr>

```

options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid composition
(range 0-1) [1]
  --comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition
(range 0-1) [1.000]
align:
  -e DOUBLE                    List matches below this E-value (range 0.0-inf)
[1.000E-03]
  --gap-open TWIN              Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN           Gap extension cost [aa:1,nucl:2]
profile:
  --mask-profile INT          Mask query sequence of profile using tantan [0,1]
[1]
  --e-profile DOUBLE          Include sequences matches with < E-value thr. into
the profile (>=0.0) [1.000E-03]
  --wg BOOL                   Use global sequence weighting for profile
calculation [0]
  --filter-msa INT            Filter msa: 0: do not filter, 1: filter [1]
  --filter-min-enable INT     Only filter MSAs with more than N sequences, 0
always filters [0]
  --max-seq-id FLOAT          Reduce redundancy of output MSA using max. pairwise
sequence identity [0.0,1.0] [0.900]
  --qid STR                   Reduce diversity of output MSAs using min.seq.
identity with query sequences [0.0,1.0]
Alternatively, can be a list of multiple
thresholds:
E.g.: 0.15,0.30,0.50 to defines filter buckets
of ]0.15-0.30] and ]0.30-0.50] [0.0]
  --qsc FLOAT                 Reduce diversity of output MSAs using min. score
per aligned residue with query sequences [-50.0,100.0] [-20.000]
  --cov FLOAT                 Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
  --diff INT                  Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
  --pseudo-cnt-mode INT       use 0: substitution-matrix or 1: context-specific
pseudocounts [0]
  --pca                      Pseudo count admixture strength []
  --pcb                      Pseudo counts: Neff at half of maximum admixture
(range 0.0-inf) []
  --profile-output-mode INT   Profile output mode: 0: binary log-odds 1: human-
readable frequencies [0]
misc:
  --allow-deletion BOOL       Allow deletions in a MSA [0]
common:
  --sub-mat TWIN              Substitution matrix file
[aa:blosom62.out,nucl:nucleotide.out]
  --db-load-mode INT          Database preload mode 0: auto, 1: fread, 2: mmap,
3: mmap+touch [0]
  --threads INT               Number of CPU-cores used (all by default) [10]

```

```
--compressed INT      Write compressed output [0]
-v INT                Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.98. result2rbh

Filter a merged result DB to retain only reciprocal best hits.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result_handling.

Usage

```
usage: mmseqs result2rbh <i:resultDB> <o:resultDB> [options]
```

Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs result2rbh <i:resultDB> <o:resultDB> [options]
```

By Eli Levy Karin

options: common:

```
--threads INT      Number of CPU-cores used (all by default) [10]
--compressed INT    Write compressed output [0]
-v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.99. result2repseq

Get representative sequences from result DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result_handling.

Usage

```
usage: mmseqs result2repseq <i:sequenceDB> <i:resultDB> <o:sequenceDb> [options]
```

Key Options

Option	Purpose
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs result2repseq <i:sequenceDB> <i:resultDB> <o:sequenceDb> [options]
```

By Milot Mirdita <milot@mirdita.de> & Martin Steinegger

<martin.steinegger@snu.ac.kr>

options: common:

```
--db-load-mode INT    Database preload mode 0: auto, 1: fread, 2: mmap, 3:
mmap+touch [0]
--compressed INT      Write compressed output [0]
--threads INT          Number of CPU-cores used (all by default) [10]
-v INT                 Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.100. result2stats

Compute statistics for each entry in a DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result_handling.

Usage

```
usage: mmseqs result2stats <i:queryDB> <i:targetDB> <i:resultDB> <o:statsDB>
[options]
```

Key Options

Option	Purpose
--stat	One of: linecount, mean, min, max, doolittle, charges, seqlen, firstline
--tsv	Return output in TSV format
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs result2stats <i:queryDB> <i:targetDB> <i:resultDB> <o:statsDB>
[options]
  By Clovis Galiez & Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --stat STR          One of: linecount, mean, min, max, doolittle, charges,
seqlen, firstline []
  --tsv BOOL          Return output in TSV format [0]
common:
  --compressed INT    Write compressed output [0]
  --threads INT        Number of CPU-cores used (all by default) [10]
  -v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching

for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.101. reverseseq

Reverse (without complement) sequences.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence_manipulation.

Usage

```
usage: mmseqs reverseseq <i:sequenceDB> <o:revSequenceDB> [options]
```

Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs reverseseq <i:sequenceDB> <o:revSequenceDB> [options]
```

By Eli Levy Karin <eli.levy.karin@gmail.com>

options: common:

```
--threads INT      Number of CPU-cores used (all by default) [10]
--compressed INT   Write compressed output [0]
-v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.102. rmdb

Remove a DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 19 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

```
usage: mmseqs rmdb <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs rmdb` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.103. search

Sensitive homology search.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family balances sensitivity against runtime by controlling candidate generation, alignment workload, and split policy. The current dependency map records 8 upstream caller(s) and 22 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control over search sensitivity, filtering, and alignment behavior for DB-to-DB runs.

Dependency entry: Open in map; functional module: `search_workflows`.

Usage

```
usage: mmseqs search <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]
```

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)
--k-score	k-mer threshold for generating similar k-mer lists
--alph-size	Alphabet size (range 2-21)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--split	Split input into N equally distributed chunks. 0: set the best split automatically
--split-mode	0: split target db; 1: split query db; 2: auto, depending on main memory

Full CLI Help Snapshot

```
usage: mmseqs search <i:queryDB> <i:targetDB> <o:alignmentDB> <tmpDir> [options]
By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid
composition (range 0-1) [1]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid
composition (range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [5.700]
  -k INT                       k-mer length (0: automatically set to optimum)
[0]
  --target-search-mode INT     target search mode (0: regular k-mer, 1:
```

```

similar k-mer) [0]
--k-score TWIN          k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
--alph-size TWIN        Alphabet size (range 2-21) [aa:21,nucl:5]
--max-seqs INT          Maximum results per query sequence allowed to
pass the prefilter (affects sensitivity) [300]
--split INT             Split input into N equally distributed chunks.
0: set the best split automatically [0]
--split-mode INT        0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
--split-memory-limit BYTE Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
--diag-score BOOL       Use ungapped diagonal scoring during prefilter
[1]
--exact-kmer-matching INT Extract only exact k-mers for matching (range
0-1) [0]
--mask INT              Mask sequences in prefilter stage with tantan:
0: w/o low complexity masking, 1: with low complexity masking [1]
--mask-prob FLOAT       Mask sequences is probablity is above threshold
[0.900]
--mask-lower-case INT   Lowercase letters will be excluded from k-mer
search 0: include region, 1: exclude region [0]
--mask-n-repeat INT     Repeat letters that occure > threshold in a rwo
[0]
--min-ungapped-score INT Accept only matches with ungapped alignment
score above threshold [15]
--spaced-kmer-mode INT  0: use consecutive positions in k-mers; 1: use
spaced k-mers [1]
--spaced-kmer-pattern STR User-specified spaced k-mer pattern []
--local-tmp STR         Path where some of the temporary files will be
created []
--disk-space-limit BYTE Set max disk space to use for reverse profile
searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the
temp folder [0]
align:
-a BOOL                Add backtrace string (convert to alignments
with mmseqs convertalis module) [0]
--alignment-mode INT    How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment [2]
--alignment-output-mode INT How to compute the alignment:
0: automatic
1: only score and end_pos
2: also start_pos and cov
3: also seq.id
4: only ungapped alignment

```

5: score only (output) cluster format [0]
 --wrapped-scoring BOOL Double the (nucleotide) query sequence during
 the scoring process to allow wrapped diagonal scoring around end and start [0]
 -e DOUBLE List matches below this E-value (range 0.0-inf)
 [1.000E-03]
 --min-seq-id FLOAT List matches above this sequence identity (for
 clustering) (range 0.0-1.0) [0.000]
 --min-aln-len INT Minimum alignment length (range 0-INT_MAX) [0]
 --seq-id-mode INT 0: alignment length 1: shorter, 2: longer
 sequence [0]
 --alt-ali INT Show up to this many alternative alignments [0]
 -c FLOAT List matches above this fraction of aligned
 (covered) residues (see --cov-mode) [0.000]
 --cov-mode INT 0: coverage of query and target
 1: coverage of target
 2: coverage of query
 3: target seq. length has to be at least x% of
 query length
 4: query seq. length has to be at least x% of
 target length
 5: short seq. needs to be at least x% of the
 other seq. length [0]
 --max-rejected INT Maximum rejected alignments before alignment
 calculation for a query is stopped [2147483647]
 --max-accept INT Maximum accepted alignments before alignment
 calculation for a query is stopped [2147483647]
 --score-bias FLOAT Score bias when computing SW alignment (in
 bits) [0.000]
 --realign BOOL Compute more conservative, shorter alignments
 (scores and E-values not changed) [0]
 --realign-score-bias FLOAT Additional bias when computing realignment
 [-0.200]
 --realign-max-seqs INT Maximum number of results to return in
 realignment [2147483647]
 --corr-score-weight FLOAT Weight of backtrace correlation score that is
 added to the alignment score [0.000]
 --gap-open TWIN Gap open cost [aa:11,nucl:5]
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]
 --zdrop INT Maximal allowed difference between score values
 before alignment is truncated (nucleotide alignment only) [40]
 --exhaustive-search-filter INT Filter result during search: 0: do not filter,
 1: filter [0]
 profile:
 --pca Pseudo count admixture strength []
 --pcb Pseudo counts: Neff at half of maximum
 admixture (range 0.0-inf) []
 --mask-profile INT Mask query sequence of profile using tantan
 [0,1] [1]
 --e-profile DOUBLE Include sequences matches with < E-value thr.

into the profile (≥ 0.0) [1.000E-01]

--wg BOOL Use global sequence weighting for profile calculation [0]

--filter-msa INT Filter msa: 0: do not filter, 1: filter [1]

--filter-min-enable INT Only filter MSAs with more than N sequences, 0 always filters [0]

--max-seq-id FLOAT Reduce redundancy of output MSA using max. pairwise sequence identity [0.0,1.0] [0.900]

--qid STR Reduce diversity of output MSAs using min.seq. identity with query sequences [0.0,1.0]

Alternatively, can be a list of multiple thresholds:

E.g.: 0.15,0.30,0.50 to defines filter buckets of [0.15-0.30] and [0.30-0.50] [0.0]

--qsc FLOAT Reduce diversity of output MSAs using min. score per aligned residue with query sequences [-50.0,100.0] [-20.000]

--cov FLOAT Filter output MSAs using min. fraction of query residues covered by matched sequences [0.0,1.0] [0.000]

--diff INT Filter MSAs by selecting most diverse set of sequences, keeping at least this many seqs in each MSA block of length 50 [1000]

--pseudo-cnt-mode INT use 0: substitution-matrix or 1: context-specific pseudocounts [0]

--profile-output-mode INT Profile output mode: 0: binary log-odds 1: human-readable frequencies [0]

--num-iterations INT Number of iterative profile search iterations [1]

--exhaustive-search BOOL For bigger profile DB, run iteratively the search by greedily swapping the search results [0]

--lca-search BOOL Efficient search for LCA candidates [0]

misc:

--taxon-list STR Taxonomy ID, possibly multiple values separated by ',' []

--prefilter-mode INT prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped [0]

--rescore-mode INT Rescore diagonals with:
0: Hamming distance
1: local alignment (score only)
2: local alignment
3: global alignment
4: longest alignment fulfilling window quality criterion [0]

--allow-deletion BOOL Allow deletions in a MSA [0]

--min-length INT Minimum codon number in open reading frames [30]

--max-length INT Maximum codon number in open reading frames [32734]

--max-gaps INT Maximum number of codons with gaps or unknown residues before an open reading frame is rejected [2147483647]

--contig-start-mode INT Contig start can be 0: incomplete, 1: complete,

2: both [2]
 --contig-end-mode INT Contig end can be 0: incomplete, 1: complete,
 2: both [2]
 --orf-start-mode INT Orf fragment can be 0: from start to stop, 1:
 from any to stop, 2: from last encountered start to stop (no start in the middle)
 [1]
 --forward-frames STR Comma-separated list of frames on the forward
 strand to be extracted [1,2,3]
 --reverse-frames STR Comma-separated list of frames on the reverse
 strand to be extracted [1,2,3]
 --translation-table INT 1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
 YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
 9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
 PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
 ALT_FLATWORM_MITOCHONDRIAL
 15) BLEPHARISMA, 16)
 CHLOROPHYCEAN_MITOCHONDRIAL, 21) TREMATODE_MITOCHONDRIAL, 22)
 SCENEDESMUS_MITOCHONDRIAL
 23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
 PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
 KARYORELICT, 28) CONDYLOSTOMA
 29) MESODINIUM, 30) PERTRICH, 31)
 BLASTOCRITHIDIA [1]
 --translate INT Translate ORF to amino acid [0]
 --use-all-table-starts BOOL Use all alternatives for a start codon in the
 genetic table, if false - only ATG (AUG) [0]
 --id-offset INT Numeric ids in index file are offset by this
 value [0]
 --sequence-overlap INT Overlap between sequences [0]
 --sequence-split-mode INT Sequence split mode 0: copy data, 1: soft link
 data and write new index, [1]
 --headers-split-mode INT Header split mode: 0: split position, 1:
 original header [0]
 --search-type INT Search type 0: auto 1: amino acid, 2:
 translated, 3: nucleotide, 4: translated nucleotide alignment [0]
 --start-sens FLOAT Start sensitivity [4.000]
 --sens-steps INT Number of search steps performed from --start-
 sens to -s [1]
 --translation-mode INT Translation AA seq from nucleotide by 0: ORFs,
 1: full reading frames [0]
 common:
 --sub-mat TWIN Substitution matrix file
 [aa:blosom62.out,nucl:nucleotide.out]
 --max-seq-len INT Maximum sequence length [65535]
 --db-load-mode INT Database preload mode 0: auto, 1: fread, 2:
 mmap, 3: mmap+touch [0]
 --threads INT Number of CPU-cores used (all by default) [10]
 --compressed INT Write compressed output [0]
 -v INT Verbosity level: 0: quiet, 1: +errors, 2:

```

+warnings, 3: +info [3]
--gpu INT                Use GPU (CUDA) if possible [0]
--gpu-server INT          Use GPU server [0]
--gpu-server-wait-timeout INT  Wait for GPU server for 0: don't wait -1: no
wait limit: >0 this many seconds [600]
--mpi-runner STR          Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
--force-reuse BOOL        Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL    Delete temporary files [0]
expert:
--filter-hits BOOL        Filter hits by seq.id. and coverage [0]
--sort-results INT        Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--create-lookup INT       Create database lookup file (can be very large)
[0]
--chain-alignments INT     Chain overlapping alignments [0]
--merge-query INT         Combine ORFs/split sequences to a single entry
[1]
--strand INT              Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]

```

examples:

```

# Search multiple FASTA against FASTA (like BLASTP, TBLASTN, BLASTX, BLASTN --
search-type 3, TBLASTX --search-type 2)
mmseqs search queryDB targetDB resultDB tmp
mmseqs convertalis queryDB targetDB resultDB result.m8

```

```

# Iterative profile search (like PSI-BLAST)
mmseqs search queryDB targetDB resultDB tmp --num-iterations 2

```

```

# Profile search against small databases (e.g. PFAM, eggNOG)
mmseqs databases PFAM pfam_db tmp
mmseqs search queryDB pfam_db resultDB tmp

```

```

# Exhaustive search against sequences or profiles (works for large DBs)
mmseqs search queryDB targetDB resultDB tmp --exhaustive-search

```

```

# Increasing sensitivity search (from 2 to 7 in 3 steps)
mmseqs search queryDB targetDB resultDB --start-sens 2 -s 7 --sens-steps 3

```

references:

```

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
- Kallenborn F, Chacon A, Hundt C, Sirelkhatim H, Didi K, Dallago C, Mirdita M,
Schmidt B, Steinegger M: GPU-accelerated homology search with MMseqs2. bioRxiv,
2024.11.13.623350 (2024)

```


6.2.104. **sequence2profile**

Turn sequence into profile by adding context specific pseudo counts.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

```
usage: mmseqs sequence2profile <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs sequence2profile` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.105. **setextendeddbtype**

Write an extended DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

```
usage: mmseqs setextendeddbtype <i:DB> [options]
```

Key Options

Option	Purpose
--extended-dbtype	Set extended dbtype 1: compressed, 2: need src, 4: context pseudoe cnts
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs setextendeddbtype <i:DB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --extended-dbtype INT    Set extended dbtype 1: compressed, 2: need src, 4:
context pseudoe cnts [0]
common:
  -v INT                  Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
```

examples:

```
# Print entries with keys 1, 2 and 3 from a sequence DB to stdout
mmseqs setextendeddbtype db --extended-dbtype 2
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.106. sortresult

Sort a result DB in the same order as the prefilter or align module.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result_handling.

Usage

```
usage: mmseqs sortresult <i:resultbDB> <o:resultDB> [options]
```

Key Options

Option	Purpose
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs sortresult <i:resultbDB> <o:resultDB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: common:
  --compressed INT    Write compressed output [0]
  --threads INT       Number of CPU-cores used (all by default) [10]
  -v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

6.2.107. splitdb

Split DB into subsets.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

```
usage: mmseqs splitdb <i:DB> <o:DB> [options]
```

Key Options

Option	Purpose
--split	Split input into N equally distributed chunks
--split-aa	Try to find the best split boundaries by entry lengths
--compressed	Write compressed output

Option	Purpose
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: mmseqs splitdb <i:DB> <o:DB> [options]

By Milot Mirdita <milot@mirdita.de>

options: common:

--split INT Split input into N equally distributed chunks [0]
 --split-aa BOOL Try to find the best split boundaries by entry lengths [0]
 --compressed INT Write compressed output [0]
 -v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info [3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.108. splitsequence

Split sequences by length.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 3 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

usage: mmseqs splitsequence <i:sequenceDB> <o:sequenceDB> [options]

Key Options

Option	Purpose
--sequence-overlap	Overlap between sequences
--sequence-split-mode	Sequence split mode 0: copy data, 1: soft link data and write new index,
--headers-split-mode	Header split mode: 0: split position, 1: original header

Option	Purpose
--max-seq-len	Maximum sequence length
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--create-lookup	Create database lookup file (can be very large)

Full CLI Help Snapshot

```
usage: mmseqs splitsequence <i:sequenceDB> <o:sequenceDB> [options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: misc:
  --sequence-overlap INT      Overlap between sequences [300]
  --sequence-split-mode INT  Sequence split mode 0: copy data, 1: soft link data
and write new index, [1]
  --headers-split-mode INT   Header split mode: 0: split position, 1: original
header [0]
common:
  --max-seq-len INT          Maximum sequence length [10000]
  --threads INT              Number of CPU-cores used (all by default) [10]
  --compressed INT          Write compressed output [0]
  -v INT                     Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
3: +info [3]
expert:
  --create-lookup INT        Create database lookup file (can be very large) [0]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

6.2.109. subtractdbs

Remove all entries from first DB occurring in second DB by key.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

usage: mmseqs subtractdbs <i:resultDBLeft> <i:resultDBRight> <o:resultDB> [options]

Key Options

Option	Purpose
-e	List matches below this E-value (range 0.0-inf)
--e-profile	Include sequences matches with < E-value thr. into the profile (>=0.0)
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs subtractdbs <i:resultDBLeft> <i:resultDBRight> <o:resultDB>
[options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>
options: align:
  -e DOUBLE          List matches below this E-value (range 0.0-inf) [1.000E-03]
profile:
  --e-profile DOUBLE Include sequences matches with < E-value thr. into the
profile (>=0.0) [1.000E-03]
common:
  --threads INT      Number of CPU-cores used (all by default) [10]
  --compressed INT   Write compressed output [0]
  -v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

6.2.110. suffixid

For each entry in a DB append the entry key to the entry itself.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

```
usage: mmseqs suffixid <i:resultDB> <o:resultDB> [options]
```

Key Options

Option	Purpose
--prefix	Use this prefix for all entries
--mapping-file	Specify a file that translates the keys of a DB to new keys, TSV format
--tsv	Return output in TSV format
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs suffixid <i:resultDB> <o:resultDB> [options]
```

By Milot Mirdita <milot@mirdita.de>

options: misc:

```
--prefix STR      Use this prefix for all entries []
--mapping-file STR Specify a file that translates the keys of a DB to new
keys, TSV format []
--tsv BOOL        Return output in TSV format [0]
```

common:

```
--threads INT     Number of CPU-cores used (all by default) [10]
--compressed INT  Write compressed output [0]
-v INT            Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.111. summarizealis

Summarize alignment result to one row (uniq. cov., cov., avg. seq. id.).

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: `result_handling`.

Usage

```
usage: mmseqs summarizealis <i:alignmentDB> <o:summerizedDB> [options]
```

Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs summarizealis <i:alignmentDB> <o:summerizedDB> [options]
```

```
By Martin Steinegger <martin.steinegger@snu.ac.kr>
```

```
options: common:
```

```
--threads INT      Number of CPU-cores used (all by default) [10]
--compressed INT   Write compressed output [0]
-v INT             Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

```
references:
```

```
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

```
- Mirdita M, von den Driesch L, Galiez C, Martin M, Soding J, Steinegger M:
Uniclust databases of clustered and deeply annotated protein sequences and
alignments. Nucleic Acids Research 45(D1), D170-D176 (2017)
```

6.2.112. summarizeheaders

Summarize FASTA headers of result DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: `result_handling`.

Usage

```
usage: mmseqs summarizeheaders <DB> [args] [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs summarizeheaders` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.113. `summarizeresult`

Extract annotations from alignment DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 2 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: `result_handling`.

Usage

```
usage: mmseqs summarizeresult <i:alignmentDB> <o:alignmentDB> [options]
```

Key Options

Option	Purpose
-a	Add backtrace string (convert to alignments with mmseqs convertalis module)
-c	List matches above this fraction of aligned (covered) residues (see <code>-cov-mode</code>)

Option	Purpose
--overlap	Maximum overlap of covered regions
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs summarizeresult <i:alignmentDB> <o:alignmentDB> [options]
  By Milot Mirdita <milot@mirdita.de> & Martin Steinegger
  <martin.steinegger@snu.ac.kr>
options: align:
  -a BOOL          Add backtrace string (convert to alignments with mmseqs
convertalis module) [0]
  -c FLOAT          List matches above this fraction of aligned (covered)
residues (see --cov-mode) [0.000]
misc:
  --overlap FLOAT   Maximum overlap of covered regions [0.000]
common:
  --threads INT     Number of CPU-cores used (all by default) [10]
  --compressed INT  Write compressed output [0]
  -v INT            Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
  - Mirdita M, von den Driesch L, Galiez C, Martin M, Soding J, Steinegger M:
Uniclust databases of clustered and deeply annotated protein sequences and
alignments. Nucleic Acids Research 45(D1), D170-D176 (2017)
```

6.2.114. summarizetabs

Extract annotations from HHblits BLAST-tab-formatted results.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

usage: mmseqs summarizetabs <DB> [args] [options]

The syntax line above is source-derived from command layer/category metadata. Run mmseqs summarizetabs locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.115. swapdb

Transpose DB with integer values in first column.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 4 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

usage: mmseqs swapdb <i:resultDB> <o:resultDB> [options]

Key Options

Option	Purpose
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: mmseqs swapdb <i:resultDB> <o:resultDB> [options]
 By Martin Steinegger <martin.steinegger@snu.ac.kr>, Clovis Galiez & Eli Levy
 Karin
 options: prefilter:
 --split-memory-limit BYTE Set max memory per split. E.g. 800B, 5K, 10M, 1G.
 Default (0) to all available system memory [0]
 common:
 --threads INT Number of CPU-cores used (all by default) [10]
 --compressed INT Write compressed output [0]
 -v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
 3: +info [3]
 references:
 - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
 for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
 (2017)

6.2.116. swapresults

Transpose prefilter/alignment DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family reshapes and exports outputs; interpretation must remain consistent with upstream scoring modes. The current dependency map records 4 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when transforming or exporting result DBs after scoring decisions are already fixed upstream.

Dependency entry: Open in map; functional module: result_handling.

Usage

usage: mmseqs swapresults <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB>
 [options]

Key Options

Option	Purpose
--split-memory-limit	Set max memory per split. E.g. 800B, 5K, 10M, 1G. Default (0) to all available system memory
-e	List matches below this E-value (range 0.0-inf)
--gap-open	Gap open cost
--gap-extend	Gap extension cost
--sub-mat	Substitution matrix file
--threads	Number of CPU-cores used (all by default)

Option	Purpose
--compressed	Write compressed output
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs swapresults <i:queryDB> <i:targetDB> <i:resultDB> <o:resultDB>
[options]
  By Martin Steinegger <martin.steinegger@snu.ac.kr>, Clovis Galiez & Eli Levy
  Karin
options: prefilter:
  --split-memory-limit BYTE Set max memory per split. E.g. 800B, 5K, 10M, 1G.
  Default (0) to all available system memory [0]
align:
  -e DOUBLE                List matches below this E-value (range 0.0-inf)
  [1.000E-03]
  --gap-open TWIN           Gap open cost [aa:11,nucl:5]
  --gap-extend TWIN        Gap extension cost [aa:1,nucl:2]
common:
  --sub-mat TWIN            Substitution matrix file
  [aa:blosum62.out,nucl:nucleotide.out]
  --threads INT             Number of CPU-cores used (all by default) [10]
  --compressed INT         Write compressed output [0]
  --db-load-mode INT        Database preload mode 0: auto, 1: fread, 2: mmap, 3:
  mmap+touch [0]
  -v INT                    Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
  3: +info [3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
  for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
  (2017)
```

6.2.117. tar2db

Convert content of tar archives to any DB.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family enforces DB contracts and storage/index integrity used by all workflows. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Dependency entry: Open in map; functional module: database.

Usage

```
usage: mmseqs tar2db <i:tar[.gz]> ... <i:tar[.gz]> <o:resultDB> [options]
```

Key Options

Option	Purpose
<code>--output-dbtype</code>	Set database type for resulting database: Amino acid sequences 0, Nucl. seq. 1, Profiles 2, Alignment result 5, Clustering result 6, Prefiltering result 7, Taxonomy result 8, Indexed database 9, cA3M MSAs 10, FASTA or A3M MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional prefiltering result 14, Offsetted headers 15
<code>--tar-include</code>	Include file names based on this regex
<code>--tar-exclude</code>	Exclude file names based on this regex
<code>--compressed</code>	Write compressed output
<code>--threads</code>	Number of CPU-cores used (all by default)
<code>-v</code>	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs tar2db <i:tar[.gz]> ... <i:tar[.gz]> <o:resultDB> [options]
```

By Milot Mirdita <milot@mirdita.de>

options: misc:

```
--output-dbtype INT  Set database type for resulting database: Amino acid
sequences 0, Nucl. seq. 1, Profiles 2, Alignment result 5, Clustering result 6,
Prefiltering result 7, Taxonomy result 8, Indexed database 9, cA3M MSAs 10, FASTA
or A3M MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional
prefiltering result 14, Offsetted headers 15 [12]
```

```
--tar-include STR    Include file names based on this regex [.*]
```

```
--tar-exclude STR    Exclude file names based on this regex [^$]
```

common:

```
--compressed INT    Write compressed output [0]
```

```
--threads INT       Number of CPU-cores used (all by default) [10]
```

```
-v INT              Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
```

```
+info [3]
```

examples:

```
# Assuming tar archive containing three aligned FASTA files:
# * folder/msa1.fa.gz * folder/msa2.fa * folder/msa3.fa
# Create a msaDB with three DB entries each containing a separate MSA
mmseqs tar2db archive.tar.gz msaDB --output-dbtype 11
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.118. taxonomy

Taxonomic classification.

Execution role: high-level API command that exposes a complete task path over MMseqs2 databases.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 2 upstream caller(s) and 15 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

usage: mmseqs taxonomy <i:queryDB> <i:targetDB> <o:taxaDB> <tmpDir> [options]

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--add-self-matches	Artificially add entries of queries with themselves (for clustering)
--seed-sub-mat	Substitution matrix file for k-mer generation
-s	Sensitivity: 1.0 faster; 4.0 fast; 7.5 sensitive
-k	k-mer length (0: automatically set to optimum)
--target-search-mode	target search mode (0: regular k-mer, 1: similar k-mer)

Option	Purpose
--k-score	k-mer threshold for generating similar k-mer lists
--alph-size	Alphabet size (range 2-21)
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
--split	Split input into N equally distributed chunks. 0: set the best split automatically
--split-mode	0: split target db; 1: split query db; 2: auto, depending on main memory

Full CLI Help Snapshot

```
usage: mmseqs taxonomy <i:queryDB> <i:targetDB> <o:taxaDB> <tmpDir> [options]
  By Milot Mirdita <milot@mirdita.de> & Martin Steinegger
<martin.steinegger@snu.ac.kr> & Eli Levy Karin <eli.levy.karin@gmail.com>
options: prefilter:
  --comp-bias-corr INT          Correct for locally biased amino acid
composition (range 0-1) [1]
  --comp-bias-corr-scale FLOAT  Correct for locally biased amino acid
composition (range 0-1) [1.000]
  --add-self-matches BOOL      Artificially add entries of queries with
themselves (for clustering) [0]
  --seed-sub-mat TWIN          Substitution matrix file for k-mer generation
[aa:VTML80.out,nucl:nucleotide.out]
  -s FLOAT                     Sensitivity: 1.0 faster; 4.0 fast; 7.5
sensitive [2.000]
  -k INT                       k-mer length (0: automatically set to optimum)
[0]
  --target-search-mode INT      target search mode (0: regular k-mer, 1:
similar k-mer) [0]
  --k-score TWIN               k-mer threshold for generating similar k-mer
lists [seq:2147483647,prof:2147483647]
  --alph-size TWIN             Alphabet size (range 2-21) [aa:21,nucl:5]
  --max-seqs INT               Maximum results per query sequence allowed to
pass the prefilter (affects sensitivity) [300]
  --split INT                  Split input into N equally distributed chunks.
0: set the best split automatically [0]
  --split-mode INT             0: split target db; 1: split query db; 2: auto,
depending on main memory [2]
  --split-memory-limit BYTE     Set max memory per split. E.g. 800B, 5K, 10M,
1G. Default (0) to all available system memory [0]
  --diag-score BOOL            Use ungapped diagonal scoring during prefilter
[1]
  --exact-kmer-matching INT     Extract only exact k-mers for matching (range
0-1) [0]
```


--mask INT Mask sequences in prefilter stage with tantan:
 0: w/o low complexity masking, 1: with low complexity masking [1]
 --mask-prob FLOAT Mask sequences is probablity is above threshold
 [0.900]
 --mask-lower-case INT Lowercase letters will be excluded from k-mer
 search 0: include region, 1: exclude region [0]
 --mask-n-repeat INT Repeat letters that occure > threshold in a rwo
 [0]
 --min-ungapped-score INT Accept only matches with ungapped alignment
 score above threshold [15]
 --spaced-kmer-mode INT 0: use consecutive positions in k-mers; 1: use
 spaced k-mers [1]
 --spaced-kmer-pattern STR User-specified spaced k-mer pattern []
 --local-tmp STR Path where some of the temporary files will be
 created []
 --disk-space-limit BYTE Set max disk space to use for reverse profile
 searches. E.g. 800B, 5K, 10M, 1G. Default (0) to all available disk space in the
 temp folder [0]
 align:
 -a B00L Add backtrace string (convert to alignments
 with mmseqs convertalis module) [0]
 --alignment-mode INT How to compute the alignment:
 0: automatic
 1: only score and end_pos
 2: also start_pos and cov
 3: also seq.id
 4: only ungapped alignment [1]
 --alignment-output-mode INT How to compute the alignment:
 0: automatic
 1: only score and end_pos
 2: also start_pos and cov
 3: also seq.id
 4: only ungapped alignment
 5: score only (output) cluster format [0]
 --wrapped-scoring B00L Double the (nucleotide) query sequence during
 the scoring process to allow wrapped diagonal scoring around end and start [0]
 -e DOUBLE List matches below this E-value (range 0.0-inf)
 [1.000E+00]
 --min-seq-id FLOAT List matches above this sequence identity (for
 clustering) (range 0.0-1.0) [0.000]
 --min-aln-len INT Minimum alignment length (range 0-INT_MAX) [0]
 --seq-id-mode INT 0: alignment length 1: shorter, 2: longer
 sequence [0]
 --alt-ali INT Show up to this many alternative alignments [0]
 -c FLOAT List matches above this fraction of aligned
 (covered) residues (see --cov-mode) [0.000]
 --cov-mode INT 0: coverage of query and target
 1: coverage of target
 2: coverage of query

query length 3: target seq. length has to be at least x% of
 target length 4: query seq. length has to be at least x% of
 other seq. length [0] 5: short seq. needs to be at least x% of the
 --max-rejected INT Maximum rejected alignments before alignment
 calculation for a query is stopped [5]
 --max-accept INT Maximum accepted alignments before alignment
 calculation for a query is stopped [30]
 --score-bias FLOAT Score bias when computing SW alignment (in
 bits) [0.000]
 --realign BOOL Compute more conservative, shorter alignments
 (scores and E-values not changed) [0]
 --realign-score-bias FLOAT Additional bias when computing realignment
 [-0.200]
 --realign-max-seqs INT Maximum number of results to return in
 realignment [2147483647]
 --corr-score-weight FLOAT Weight of backtrace correlation score that is
 added to the alignment score [0.000]
 --gap-open TWIN Gap open cost [aa:11,nucl:5]
 --gap-extend TWIN Gap extension cost [aa:1,nucl:2]
 --zdrop INT Maximal allowed difference between score values
 before alignment is truncated (nucleotide alignment only) [40]
 --exhaustive-search-filter INT Filter result during search: 0: do not filter,
 1: filter [0]
 profile:
 --pca Pseudo count admixture strength []
 --pcb Pseudo counts: Neff at half of maximum
 admixture (range 0.0-inf) []
 --mask-profile INT Mask query sequence of profile using tantan
 [0,1] [1]
 --e-profile DOUBLE Include sequences matches with < E-value thr.
 into the profile (>=0.0) [1.000E-03]
 --wg BOOL Use global sequence weighting for profile
 calculation [0]
 --filter-msa INT Filter msa: 0: do not filter, 1: filter [1]
 --filter-min-enable INT Only filter MSAs with more than N sequences, 0
 always filters [0]
 --max-seq-id FLOAT Reduce redundancy of output MSA using max.
 pairwise sequence identity [0.0,1.0] [0.900]
 --qid STR Reduce diversity of output MSAs using min.seq.
 identity with query sequences [0.0,1.0]
 Alternatively, can be a list of multiple
 thresholds:
 E.g.: 0.15,0.30,0.50 to defines filter buckets
 of]0.15-0.30] and]0.30-0.50] [0.0]
 --qsc FLOAT Reduce diversity of output MSAs using min.
 score per aligned residue with query sequences [-50.0,100.0] [-20.000]

```

--cov FLOAT          Filter output MSAs using min. fraction of query
residues covered by matched sequences [0.0,1.0] [0.000]
--diff INT           Filter MSAs by selecting most diverse set of
sequences, keeping at least this many seqs in each MSA block of length 50 [1000]
--pseudo-cnt-mode INT use 0: substitution-matrix or 1: context-
specific pseudocounts [0]
--profile-output-mode INT Profile output mode: 0: binary log-odds 1:
human-readable frequencies [0]
--exhaustive-search BOOL For bigger profile DB, run iteratively the
search by greedily swapping the search results [0]
--lca-search BOOL      Efficient search for LCA candidates [0]
misc:
--orf-filter INT       Prefilter query ORFs with non-selective search
Only used during nucleotide-vs-protein
classification
NOTE: Consider disabling when classifying short
reads [1]
--orf-filter-e DOUBLE  E-value threshold used for query ORF
prefiltering [1.000E+02]
--orf-filter-s FLOAT   Sensitivity used for query ORF prefiltering
[2.000]
--lca-mode INT         LCA Mode 1: single search LCA , 2/3:
approximate 2bLCA, 4: top hit [3]
--tax-output-mode INT  0: output LCA, 1: output alignment 2: output
both [0]
--majority FLOAT       minimal fraction of agreement among
taxonomically assigned sequences of a set [0.500]
--vote-mode INT        Mode of assigning weights to compute majority.
0: uniform, 1: minus log E-value, 2: score [1]
--lca-ranks STR         Add column with specified ranks (',' separated)
[]
--tax-lineage INT       0: don't show, 1: add all lineage names, 2: add
all lineage taxids [0]
--blacklist STR         Comma separated list of ignored taxa in LCA
computation [12908:unclassified sequences,28384:other sequences]
--taxon-list STR        Taxonomy ID, possibly multiple values separated
by ',' []
--prefilter-mode INT    prefilter mode: 0: kmer/ungapped 1: ungapped,
2: nofilter, 3: ungapped&gapped [0]
--rescore-mode INT      Rescore diagonals with:
0: Hamming distance
1: local alignment (score only)
2: local alignment
3: global alignment
4: longest alignment fulfilling window quality
criterion [0]
--allow-deletion BOOL   Allow deletions in a MSA [0]
--min-length INT        Minimum codon number in open reading frames
[30]

```

```

--max-length INT                Maximum codon number in open reading frames
[32734]
--max-gaps INT                  Maximum number of codons with gaps or unknown
residues before an open reading frame is rejected [2147483647]
--contig-start-mode INT         Contig start can be 0: incomplete, 1: complete,
2: both [2]
--contig-end-mode INT          Contig end can be 0: incomplete, 1: complete,
2: both [2]
--orf-start-mode INT           Orf fragment can be 0: from start to stop, 1:
from any to stop, 2: from last encountered start to stop (no start in the middle)
[1]
--forward-frames STR           Comma-separated list of frames on the forward
strand to be extracted [1,2,3]
--reverse-frames STR           Comma-separated list of frames on the reverse
strand to be extracted [1,2,3]
--translation-table INT         1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)
PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16)
CHLOROPHYCEAN_MITOCHONDRIAL, 21) TREMATODE_MITOCHONDRIAL, 22)
SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31)
BLASTOCRITHIDIA [1]
--translate INT                Translate ORF to amino acid [0]
--use-all-table-starts BOOL    Use all alternatives for a start codon in the
genetic table, if false - only ATG (AUG) [0]
--id-offset INT                Numeric ids in index file are offset by this
value [0]
--sequence-overlap INT         Overlap between sequences [0]
--sequence-split-mode INT      Sequence split mode 0: copy data, 1: soft link
data and write new index, [1]
--headers-split-mode INT       Header split mode: 0: split position, 1:
original header [0]
--search-type INT              Search type 0: auto 1: amino acid, 2:
translated, 3: nucleotide, 4: translated nucleotide alignment [0]
--translation-mode INT         Translation AA seq from nucleotide by 0: ORFs,
1: full reading frames [0]
common:
--compressed INT              Write compressed output [0]
--threads INT                  Number of CPU-cores used (all by default) [10]
-v INT                         Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]
--sub-mat TWIN                 Substitution matrix file
[aa:blosum62.out,nuc:nucleotide.out]

```

```

--max-seq-len INT           Maximum sequence length [65535]
--db-load-mode INT          Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
--gpu INT                   Use GPU (CUDA) if possible [0]
--gpu-server INT            Use GPU server [0]
--gpu-server-wait-timeout INT Wait for GPU server for 0: don't wait -1: no
wait limit: >0 this many seconds [600]
--mpi-runner STR            Use MPI on compute cluster with this MPI
command (e.g. "mpirun -np 42") []
--force-reuse BOOL          Reuse tmp filse in tmp/latest folder ignoring
parameters and version changes [0]
--remove-tmp-files BOOL     Delete temporary files [0]
expert:
--filter-hits BOOL          Filter hits by seq.id. and coverage [0]
--sort-results INT          Sort results: 0: no sorting, 1: sort by E-value
(Alignment) or seq.id. (Hamming) [0]
--create-lookup INT         Create database lookup file (can be very large)
[0]
--chain-alignments INT       Chain overlapping alignments [0]
--merge-query INT           Combine ORFs/split sequences to a single entry
[1]
--strand INT                Strand selection only works for DNA/DNA search
0: reverse, 1: forward, 2: both [1]

```

examples:

```

# Download a sequence database with taxonomy information
mmseqs databases UniProtKB/Swiss-Prot swissprotDB tmp

# Assign taxonomy based on 2bLCA
mmseqs taxonomy queryDB swissprotDB result tmp

# Assign taxonomy based on top hit
mmseqs taxonomy queryDB swissprotDB result tmp --lca-mode 4

# Assign taxonomy without ORF prefilter
# Classifies higher percentage for short nucleotide input (e.g. short reads) at
the cost of speed
mmseqs taxonomy queryNuclDB swissprotDB result tmp --orf-filter 0

# Create a Krona report
mmseqs taxonomyreport swissprotDB result report.html --report-mode 1

```

references:

```

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
- Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and
sensitive taxonomic assignment to metagenomic contigs. Bioinformatics, btab184
(2021)

```

6.2.119. taxonomyreport

Create a taxonomy report in Kraken or Krona format.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family maps sequence evidence into taxonomy labels and reports under explicit aggregation rules. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: taxonomy.

Usage

```
usage: mmseqs taxonomyreport <i:seqTaxDB> <i:taxResultDB/resultDB/sequenceDB>
<o:taxonomyReport> [options]
```

Key Options

Option	Purpose
--report-mode	Taxonomy report mode 0: Kraken 1: Krona
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs taxonomyreport <i:seqTaxDB> <i:taxResultDB/resultDB/sequenceDB>
<o:taxonomyReport> [options]
  By Milot Mirdita <milot@mirdita.de> & Florian Breitwieser <florian.bw@gmail.com>
options: misc:
  --report-mode INT    Taxonomy report mode 0: Kraken 1: Krona [0]
common:
  --threads INT        Number of CPU-cores used (all by default) [10]
  -v INT               Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]

references:
  - Mirdita M, Steinegger M, Breitwieser F, Soding J, Levy Karin E: Fast and
sensitive taxonomic assignment to metagenomic contigs. Bioinformatics, btab184
(2021)
```

6.2.120. touchdb

Preload DB into memory (page cache).

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

```
usage: mmseqs touchdb <i:DB> [options]
```

Key Options

Option	Purpose
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs touchdb <i:DB> [options]
```

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: common:

```
--threads INT    Number of CPU-cores used (all by default) [10]
-v INT           Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]
```

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.121. transitivealign

Transfer alignments via transitivity.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family computes pair quality and coordinates and usually dominates per-pair compute cost after prefiltering. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: alignment.

Usage

```
usage: mmseqs transitivealign <inputDB(s)> <outputDB> [options]
```

The syntax line above is source-derived from command layer/category metadata. Run `mmseqs transitivealign` locally for exact positional arguments in your build.

Key Options

Local CLI option snapshots are not available for this command. Use the dependency entry and calling workflow source files to recover parameter bundles for your runtime path.

Full CLI Help Snapshot

Note

This page keeps a source-derived summary because no local help snapshot was found for this command.

6.2.122. `translateaa`

Translate proteins to lexicographically lowest codons.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: `sequence_manipulation`.

Usage

usage: `mmseqs translateaa <i:sequenceDB> <o:sequenceDB> [options]`

Key Options

Option	Purpose
<code>--threads</code>	Number of CPU-cores used (all by default)
<code>--compressed</code>	Write compressed output
<code>-v</code>	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: `mmseqs translateaa <i:sequenceDB> <o:sequenceDB> [options]`

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: common:

<code>--threads INT</code>	Number of CPU-cores used (all by default) [10]
<code>--compressed INT</code>	Write compressed output [0]
<code>-v INT</code>	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

[3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11), 1026-1028 (2017)

6.2.123. translatenucs

Translate nucleotides to proteins.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family transforms sequence space before or after major compute stages. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: sequence_manipulation.

Usage

usage: mmseqs translatenucs <i:sequenceDB> <o:sequenceDB> [options]

Key Options

Option	Purpose
--translation-table	1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3) YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
--add-orf-stop	Add stop codon “*” at complete start and end
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
--compressed	Write compressed output
--threads	Number of CPU-cores used (all by default)

Full CLI Help Snapshot

usage: mmseqs translatenucs <i:sequenceDB> <o:sequenceDB> [options]

By Milot Mirdita <milot@mirdita.de>

options: misc:

--translation-table INT 1) CANONICAL, 2) VERT_MITOCHONDRIAL, 3)
YEAST_MITOCHONDRIAL, 4) MOLD_MITOCHONDRIAL, 5) INVERT_MITOCHONDRIAL, 6) CILIATE
9) FLATWORM_MITOCHONDRIAL, 10) EUPLOTID, 11)

```

PROKARYOTE, 12) ALT_YEAST, 13) ASCIDIAN_MITOCHONDRIAL, 14)
ALT_FLATWORM_MITOCHONDRIAL
15) BLEPHARISMA, 16) CHLOROPHYCEAN_MITOCHONDRIAL, 21)
TREMATODE_MITOCHONDRIAL, 22) SCENEDESMUS_MITOCHONDRIAL
23) THRAUSTOCHYTRIUM_MITOCHONDRIAL, 24)
PTEROBRANCHIA_MITOCHONDRIAL, 25) GRACILIBACTERIA, 26) PACHYSOLEN, 27)
KARYORELICT, 28) CONDYLOSTOMA
29) MESODINIUM, 30) PERTRICH, 31) BLASTOCRITHIDIA [1]
--add-orf-stop BOOL      Add stop codon '*' at complete start and end [0]
common:
-v INT                   Verbosity level: 0: quiet, 1: +errors, 2: +warnings,
3: +info [3]
--compressed INT         Write compressed output [0]
--threads INT            Number of CPU-cores used (all by default) [10]

```

6.2.124. tsv2db

Execution role: low-level command used for DB management, conversion, and pipeline composition.

Typical use case: choose this command when preparing or restructuring DB artifacts to satisfy downstream module contracts.

Usage

Key Options

Option	Purpose
	MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional prefiltering result 14, Offsetted headers 15
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs tsv2db <i:tsvFile> <o:resultDB> [options]
  By Milot Mirdita <milot@mirdita.de>
options: prefilter:
  --add-self-matches BOOL  Artificially add entries of queries with themselves
(for clustering) [0]
misc:
  --output-dbtype INT      Set database type for resulting database: Amino acid
sequences 0, Nucl. seq. 1, Profiles 2, Alignment result 5, Clustering result 6,
Prefiltering result 7, Taxonomy result 8, Indexed database 9, cA3M MSAs 10, FASTA
or A3M MSAs 11, Generic database 12, Omit dbtype file 13, Bi-directional
prefiltering result 14, Offsetted headers 15 [12]
common:
  --compressed INT        Write compressed output [0]
  -v INT                  Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]

references:
  - Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
```

6.2.125. tsv2exprofiledb

Create a expandable profile db from TSV files.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family preserves profile semantics across conversion and search steps. The current dependency map records 0 upstream caller(s) and 5 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: profiles.

Usage

usage: mmseqs tsv2exprofiledb <i:tsvFilesBase> <o:exprofileDB> [options]

Key Options

Option	Purpose
--gpu	Use GPU (CUDA) if possible
--threads	Number of CPU-cores used (all by default)
--compressed	Write compressed output
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: mmseqs tsv2exprofiledb <i:tsvFilesBase> <o:exprofileDB> [options]

By Milot Mirdita <milot@mirdita.de>

options: common:

--gpu INT Use GPU (CUDA) if possible [0]
--threads INT Number of CPU-cores used (all by default) [10]
--compressed INT Write compressed output [1]
-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info
[3]

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028 (2017)

6.2.126. ungappedprefilter

Optimal diagonal score search.

Execution role: core compute module typically called by workflows and advanced custom pipelines.

This command family is the main acceleration gate that prunes candidate pairs before expensive alignment. The current dependency map records 1 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command in custom pipelines that must expose candidate-generation behavior before alignment stages.

Dependency entry: Open in map; functional module: prefiltering.

Usage

usage: mmseqs ungappedprefilter <i:queryDB> <i:targetDB> <o:prefilterDB> [options]

Key Options

Option	Purpose
--comp-bias-corr	Correct for locally biased amino acid composition (range 0-1)
--comp-bias-corr-scale	Correct for locally biased amino acid composition (range 0-1)
--min-ungapped-score	Accept only matches with ungapped alignment score above threshold
--max-seqs	Maximum results per query sequence allowed to pass the prefilter (affects sensitivity)
-c	List matches above this fraction of aligned (covered) residues (see --cov-mode)
-e	List matches below this E-value (range 0.0-inf)
--cov-mode	0: coverage of query and target
--taxon-list	Taxonomy ID, possibly multiple values separated by ‘,’
--prefilter-mode	prefilter mode: 0: kmer/ungapped 1: ungapped, 2: nofilter, 3: ungapped&gapped
--sub-mat	Substitution matrix file
--db-load-mode	Database preload mode 0: auto, 1: fread, 2: mmap, 3: mmap+touch
--gpu	Use GPU (CUDA) if possible

Full CLI Help Snapshot

usage: mmseqs ungappedprefilter <i:queryDB> <i:targetDB> <o:prefilterDB>

[options]

By Milot Mirdita <milot@mirdita.de> & Martin Steinegger

<martin.steinegger@snu.ac.kr>

options: prefilter:

--comp-bias-corr INT Correct for locally biased amino acid composition (range 0-1) [1]
--comp-bias-corr-scale FLOAT Correct for locally biased amino acid composition (range 0-1) [1.000]
--min-ungapped-score INT Accept only matches with ungapped alignment score above threshold [15]
--max-seqs INT Maximum results per query sequence allowed to pass the prefilter (affects sensitivity) [300]

align:

-c FLOAT List matches above this fraction of aligned (covered) residues (see --cov-mode) [0.000]
-e DOUBLE List matches below this E-value (range 0.0-inf)

```

[1.000E-03]
--cov-mode INT          0: coverage of query and target
                        1: coverage of target
                        2: coverage of query
                        3: target seq. length has to be at least x% of
query length
                        4: query seq. length has to be at least x% of
target length
                        5: short seq. needs to be at least x% of the
other seq. length [0]
misc:
--taxon-list STR        Taxonomy ID, possibly multiple values separated
by ',' []
--prefilter-mode INT    prefilter mode: 0: kmer/ungapped 1: ungapped, 2:
nofilter, 3: ungapped&gapped [0]
common:
--sub-mat TWIN          Substitution matrix file
[aa:blosum62.out,nucl:nucleotide.out]
--db-load-mode INT      Database preload mode 0: auto, 1: fread, 2:
mmap, 3: mmap+touch [0]
--gpu INT               Use GPU (CUDA) if possible [0]
--gpu-server INT        Use GPU server [0]
--gpu-server-wait-timeout INT Wait for GPU server for 0: don't wait -1: no
wait limit: >0 this many seconds [600]
--threads INT           Number of CPU-cores used (all by default) [10]
--compressed INT        Write compressed output [0]
-v INT                  Verbosity level: 0: quiet, 1: +errors, 2:
+warnings, 3: +info [3]

references:
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)
- Kallenborn F, Chacon A, Hundt C, Sirelkhatim H, Didi K, Dallago C, Mirdita M,
Schmidt B, Steinegger M: GPU-accelerated homology search with MMseqs2. bioRxiv,
2024.11.13.623350 (2024)

```

6.2.127. unpackdb

Unpack a DB into separate files.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

```
usage: mmseqs unpackdb <i:DB> <o:outDir> [options]
```

Key Options

Option	Purpose
--unpack-name-mode	Name unpacked files by 0: DB key, 1: accession (through .lookup)
--unpack-suffix	File suffix for unpacked files.
--threads	Number of CPU-cores used (all by default)
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

```
usage: mmseqs unpackdb <i:DB> <o:outDir> [options]
```

```
By Milot Mirdita <milot@mirdita.de>
```

```
options: misc:
```

```
--unpack-name-mode INT    Name unpacked files by 0: DB key, 1: accession  
(through .lookup) [1]
```

```
--unpack-suffix STR      File suffix for unpacked files.  
                          Add .gz suffix to write compressed files. []
```

```
common:
```

```
--threads INT            Number of CPU-cores used (all by default) [10]
```

```
-v INT                   Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:  
+info [3]
```

```
references:
```

```
- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching  
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028  
(2017)
```

6.2.128. view

Print DB entries given in `-id-list` to stdout.

Execution role: low-level command used for DB management, conversion, and pipeline composition.

This command family provides compositional utilities for custom pipelines, migration tasks, and diagnostics. The current dependency map records 0 upstream caller(s) and 0 downstream call(s), which indicates how broadly parameter changes can propagate.

Typical use case: choose this command when you need explicit control of this stage instead of relying on workflow defaults.

Dependency entry: Open in map; functional module: utilities.

Usage

usage: mmseqs view <i:DB> [options]

Key Options

Option	Purpose
--id-list	Entries to be printed separated by ‘;
--id-mode	Select DB entries based on 0: database keys, 1: FASTA identifiers (.lookup)
--idx-entry-type	0: sequence, 1: src sequence, 2: header, 3: src header
-v	Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3: +info

Full CLI Help Snapshot

usage: mmseqs view <i:DB> [options]

By Martin Steinegger <martin.steinegger@snu.ac.kr>

options: misc:

--id-list STR Entries to be printed separated by ',' []
--id-mode INT Select DB entries based on 0: database keys, 1: FASTA
identifiers (.lookup) [0]
--idx-entry-type INT 0: sequence, 1: src sequence, 2: header, 3: src header
[0]

common:

-v INT Verbosity level: 0: quiet, 1: +errors, 2: +warnings, 3:
+info [3]

examples:

Print entries with keys 1, 2 and 3 from a sequence DB to stdout
mmseqs view sequencedb --id-list 1,2,3

references:

- Steinegger M, Soding J: MMseqs2 enables sensitive protein sequence searching
for the analysis of massive data sets. Nature Biotechnology, 35(11), 1026-1028
(2017)

7. MMseqs2 Dependency Map

This file is generated from MMseqs2/src/MMseqsBase.cpp and workflow scripts.

Metric	Value
Total visible commands	128

n/a in connection fields means no direct edge was resolved by static extraction.

7.1. Easy Workflows

7.1.1. **easy-cluster**

Slower, sensitive clustering.

Aspect	Value
Layer	workflow
Category flags	COMMAND_EASY
Calls	cluster, createdb, createseqfiledb, createtsv, result2flat, result2repseq, rmdb
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.1.2. **easy-linclust**

Fast linear time cluster, less sensitive clustering.

Aspect	Value
Layer	workflow
Category flags	COMMAND_EASY
Calls	createdb, createseqfiledb, createtsv, linclust, result2flat, result2repseq, rmdb
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.1.3. **easy-linsearch**

Fast, less sensitive homology search.

Aspect	Value
Layer	workflow
Category flags	COMMAND_EASY COMMAND_EXPERT
Calls	convertalis, createdb, createlinindex, linsearch, rmdb, search, summarizeresult

Aspect	Value
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.1.4. **easy-rbh**

Find reciprocal best hit.

Aspect	Value
Layer	workflow
Category flags	COMMAND_EASY
Calls	convertalis, createdb, rbh, rmdb
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.1.5. **easy-search**

Sensitive homology search.

Aspect	Value
Layer	workflow
Category flags	COMMAND_EASY
Calls	convertalis, createdb, createlinindex, linsearch, rmdb, search, summarizeresult
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.1.6. **easy-taxonomy**

Taxonomic classification.

Aspect	Value
Layer	workflow

Aspect	Value
Category flags	COMMAND_EASY
Calls	addtaxonomy, convertalis, createdb, createtsv, filterdb, lca, rmdb, summarizealis, swapresults, taxonomy, taxonomyreport
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.2. Search Workflows

7.2.1. linsearch

Fast, less sensitive homology search.

Aspect	Value
Layer	high_level_api
Category flags	COMMAND_MAIN COMMAND_EXPERT
Calls	align, concatdbs, extractorfs, filterdb, kmersearch, offsetalignment, rescorediagonal, rmdb, swapresults
Called by	easy-linsearch, easy-search
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.2.2. map

Map nearly identical sequences.

Aspect	Value
Layer	high_level_api
Category flags	COMMAND_MAIN
Calls	search
Called by	n/a
Workflow scripts	n/a

Aspect	Value
Command reference	Open page
Functional module entry	Open module page

7.2.3. rbh

Reciprocal best hit search.

Aspect	Value
Layer	high_level_api
Category flags	COMMAND_MAIN
Calls	filterdb, mergedbs, result2rbh, rmdb, search, swapresults
Called by	easy-rbh
Workflow scripts	easyrbh.sh
Command reference	Open page
Functional module entry	Open module page

7.2.4. search

Sensitive homology search.

Aspect	Value
Layer	high_level_api
Category flags	COMMAND_MAIN
Calls	align, createsubdb, expand2profile, expandaln, extractframes, extractorfs, filterresult, lcaalign, mergedbs, mvdb, offsetalignment, prefilter, profile2consensus, rescorediagonal, result2profile, result2stats, rmdb, search, splitsequence, subtractdb, swapresults, ungappedprefilter
Called by	clusterupdate, easy-linsearch, easy-search, map, multihitsearch, rbh, search, taxonomy
Workflow scripts	enrich.sh, iterativepp.sh, map.sh, multihitsearch.sh, rbh.sh, taxonomy.sh, update_clustering.sh

Aspect	Value
Command reference	Open page
Functional module entry	Open module page

7.3. Clustering

7.3.1. clust

Cluster result by Set-Cover/Connected-Component/Greedy-Incremental.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_CLUSTER
Calls	n/a
Called by	cluster, linclust
Workflow scripts	cascaded_clustering.sh, clustering.sh, linclust.sh, nucleotide_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.3.2. cluster

Slower, sensitive clustering.

Aspect	Value
Layer	high_level_api
Category flags	COMMAND_MAIN
Calls	align, clust, clusthash, concatdb, createsubdb, extractframes, filterdb, linclust, mergeclusters, mergedbs, mvdb, offsetalignment, prefilter, rescorediagonal, rmdb, subtractdb, swapdb, tsv2db
Called by	clusterupdate, easy-cluster
Workflow scripts	update_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.3.3. clusterupdate

Update previous clustering with new sequences.

Aspect	Value
Layer	high_level_api
Category flags	COMMAND_MAIN
Calls	cluster, concatdbs, createsubdb, diffseqdbs, filterdb, mergedbs, mvdb, prefixid, renamedbkeys, result2repseq, rmdb, search, swapdb
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.3.4. clusthash

Hash-based clustering of equal length sequences.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_CLUSTER
Calls	n/a
Called by	cluster
Workflow scripts	clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.3.5. linclust

Fast, less sensitive clustering.

Aspect	Value
Layer	high_level_api
Category flags	COMMAND_MAIN
Calls	align, clust, createsubdb, filterdb, kmermatcher, mergeclusters, rescorediagonal, rmdb

Aspect	Value
Called by	cluster, easy-linclud
Workflow scripts	cascaded_clustering.sh, nucleotide_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.3.6. mergeclusters

Merge multiple cascaded clustering steps.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_CLUSTER
Calls	n/a
Called by	cluster, linclud
Workflow scripts	cascaded_clustering.sh, clustering.sh, linclud.sh, nucleotide_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.3.7. pickconsensusrep

Select new representatives for each cluster based on consensus.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_CLUSTER
Calls	align, msa2profile, prefixid, renamedbkeys, result2msa, rmdb, tsv2db
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.4. Prefiltering

7.4.1. countkmer

Count k-mers.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SPECIAL
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.4.2. gappedprefilter

Optimal Smith-Waterman-based prefiltering (slow).

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_PREFILTER
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.4.3. kmermatcher

Find bottom-m-hashed k-mer matches within sequence DB.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_PREFILTER
Calls	n/a
Called by	linclust
Workflow scripts	linclust.sh
Command reference	Open page

Aspect	Value
Functional module entry	Open module page

7.4.4. **kmersearch**

Find bottom-m-hashed k-mer matches between target and query DB.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_PREFILTER
Calls	n/a
Called by	linsearch
Workflow scripts	linsearch.sh
Command reference	Open page
Functional module entry	Open module page

7.4.5. **prefilter**

Double consecutive diagonal k-mer search.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_PREFILTER
Calls	n/a
Called by	cluster, search, taxonomy
Workflow scripts	blastp.sh, blastpgp.sh, cascaded_clustering.sh, clustering.sh, enrich.sh, iterativepp.sh, nucleotide_clustering.sh, searchslicedtargetprofile.sh, searchtargetprofile.sh, taxpercontig.sh, translated_search.sh
Command reference	Open page
Functional module entry	Open module page

7.4.6. **ungappedprefilter**

Optimal diagonal score search.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_PREFILTER
Calls	n/a
Called by	search
Workflow scripts	blastp.sh, blastpgp.sh
Command reference	Open page
Functional module entry	Open module page

7.5. Alignment

7.5.1. align

Optimal gapped local alignment.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_ALIGNMENT
Calls	n/a
Called by	cluster, linclust, linsearch, pickconsensusrep, search
Workflow scripts	iterativepp.sh, nucleotide_clustering.sh, pickconsensusrep.sh, searchslicedtargetprofile.sh
Command reference	Open page
Functional module entry	Open module page

7.5.2. alignall

Within-result all-vs-all gapped local alignment.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_ALIGNMENT
Calls	n/a
Called by	n/a
Workflow scripts	n/a

Aspect	Value
Command reference	Open page
Functional module entry	Open module page

7.5.3. **alignbykmer**

Heuristic gapped local k-mer based alignment.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_ALIGNMENT
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.5.4. **expandaln**

Expand an alignment result based on another.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_PROFILE_PROFILE
Calls	n/a
Called by	search
Workflow scripts	enrich.sh, iterativepp.sh
Command reference	Open page
Functional module entry	Open module page

7.5.5. **fwbw**

Forward Backward Alignment.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_ALIGNMENT
Calls	n/a

Aspect	Value
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.5.6. **offsetalignment**

Offset alignment by ORF start position.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_RESULT
Calls	n/a
Called by	cluster, linsearch, search
Workflow scripts	blastn.sh, linsearch.sh, nucleotide_clustering.sh, translated_search.sh
Command reference	Open page
Functional module entry	Open module page

7.5.7. **proteinaln2nucl**

Transform protein alignments to nucleotide alignments.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_RESULT
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.5.8. **rescorediagonal**

Compute sequence identity for diagonal.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_ALIGNMENT
Calls	n/a
Called by	cluster, linclust, linsearch, search, taxonomy
Workflow scripts	linclust.sh, linsearch.sh, nucleotide_clustering.sh, taxpercontig.sh
Command reference	Open page
Functional module entry	Open module page

7.5.9. **transitivealign**

Transfer alignments via transitivity.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_ALIGNMENT
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.6. Profiles

7.6.1. **convertca3m**

Convert a cA3M DB to a result DB.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_PROFILE_PROFILE
Calls	n/a
Called by	n/a
Workflow scripts	n/a

Aspect	Value
Command reference	Open page
Functional module entry	Open module page

7.6.2. **convertmsa**

Convert Stockholm/PFAM MSA file to a MSA DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION
Calls	n/a
Called by	databases
Workflow scripts	databases.sh
Command reference	Open page
Functional module entry	Open module page

7.6.3. **convertprofiledb**

Convert a HH-suite HHM DB to a profile DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_PROFILE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.6.4. **expand2profile**

Expand an alignment result based on another and create a profile.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_PROFILE_PROFILE
Calls	n/a

Aspect	Value
Called by	search
Workflow scripts	iterativepp.sh
Command reference	Open page
Functional module entry	Open module page

7.6.5. msa2profile

Convert a MSA DB to a profile DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_PROFILE COMMAND_DATABASE_CREATION
Calls	n/a
Called by	databases, pickconsensusrep
Workflow scripts	databases.sh, pickconsensusrep.sh
Command reference	Open page
Functional module entry	Open module page

7.6.6. msa2result

Convert a MSA DB to a profile DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_PROFILE COMMAND_EXPERT
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.6.7. pairaln

Pair sequences to match best protein A and B from a species.

Aspect	Value
Layer	low_level_api

Aspect	Value
Category flags	COMMAND_EXPERT
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.6.8. **profile2consensus**

Extract consensus sequence DB from a profile DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_PROFILE
Calls	n/a
Called by	search
Workflow scripts	iterativepp.sh
Command reference	Open page
Functional module entry	Open module page

7.6.9. **profile2neff**

Convert a profile DB to a tab-separated list of Neff scores.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_PROFILE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.6.10. **profile2pssm**

Convert a profile DB to a tab-separated PSSM file.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_PROFILE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.6.11. **profile2repseq**

Extract representative sequence DB from a profile DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_PROFILE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.6.12. **result2profile**

Compute profile DB from a result DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_PROFILE
Calls	n/a
Called by	search
Workflow scripts	blastpgp.sh, enrich.sh
Command reference	Open page
Functional module entry	Open module page

7.6.13. **sequence2profile**

Turn sequence into profile by adding context specific pseudo counts.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_PROFILE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.6.14. tsv2exprofiledb

Create a expandable profile db from TSV files.

Aspect	Value
Layer	mid_level_api
Category flags	COMMAND_PROFILE_PROFILE
Calls	aliasdb, compress, mvdb, rmdb, tsv2db
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.7. Database

7.7.1. aliasdb

Create relative symlink of DB to another name in the same folder.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_STORAGE
Calls	n/a
Called by	tsv2exprofiledb
Workflow scripts	tsv2exprofiledb.sh
Command reference	Open page
Functional module entry	Open module page

7.7.2. concatdbs

Concatenate two DBs, giving new IDs to entries from 2nd DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SET
Calls	n/a
Called by	cluster, clusterupdate, linsearch
Workflow scripts	linsearch.sh, nucleotide_clustering.sh, update_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.7.3. cpdb

Copy a DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_STORAGE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.7.4. createdb

Convert FASTA/Q file(s) to a sequence DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION
Calls	n/a
Called by	databases, easy-cluster, easy-linclud, easy-linsearch, easy-rbh, easy-search, easy-taxonomy, multihitdb

Aspect	Value
Workflow scripts	databases.sh, easycluster.sh, easyrbh.sh, easysearch.sh, easytaxonomy.sh, multihitdb.sh
Command reference	Open page
Functional module entry	Open module page

7.7.5. createindex

Store precomputed index on disk to reduce search overhead.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION
Calls	extractframes, extractorfs, rmdb, splitsequence
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.7.6. createlinindex

Create linsearch index.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION COMMAND_EXPERT
Calls	extractframes, extractorfs, rmdb, splitsequence
Called by	easy-linsearch, easy-search
Workflow scripts	easysearch.sh
Command reference	Open page
Functional module entry	Open module page

7.7.7. createsubdb

Create a subset of a DB from list of DB keys.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SET
Calls	n/a
Called by	cluster, clusterupdate, linclust, search, taxonomy
Workflow scripts	blastp.sh, cascaded_clustering.sh, clustering.sh, linclust.sh, nucleotide_clustering.sh, taxpercontig.sh, translated_search.sh, update_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.7.8. databases

List and download databases.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION
Calls	convertmsa, createdb, createtaxdb, msa2profile, nrtotaxmapping, prefixid, rmdb, tar2db
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.7.9. db2tar

Archive contents of a DB to a tar archive.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION COMMAND_EXPERT
Calls	n/a
Called by	n/a

Aspect	Value
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.7.10. **lndb**

Symlink a DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_STORAGE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.7.11. **mergedbs**

Merge entries from multiple DBs.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SET
Calls	n/a
Called by	cluster, clusterupdate, rbh, search
Workflow scripts	blastp.sh, blastpgp.sh, cascaded_clustering.sh, enrich.sh, iterativepp.sh, rbh.sh, searchslicedtargetprofile.sh, update_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.7.12. **mvdb**

Move a DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_STORAGE
Calls	n/a
Called by	cluster, clusterupdate, search, taxonomy, tsv2exprofiledb
Workflow scripts	blastp.sh, cascaded_clustering.sh, searchslicedtargetprofile.sh, taxonomy.sh, tsv2exprofiledb.sh, update_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.7.13. renamedbkeys

Create a new DB with original keys renamed.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DB
Calls	n/a
Called by	clusterupdate, pickconsensusrep
Workflow scripts	update_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.7.14. rmdb

Remove a DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_STORAGE
Calls	n/a
Called by	cluster, clusterupdate, createindex, createlinindex, databases, easy-cluster, easy-linclud, easy-linsearch, easy-rbh, easy-search, easy-taxonomy,

Aspect	Value
	linclust, linsearch, multihitsearch, pickconsensusrep, rbh, search, taxonomy, tsv2exprofiledb
Workflow scripts	blastn.sh, blastp.sh, blastpgp.sh, cascaded_clustering.sh, clustering.sh, createindex.sh, databases.sh, easycluster.sh, easyrbh.sh, easysearch.sh, easytaxonomy.sh, iterativepp.sh, linclust.sh, linsearch.sh, multihitsearch.sh, nucleotide_clustering.sh, pickconsensusrep.sh, rbh.sh, searchsllicedtargetprofile.sh, searchtargetprofile.sh, taxonomy.sh, taxpercontig.sh, translated_search.sh, tsv2exprofiledb.sh, update_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.7.15. **splitdb**

Split DB into subsets.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SET
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.7.16. **splitsequence**

Split sequences by length.

Aspect	Value
Layer	low_level_api

Aspect	Value
Category flags	COMMAND_SEQUENCE
Calls	n/a
Called by	createindex, createlindex, search
Workflow scripts	blastn.sh, createindex.sh
Command reference	Open page
Functional module entry	Open module page

7.7.17. **subtractdbs**

Remove all entries from first DB occurring in second DB by key.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SET
Calls	n/a
Called by	cluster, search
Workflow scripts	blastpgp.sh, cascaded_clustering.sh, enrich.sh, iterativepp.sh, nucleotide_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.7.18. **swapdb**

Transpose DB with integer values in first column.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DB
Calls	n/a
Called by	cluster, clusterupdate, multihitdb, taxonomy
Workflow scripts	cascaded_clustering.sh, multihitdb.sh, taxpercontig.sh, update_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.7.19. tar2db

Convert content of tar archives to any DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION COMMAND_EXPERT
Calls	n/a
Called by	databases
Workflow scripts	databases.sh
Command reference	Open page
Functional module entry	Open module page

7.7.20. tsv2db

Convert a TSV file to any DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DATABASE_CREATION COMMAND_EXPERT
Calls	n/a
Called by	cluster, multihitdb, pickconsensusrep, tsv2exprofiledb
Workflow scripts	cascaded_clustering.sh, multihitdb.sh, pickconsensusrep.sh, tsv2exprofiledb.sh
Command reference	Open page
Functional module entry	Open module page

7.8. Result Handling

7.8.1. convert2fasta

Convert sequence DB to FASTA format.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_FORMAT_CONVERSION
Calls	n/a
Called by	n/a

Aspect	Value
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.8.2. **convertalis**

Convert alignment DB to BLAST-tab, SAM or custom format.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_FORMAT_CONVERSION
Calls	n/a
Called by	easy-linsearch, easy-rbh, easy-search, easy-taxonomy
Workflow scripts	easyrbh.sh, easysearch.sh, easytaxonomy.sh
Command reference	Open page
Functional module entry	Open module page

7.8.3. **createseqfiledb**

Create a DB of unaligned FASTA entries.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_FORMAT_CONVERSION COMMAND_EXPERT
Calls	n/a
Called by	easy-cluster, easy-lincludst
Workflow scripts	easycluster.sh
Command reference	Open page
Functional module entry	Open module page

7.8.4. **createtsv**

Convert result DB to tab-separated flat file.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_FORMAT_CONVERSION
Calls	n/a
Called by	easy-cluster, easy-linclud, easy-taxonomy
Workflow scripts	easycluster.sh, easytaxonomy.sh
Command reference	Open page
Functional module entry	Open module page

7.8.5. extractdomains

Extract highest scoring alignment regions for each sequence from BLAST-tab file.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SPECIAL
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.8.6. filterresult

Pairwise alignment result filter.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_RESULT
Calls	n/a
Called by	search
Workflow scripts	searchslicedtargetprofile.sh
Command reference	Open page
Functional module entry	Open module page

7.8.7. **result2dnamsa**

Compute MSA DB with out insertions in the query for DNA sequences.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_RESULT
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.8.8. **result2flat**

Create flat file by adding FASTA headers to DB entries.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_FORMAT_CONVERSION COMMAND_EXPERT
Calls	n/a
Called by	easy-cluster, easy-linclud
Workflow scripts	easycluster.sh
Command reference	Open page
Functional module entry	Open module page

7.8.9. **result2msa**

Compute MSA DB from a result DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_RESULT
Calls	n/a
Called by	pickconsensusrep
Workflow scripts	pickconsensusrep.sh
Command reference	Open page

Aspect	Value
Functional module entry	Open module page

7.8.10. **result2rbh**

Filter a merged result DB to retain only reciprocal best hits.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_RESULT
Calls	n/a
Called by	rbh
Workflow scripts	rbh.sh
Command reference	Open page
Functional module entry	Open module page

7.8.11. **result2repseq**

Get representative sequences from result DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_RESULT
Calls	n/a
Called by	clusterupdate, easy-cluster, easy-linclud
Workflow scripts	easycluster.sh, update_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.8.12. **result2stats**

Compute statistics for each entry in a DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_RESULT
Calls	n/a
Called by	multihitdb, search

Aspect	Value
Workflow scripts	multihitdb.sh, searchslicedtargetprofile.sh
Command reference	Open page
Functional module entry	Open module page

7.8.13. sortresult

Sort a result DB in the same order as the prefilter or align module.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_RESULT
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.8.14. summarizealis

Summarize alignment result to one row (uniq. cov., cov., avg. seq. id.).

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_RESULT
Calls	n/a
Called by	easy-taxonomy
Workflow scripts	easytaxonomy.sh
Command reference	Open page
Functional module entry	Open module page

7.8.15. summarizeheaders

Summarize FASTA headers of result DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SPECIAL

Aspect	Value
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.8.16. **summarizerresult**

Extract annotations from alignment DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_RESULT
Calls	n/a
Called by	easy-linsearch, easy-search
Workflow scripts	easysearch.sh
Command reference	Open page
Functional module entry	Open module page

7.8.17. **swapresults**

Transpose prefilter/alignment DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_RESULT
Calls	n/a
Called by	easy-taxonomy, linsearch, rbh, search
Workflow scripts	easytaxonomy.sh, linsearch.sh, rbh.sh, searchslcedtargetprofile.sh, searchtargetprofile.sh
Command reference	Open page
Functional module entry	Open module page

7.9. Sequence Manipulation

7.9.1. **extractalignedregion**

Extract aligned sequence region from query.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SEQUENCE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.9.2. **extractframes**

Extract frames from a nucleotide sequence DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SEQUENCE
Calls	n/a
Called by	cluster, createindex, createlinindex, search
Workflow scripts	blastn.sh, createindex.sh, nucleotide_clustering.sh, translated_search.sh
Command reference	Open page
Functional module entry	Open module page

7.9.3. **extractorfs**

Six-frame extraction of open reading frames.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SEQUENCE
Calls	n/a

Aspect	Value
Called by	createindex, createlinindex, linsearch, multihitdb, search, taxonomy
Workflow scripts	createindex.sh, multihitdb.sh, taxpercontig.sh, translated_search.sh
Command reference	Open page
Functional module entry	Open module page

7.9.4. masksequence

Soft mask sequence DB using tantan.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SEQUENCE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.9.5. orftocontig

Write ORF locations in alignment format.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SEQUENCE
Calls	n/a
Called by	multihitdb
Workflow scripts	multihitdb.sh
Command reference	Open page
Functional module entry	Open module page

7.9.6. recoverlongestorf

Recover longest ORF for taxonomy annotation after elimination.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_EXPERT
Calls	n/a
Called by	taxonomy
Workflow scripts	taxpercontig.sh
Command reference	Open page
Functional module entry	Open module page

7.9.7. **reverseseq**

Reverse (without complement) sequences.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SEQUENCE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.9.8. **translateaa**

Translate proteins to lexicographically lowest codons.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SEQUENCE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.9.9. **translatenucs**

Translate nucleotides to proteins.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SEQUENCE
Calls	n/a
Called by	multihitdb
Workflow scripts	multihitdb.sh
Command reference	Open page
Functional module entry	Open module page

7.10. Taxonomy

7.10.1. addtaxonomy

Add taxonomic labels to result DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_TAXONOMY COMMAND_EXPERT
Calls	n/a
Called by	easy-taxonomy
Workflow scripts	easytaxonomy.sh
Command reference	Open page
Functional module entry	Open module page

7.10.2. aggregatetax

Aggregate multiple taxon labels to a single label.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_TAXONOMY
Calls	n/a
Called by	taxonomy
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.10.3. aggregatetaxweights

Aggregate multiple taxon labels to a single label.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_TAXONOMY
Calls	n/a
Called by	taxonomy
Workflow scripts	taxpercontig.sh
Command reference	Open page
Functional module entry	Open module page

7.10.4. createbintaxmapping

Create binary taxonomy mapping from tabular taxonomy mapping.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_TAXONOMY COMMAND_EXPERT
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.10.5. createbintaxonomy

Create binary taxonomy from NCBI input.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_TAXONOMY COMMAND_EXPERT
Calls	n/a
Called by	createtaxdb
Workflow scripts	createtaxdb.sh
Command reference	Open page
Functional module entry	Open module page

7.10.6. **createdmptaxonomy**

Create dmp files from binary taxonomy.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_TAXONOMY COMMAND_EXPERT
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.10.7. **createtaxdb**

Add taxonomic labels to sequence DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_TAXONOMY
Calls	createbintaxonomy
Called by	databases
Workflow scripts	databases.sh
Command reference	Open page
Functional module entry	Open module page

7.10.8. **filtertaxdb**

Filter taxonomy result database.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_TAXONOMY
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.10.9. **filtertaxseqdb**

Filter taxonomy sequence database.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_TAXONOMY
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.10.10. **lca**

Compute the lowest common ancestor.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_TAXONOMY
Calls	n/a
Called by	easy-taxonomy, taxonomy
Workflow scripts	taxonomy.sh
Command reference	Open page
Functional module entry	Open module page

7.10.11. **lcaalign**

Efficient gapped alignment for lca computation.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_TAXONOMY
Calls	n/a
Called by	search
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.10.12. majoritylca

Compute the lowest common ancestor using majority voting.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_TAXONOMY COMMAND_EXPERT
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.10.13. nrtotaxmapping

Create taxonomy mapping for NR database.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SPECIAL
Calls	n/a
Called by	databases
Workflow scripts	databases.sh
Command reference	Open page
Functional module entry	Open module page

7.10.14. taxonomy

Taxonomic classification.

Aspect	Value
Layer	high_level_api
Category flags	COMMAND_MAIN
Calls	aggregatetax, aggregatetaxweights, createsubdb, extractorfs, filterdb, lca, mergeresultsbyset, mvdb, prefilter, recoverlongestorf, rescorediagonal, rmdb, search, swapdb, taxonomy
Called by	easy-taxonomy, taxonomy

Aspect	Value
Workflow scripts	<code>easytaxonomy.sh</code> , <code>taxpercontig.sh</code>
Command reference	Open page
Functional module entry	Open module page

7.10.15. **taxonomyreport**

Create a taxonomy report in Kraken or Krona format.

Aspect	Value
Layer	<code>low_level_api</code>
Category flags	<code>COMMAND_TAXONOMY COMMAND_FORMAT_CONVERSION</code>
Calls	n/a
Called by	<code>easy-taxonomy</code>
Workflow scripts	<code>easytaxonomy.sh</code>
Command reference	Open page
Functional module entry	Open module page

7.11. Multi Hit

7.11.1. **besthitperset**

For each set of sequences compute the best element and update p-value.

Aspect	Value
Layer	<code>high_level_api</code>
Category flags	<code>COMMAND_MULTIHIT</code>
Calls	n/a
Called by	<code>multihitsearch</code>
Workflow scripts	<code>multihitsearch.sh</code>
Command reference	Open page
Functional module entry	Open module page

7.11.2. **combinepvalperset**

For each set compute the combined p-value.

Aspect	Value
Layer	<code>high_level_api</code>

Aspect	Value
Category flags	COMMAND_MULTIHIT
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.11.3. **mergeresultsbyset**

Merge results from multiple ORFs back to their respective contig.

Aspect	Value
Layer	high_level_api
Category flags	COMMAND_MULTIHIT
Calls	n/a
Called by	multihitsearch, taxonomy
Workflow scripts	multihitsearch.sh, taxpercontig.sh
Command reference	Open page
Functional module entry	Open module page

7.11.4. **multihitdb**

Create sequence DB for multi hit searches.

Aspect	Value
Layer	high_level_api
Category flags	COMMAND_MULTIHIT
Calls	createdb, extractorfs, filterdb, orftocontig, result2stats, swapdb, translatenucs, tsv2db
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.11.5. multihitsearch

Search with a grouped set of sequences against another grouped set.

Aspect	Value
Layer	high_level_api
Category flags	COMMAND_MULTIHIT
Calls	besthitperset, mergeresultsbyset, rmdb, search
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.12. Utilities

7.12.1. apply

Execute given program on each DB entry.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DB
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.12.2. compress

Compress DB entries.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_STORAGE
Calls	n/a
Called by	tsv2exprofiledb
Workflow scripts	tsv2exprofiledb.sh

Aspect	Value
Command reference	Open page
Functional module entry	Open module page

7.12.3. **convertkb**

Convert UniProtKB data to a DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SPECIAL
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.12.4. **decompress**

Decompress DB entries.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_STORAGE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.12.5. **diffseqdbs**

Compute diff of two sequence DBs.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SPECIAL
Calls	n/a

Aspect	Value
Called by	clusterupdate
Workflow scripts	update_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.12.6. filterdb

DB filtering by given conditions.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DB
Calls	n/a
Called by	cluster, clusterupdate, easy-taxonomy, linclust, linsearch, multihitdb, rbh, taxonomy
Workflow scripts	cascaded_clustering.sh, easytaxonomy.sh, linclust.sh, linsearch.sh, multihitdb.sh, rbh.sh, taxonomy.sh, taxpercontig.sh, update_clustering.sh
Command reference	Open page
Functional module entry	Open module page

7.12.7. gff2db

Extract regions from a sequence database based on a GFF3 file.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SPECIAL
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.12.8. gpuserver

Start a GPU server.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_STORAGE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.12.9. maskbygff

Mask out sequence regions in a sequence DB by features selected from a GFF3 file.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SPECIAL
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.12.10. prefixid

For each entry in a DB prepend the entry key to the entry itself.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DB
Calls	n/a
Called by	clusterupdate, databases, pickconsensusrep
Workflow scripts	databases.sh, pickconsensusrep.sh, update_clustering.sh
Command reference	Open page

Aspect	Value
Functional module entry	Open module page

7.12.11. **setextendeddbtype**

Write an extended DB.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DB
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.12.12. **suffixid**

For each entry in a DB append the entry key to the entry itself.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DB
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.12.13. **summarizetabs**

Extract annotations from HHblits BLAST-tab-formatted results.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_SPECIAL
Calls	n/a
Called by	n/a

Aspect	Value
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.12.14. **touchdb**

Preload DB into memory (page cache).

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_STORAGE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.12.15. **unpackdb**

Unpack a DB into separate files.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_STORAGE
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page

7.12.16. **view**

Print DB entries given in -id-list to stdout.

Aspect	Value
Layer	low_level_api
Category flags	COMMAND_DB

Aspect	Value
Calls	n/a
Called by	n/a
Workflow scripts	n/a
Command reference	Open page
Functional module entry	Open module page